Conviction Formation Theory

A cycle of essays about what can carry thinking and action amid competing convictions and standards, without certainty or a final foundation.

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I. Forming And Revising Convictions

A first cycle of essays examining how convictions form, how inquiry can reorganize them, and how finite agents participate without fully controlling what becomes convincing.

Conviction Formation

1. The Predicament of Finite Orientation

We live among people who understand the world differently.

Sometimes the differences are small. Two engineers disagree about the likely cause of a failure. Two friends remember a conversation differently. A doctor and patient assign different weight to the possible side effects of a treatment.

Sometimes the differences reach much further.

Religions disagree about what exists and what ultimately matters. Philosophical traditions disagree about knowledge, morality, mind, freedom, and reality. Political traditions disagree about justice and legitimate authority. Scientific theories can disagree about how observed phenomena should be explained. Cultures and institutions differ in their expectations, categories, and practices. Individual people inherit different histories and develop different ways of understanding what happens around them.

People can disagree about what evidence is, which authorities deserve trust, what constitutes a good explanation, which distinctions matter, what should be regarded as morally relevant, or which procedures are legitimate.

But people with very different comprehensive views also share large regions of practice. They may use the same arithmetic, repair the same machines, recognize the same traffic signs, measure temperatures with the same thermometer, follow the same legal procedure, or discover the same bug in a computer program. A devout Christian, a Buddhist, an atheist, and someone who has never cared much about religion need not develop different methods for determining whether a door is locked.

Some things settle easily between people who disagree deeply elsewhere. Other things remain disputed among people who otherwise understand one another very well.

The practical difficulty is that we cannot first settle all of this.

A person considering whether to leave a marriage cannot postpone the decision

until moral philosophy has been completed. A judge cannot wait until philosophers agree on the nature of truth and responsibility. Someone deciding whether to trust a warning must act before possessing a final theory of testimony. A voter cannot inspect every relevant fact and resolve every dispute about political legitimacy before making a choice. A scientist working between competing explanations cannot stand outside all theoretical commitments before interpreting an experiment.

We may have criteria, methods, arguments, and standards that convince us strongly. But their authority is not simply available from a final standpoint outside every further question. They too can be challenged, defended, revised, or rejected.

We believe and doubt. We inquire. We trust and distrust. We decide what deserves attention, make commitments, coordinate with others, and revise our judgments.

We act, and sometimes discover later that we acted badly.

The problem is not merely that human beings lack certainty. It is that finite agents must orient themselves while many of the larger questions that could ground their orientation remain unsettled.

One response would be to search for the correct comprehensive framework first.

That project may be worthwhile.

But it creates a practical problem of its own. For the search itself we already need to judge evidence, understand arguments, trust some procedures, reject others, decide which questions matter, and continue living while the search proceeds. We cannot suspend every conviction in order to investigate conviction from nowhere.

What then can finite agents investigate and do while these larger disputes remain unresolved?

People become convinced of things. Some matters seem obvious to them. Others remain doubtful. Some judgments survive criticism for decades. Others disappear after one conversation. Some ways of reasoning exert enormous force. Some arguments leave one person convinced and another untouched. New experiences reorganize old judgments. Questions reopen matters that had seemed settled.

These processes happen whether or not we possess a final philosophical explanation of them.

A theory that cannot accommodate the fact that people become convinced differently under different orientations would fail to describe the phenomenon it is meant to explain.

Thus instead of first asking what final standard should govern conviction, we can decide to examine how conviction actually forms, acquires force, stabilizes, changes, and participates in orientation and action.

We can ask how far we can proceed before the larger questions are settled.

This does not replace the question, “Is this true?” with the question, “Does this convince me?”

These are different questions and they deserve different treatment.

But choosing a preferred framework of truth, justification, rationality, or inquiry does not make the phenomenon of conviction disappear. Its arguments, standards, and procedures must themselves acquire convincing force for finite agents, and can also become objects of investigation.

Conviction formation can become an object of investigation even while disputes about truth, justification, rationality, and inquiry continue.

2. What Is a Conviction?

Suppose someone tells you: “The meeting starts at ten.”

You may repeat the sentence without believing it. You may have been asked to pass the message along.

You may act on it without being convinced. Maybe your manager told you to arrive at ten and you obey, although you privately suspect the meeting actually starts at eleven.

You may agree for strategic reasons. An argument about the starting time can have become tiresome and you say, “Fine, you’re right,” merely to end it.

You may also genuinely expect the meeting to begin at ten. You arrange your morning around it. You become surprised when nobody appears. You begin looking for an explanation.

The last is the kind of case that interests us here.

A conviction is something a person is convinced of. But that only moves the problem further, to a definition of *being convinced*.

We can begin with a clear central case:

A person who is convinced of something ordinarily relies on it as a basis for further expectation, interpretation, reasoning, judgment, or action without even getting the idea that there is something to settle at that moment.

That means being convinced is not directly observable in another person. No single behavior reveals it. People can speak and act under instruction, coercion, convention, habit, strategy, or on the basis of assumptions they do not themselves believe. What we observe instead are patterns of reliance, expectation, surprise, reconsideration, and action from which conviction can be inferred.

Convictions can shape interpretation and reasoning, guide judgment and action, and affect what appears surprising, threatening, relevant, possible, admirable, or absurd.

But they do not need to be simply present or absent.

Sometimes the language of degree seems natural.

"I think Peter did it, but I'm not sure."

"I'm almost certain the train leaves at six."

"I have a vague suspicion that something is wrong."

In such cases, how confident we are and whether we rely on something can come apart. We may rely on something while remaining uncertain about it, or be highly confident about something that matters little for what we do next.

Conviction also need not be consistent throughout a person. Different parts of their orientation can support incompatible patterns of reliance at the same time.

Still, these cases resemble the clear case closely enough that it remains useful to call them convictions.

The word *belief* will remain useful in its ordinary broad sense. *Conviction* will be used when the convincing force of something, or the way it participates in further orientation, is specifically at issue.

Convictions need not be explicitly formulated.

You may never have stated that a certain friend is dependable. Yet you call them first when something goes wrong.

A skilled programmer may rely on distinctions that would take considerable effort to put into words. Someone walking through a familiar city may expect a street to continue behind a building without ever explicitly thinking, "The street continues behind that building."

Much of what guides us operates without first becoming a sentence.

This also limits how directly we can inspect it. We sometimes discover what mattered to us only after acting, hesitating, regretting, or being challenged.

We do not have complete access to conviction formation, but we seem to be able to observe enough of it to investigate it.

3. Conviction Is Not Simply Chosen

There is an easy experiment.

Take a statement you currently find implausible.

Suppose, for example, that you do not believe ancient extraterrestrials visited Earth and helped build early civilizations.

Now decide to believe it.

Not to imagine it. Or to entertain the possibility. Or to repeat the sentence.

Not to play the role of someone who believes it.

Simply become convinced that it happened.

For many people, nothing much follows from the decision.

They can say, "Ancient aliens built the pyramids," write it down, or imagine how the world would look if it were true.

They could even spend an afternoon behaving as if they believed it.

But the proposition still does not become convincing merely because they have issued themselves an instruction.

If you already believe the ancient-aliens claim, reverse the experiment. Decide that it is false.

The same pattern is easy to reproduce with other examples. Clear counterexamples are much harder to find.

More than that, the idea of simply deciding to become convinced of something, without changing anything else, is difficult even to make sense of as an ordinary act of choice.

Could you decide to become convinced that a spider is cuter than a ladybug?

Our practices of persuasion, inquiry, teaching, and even manipulation seem to assume the opposite: we try to change what someone encounters, notices, considers, or experiences so that conviction may change.

This gives us the fundamental claim of Conviction Formation Theory:

Becoming convinced is not under direct voluntary control.

This is a substantive claim, and it can be challenged. A genuine case in which a person is not convinced of something, deliberately decides to become convinced of it, and becomes convinced as a consequence of that decision would count against it.

More concretely, a counterexample would have roughly this form:

A person is not convinced of P.
They deliberately decide to become convinced of P.
They then become convinced of P as a consequence of that decision, with becoming convinced of P itself under their voluntary control.

Intermediate psychological changes would not by themselves disqualify such a case. A decision might redirect attention, alter how something is interpreted, or set other processes in motion.

The question is whether the person can intentionally bring about becoming convinced of P by deciding to do so, rather than merely alter conditions under which conviction may later form without controlling the result.

In an actual case, distinguishing these possibilities may not always be easy. Particular cases would have to be decided on the evidence.

A convincing counterexample would be interesting beyond Conviction Formation Theory. If people can sometimes bring conviction under direct voluntary control, even under restricted conditions, that would reveal a capacity whose boundaries would immediately become worth investigating. Could it be learned or trained? Does it occur only under unusual conditions? Could someone acquire a useful conviction they currently lack by fiat, simply because having it would help them? These questions sound strange precisely because the capacity itself currently seems strange. A clear counterexample would make them empirical questions.

This matters for the problem with which we began. We cannot solve disagreement simply by deciding which convictions, criteria, or standards should govern us and then installing them by choice. Whatever role they acquire in our thinking, they too have to become sufficiently convincing to operate there.

The distinction becomes especially clear in ordinary perception.

Look outside on a clear day.

Suppose the sky appears blue.

Now decide that it looks red.

You can say that it is red, imagine a red sky, close your eyes, or look through a colored filter. You can remind yourself that illumination, atmospheric conditions, and human vision are complicated.

But while you continue looking under ordinary conditions, you do not simply choose what appears.

Under ordinary conditions, what we perceive strongly constrains what we become convinced of about our immediate surroundings.

We can decide where to look, inspect something again, move closer, switch on a light, or ask someone else what they see.

But once we look, that part of what follows is not decided by us.

Something appears.

We can deliberately initiate some of the processes under which convictions form, but we do not directly determine their result.

Mental arithmetic offers another familiar example.

Calculate 7 + 5.

You can decide to perform the calculation.

You can also decide whether to use your fingers, count upward, remember an arithmetic fact, or check with another method.

But you do not choose which result will strike you as correct.

You can write "13".

That does not make 13 convincing.

The influence we can exert is indirect.

Consider the ancient-aliens example again. Instead of issuing yourself the command "Believe this," change the conditions. For weeks, read only material defending the idea. Watch sympathetic documentaries. Spend time with people who already accept it. Avoid criticism. Encounter the same examples repeatedly.

Maybe nothing changes. But maybe what at first seemed absurd could begin to seem possible, then even convincing.

The same indirect structure appears in very different settings, but the conditions matter. In the ancient-aliens example, influences that might push the other way are deliberately reduced. In other situations, they remain present.

A mathematics teacher chooses examples and exercises but cannot choose what the student will understand. A scientist can run another experiment without being able to control what the result will convince her of. Someone angry can postpone a message until morning because what seems convincing in anger may not remain convincing later.

In all these cases, people alter conditions under which conviction may later change, without controlling all the other influences at work: information, attention, exposure, practice, social surroundings, timing, questions, procedures, or physical circumstances. None guarantees a particular result.

Lack of direct control is not lack of influence.

We can participate in the conditions under which conviction forms without choosing the result.

4. Conviction Forms Against a Background

Imagine receiving a short message from a close friend: "Fine."

What does it mean?

There is no answer in the word alone.

After proposing a meeting at seven, it may signal agreement. After an hour of arguing, it may sound irritated. A friend who usually writes in short messages

may make nothing of it, while the same word from someone normally warm and expansive may seem alarming.

A tiny piece of information enters an already structured situation.

The situation helps determine what the message can become evidence for.

Conviction forms against such a background. When something new occurs, much is already in place.

We have previous convictions, expectations, language, concepts, memories, skills, habits, and relationships. We have distinctions, ways of interpreting situations, social practices, physical circumstances, questions already occupying us, and practical commitments.

New information enters this background and may confirm, disturb, extend, or reorganize parts of it.

This presently operative background through which a person finds their bearings can be called *orientation*.

It may include convictions, but it is broader than conviction. An unresolved question, a practical skill, an unformulated expectation, a perceived possibility, or a commitment such as having promised to meet someone tomorrow can all be part of someone's orientation. So can memories and relationships that make some interpretations immediately plausible and others remote.

Orientation is an organizing term for these already operative conditions.

This matters because new convictions depend partly on what is already there.

Consider a toddler and an astronomer looking at the night sky.

The same visible points of light can participate in very different orientations.

The astronomer may see a planet, recognize a constellation, notice atmospheric distortion, or infer something from apparent movement.

The toddler does not merely possess fewer propositions about the same scene. Different distinctions are available, other questions can arise, and different things may become salient.

Or consider a sentence in an unfamiliar language.

For someone who cannot read it, the marks may be recognizable as writing without carrying much further structure.

For a fluent reader, meaning can appear almost automatically.

Training has changed what the same marks can do.

New information does not arrive into an empty space. It is encountered through the active elements of an existing orientation.

Stories

One especially important way human beings organize situations is through stories.

Suppose you see a colleague leave a meeting abruptly.

You might connect the event with what happened earlier.

Someone may have insulted her.

She may have received an urgent message.

Perhaps she was angry about a decision.

Or she simply had another appointment.

The event acquires different meanings depending on the surrounding story.

Stories can organize who is involved, what happened, how events connect, what preceded them, what people intended, what might happen next, and what matters about all of it.

This is not restricted to polished narratives.

Much everyday understanding has this form.

Someone says:

"He has been distant since Monday."

"We were getting along until that argument."

"She promised to call, but then this happened."

The events are being connected into a provisional whole that makes further expectation and action possible.

Stories can participate strongly in conviction formation. If you become convinced that a colleague deliberately humiliated you, later events may be interpreted against that background. A neutral remark can acquire significance. An apology may be treated as insincere. A forgotten meeting may become further evidence.

We often organize particular situations this way. A disagreement with a friend becomes part of the story of that relationship. A difficult year becomes part of the story of a life. Events at work become part of a story about an institution, a profession, or how things are done there. These stories often connect with broader ones about groups, places, societies, or a life as a whole.

Other parts of orientation need not be organized primarily as stories. Navigating a room may rely mainly on spatial orientation. Performing a familiar task can be procedural. A mathematical proof may be represented symbolically or diagrammatically. An affective response may arise before any story about it becomes available.

Orientation is the broader term. Stories are one important way in which parts of it can be organized and connected.

5. Recurring Conditions of Conviction Formation

We can now look more closely at some conditions under which convictions form and acquire force.

What follows is not an exhaustive list.

Some are ordinary observations. Some are plausible interpretations of those observations. Their interaction is likely to vary substantially between persons, domains, and situations.

The purpose is to identify recurring patterns worth investigating.

Attention and Salience

Imagine walking through a city with an architect.

You notice shops, people, traffic, perhaps whether the street feels pleasant.

The architect notices window proportions, materials, load-bearing structures, the transition between old and new buildings.

Walk the same street with a security specialist and different things may stand out again.

Entrances, blind corners, lighting, and crowd movement for example.

Nothing requires the three people to disagree.

They may simply be attending to different things.

Reasoning does not usually begin from a neutral inventory of everything present.

Something first becomes relevant enough to enter the process.

A remark that previously passed unnoticed becomes an insult.

Cracks in a wall may suddenly matter once you know what structural movement looks like.

A teacher can point out a musical detail that you had never noticed before.

And after learning what a symptom can indicate, it may become alarming.

Attention can sometimes be directed deliberately.

Look here. Compare these two. Listen to the bass line. Check what happens immediately before the error.

But salience can also be produced by events themselves.

A loud noise interrupts. Pain captures attention. A frightening possibility becomes difficult not to consider.

Texts, diagrams, interfaces, environments, and other people can all reorganize what stands out.

Differences in conviction begin before explicit disagreement.

One person sees a problem. Another does not yet see anything there to disagree about.

What becomes salient can alter what becomes available for further conviction formation.

Perception and Affect

Some convictions acquire force without first passing through an explicit argument.

You look outside and see that it is raining. You hear a crash and become convinced that something fell.

In other cases, affect participates in what becomes convincing. A remark strikes you as cruel. Someone you trusted suddenly seems threatening. A piece of music strikes you as beautiful.

These are distinguishable cases. Perception, affect, evaluation, and interpretation may contribute in different ways. What they share is only that explicit reasoning need not come first.

Pain, fear, disgust, attraction, empathy, and anger can change what becomes salient and how a situation is interpreted before any explicit argument is made. Anger, for example, can temporarily make hostile interpretations seem compelling and alternatives remote. When it subsides, some of that convincing force may disappear with it.

Experience, Repetition, and Practice

Convictions can acquire stability through repeated experience.

A friend repeatedly keeps promises.

The same instrument gives the expected reading again and again.

A familiar procedure continues to produce the expected result.

Trust can grow without any single decisive event.

But repetition does not strengthen conviction by itself.

Exposure can produce boredom or resistance, something initially striking can disappear into the background, and repeated failure can weaken confidence instead.

What matters is what happens through repeated encounters within a particular orientation. Repeated success can stabilize a conviction. Repeated failure can weaken one. Neither result follows from repetition alone.

Experience and practice can also change the orientation against which later convictions form.

Consider learning to drive.

At first, many things require deliberate attention.

Mirrors.

Clutch.

Gear.

Speed.

Lane position.

Other vehicles.

With practice, distinctions that once demanded effort become readily available, and some responses become almost automatic.

Something similar happens in more specialized practices.

An experienced radiologist may notice distinctions in an image that a beginner misses. Similar changes occur in other learned practices: a musician begins to recognize relations that once passed unnoticed, or a mathematician sees structures that previously required laborious reconstruction.

Practice has not merely added information. It has changed what can be noticed, recognized, or handled readily.

Whatever the underlying cognitive explanation, experience and learned practices can alter both existing convictions and the conditions under which later conviction forms.

Interaction and Social Reinforcement

Convictions also form between people.

You tell someone what happened.

They believe you.

Or they look doubtful.

Someone points out an alternative you had not considered.

A group may laugh at a remark.

The teacher corrects your answer.

A colleague says, “That is exactly what I saw.”

Or a friend asks, “Are you sure?”

Such responses can reinforce, destabilize, redirect, or leave conviction untouched.

People encounter not only explicit instruction, but approval, correction, imitation, surprise, disapproval, and expectations about what “people like us” do.

Over time, some of these patterns can become deeply familiar.

Similar processes continue in adult life.

Professions and institutions preserve methods and procedures. Religious, scientific, familial, and political communities can carry texts, practices, expectations, categories, and standards across long periods.

Convictions can acquire part of their stability through participation in durable social structures.

And when those structures change, conviction may change with them.

Integration

Some convictions are hard to disturb because many parts of a person’s orientation support them at once.

Suppose someone has been convinced for decades that close friendship is important.

That conviction may connect with memories, habits, plans, identity, relationships, moral judgments, and expectations about a worthwhile life.

Challenging it is not like replacing one isolated item in a list.

Many things may have to move.

A political or religious conviction may become connected with relationships, identity, memories, practices, and ways of interpreting events. Convictions about oneself can similarly affect which ambitions seem possible and which failures seem intelligible.

This suggests a strong hypothesis:

Convictions integrated with many active parts of orientation may often resist local disturbance more strongly than relatively isolated convictions.

But the hypothesis should not be mistaken for a principle of justification.

Integration can support false convictions.

A conspiracy theory can integrate observations, identity, community, and distrust into a highly resilient structure.

Removing one piece may accomplish little because the rest continues to hold.

This is a hypothesis about persistence, not about merit. A conviction does not become better justified merely because many parts of a person's orientation support it.

That distinction matters because the same conditions can play different roles. Reasons, evidence, methods, and criteria may give us grounds for judging one conviction better supported than another. They may also participate in changing what someone is actually convinced of. Since conviction is not under direct voluntary control, these two questions cannot simply be collapsed: what should convince us, and what in fact becomes convincing, are related but distinct.

Here we are concerned with the second question. We can investigate why an integrated conviction survives local challenge without treating that survival as evidence in its favor.

Integration may explain persistence. It does not vindicate what persists.

6. Stabilization

The previous examples repeatedly used words such as stable, resilient, and settled.

A conviction is stabilized insofar as it continues to operate despite disturbances that might otherwise have changed it, over some period and under some conditions.

Stability is local rather than absolute.

You can remain convinced that a colleague is dependable until an unexplained event occurs. Political convictions may survive criticism within one social environment and weaken after years spent in another. A moral judgment that seems settled in calm conditions may become less secure under fear or anger.

A conviction is stable for this person. Under these conditions. Against these disturbances. Over this period.

Different stabilizing conditions may also combine. Perception may repeatedly confirm what is expected, while habit can reduce occasions for reconsideration. A social group may reinforce a conviction. Formal arguments can connect convictions with other accepted premises. Repeated practical success may strengthen reliance. Identity can make alternatives difficult to integrate.

Stabilization is not certainty. A person can retain doubts while continuing to rely on a conviction.

Nor is stabilization permanence. Even convictions that survive for decades can change.

Stability tells us something about how a conviction persists, not about its merit.

7. Truth and Conviction Formation

At this point we can return briefly to the problem with which we began.

Different frameworks make different claims about what is true.

They may also disagree about how truth should be understood, which evidence matters, or what procedures deserve authority.

Conviction Formation Theory does not set out to settle these disputes.

But it still matters which questions about truth bear on the present investigation.

Several different issues have sometimes been grouped together under the word *truth*. Separating them helps show which of those issues matter here.

First, there is an ordinary, pretheoretic use of truth in which things can be a certain way independently of what someone happens to believe, and what we say can get them right or wrong.

If I am convinced that my keys are on the table and they are actually in my coat, my conviction does not relocate the keys.

Second, there are philosophical theories of truth.

Correspondence theories, deflationary theories, pragmatic theories, pluralist theories, and others attempt to explain what truth consists in or what truth-talk does.

Third, there is a stronger epistemic hope sometimes attached to truth: that appeal to truth might itself provide finite agents with a final, non-circular procedure telling them what to believe, or a foundation from which inquiry can be conclusively settled.

Whether such a foundation is available is a different problem. Nothing here requires us to assume that it is. Nothing shows that it is not.

The methodological claim is:

Many questions concerning conviction formation can be investigated without first settling a theory of truth or obtaining a final epistemic foundation.

This applies to Conviction Formation Theory itself. A reader cannot simply decide to become convinced by it. The distinctions and arguments offered here must acquire enough force to be taken up, examined, and used. They need not first be established as the final truth about conviction formation before that can happen. A framework can become useful in inquiry while remaining open to criticism, revision, or rejection.

Being convincing does not thereby make Conviction Formation Theory true. Its substantive claims can still turn out to be untenable. But their truth need not be settled in advance for the investigation to begin.

We can keep two questions apart:

How does this become convincing?

and

Is it true?

The present investigation addresses the first, but not the second.

But the idea that our convictions can sometimes become better constrained because they answer to how things are may itself be a very convincing explanation of some of our abilities to form and correct convictions.

We do not need to possess a detailed theory of correspondence that convinces us in every respect for that explanation to have force.

Conviction can outrun theory here too.

8. Destabilization and Change

Convictions do not only form and stabilize.

They weaken, fracture, become qualified or reorganized, and sometimes disappear.

The processes are often easiest to see when something that once felt obvious stops doing so.

Contradiction and Tension

Suppose Peter is accused of a crime.

A witness identifies him at the scene.

But Peter's friend insists that Peter was with him somewhere else at the relevant time.

If both claims initially convince you, they cannot remain comfortably side by side once the conflict becomes salient.

The witness may have been mistaken, or the friend may be lying. Perhaps the timeline was misunderstood, Peter left and returned, or one of the reports was described inaccurately.

The contradiction does not determine which conviction should change.

It creates pressure.

People respond to such pressure in different ways.

They may revise or qualify one conviction, reinterpret the situation, compartmentalize the conflict, avoid it, or abandon one of the convictions.

Sometimes they tolerate inconsistency for a long time until circumstances force it into attention.

A contradiction that is never noticed need not destabilize anything.

A contradiction that becomes salient may reorganize an entire field.

Formal systems make some contradictions difficult to ignore because the conflicting consequences can be explicitly reconstructed. Everyday life is often less tidy.

Questions

A question can also destabilize without directly asserting the opposite of a conviction.

Suppose you have become convinced that a colleague ignored you deliberately.

Then someone asks: "What else could explain it?"

All of a sudden, alternatives may reopen.

The colleague may not have seen you, may have been distracted, or perhaps was responding to something else.

A question can turn what had been treated as settled back into something examinable, make an assumption visible, redirect attention, or create space around a conclusion.

This does not mean questioning always weakens conviction.

A challenge may strengthen it.

Unanswered questions may disappear.

Good questions may expose an overlooked difficulty, while a bad one merely distracts.

The role of questions is large enough to require separate treatment.

For now, it is enough that questions can reopen parts of orientation that had become relatively closed.

Failed Practice

Some convictions weaken in use.

Suppose you are convinced that a particular shortcut saves time.

You take it repeatedly.

Again and again, traffic makes the route slower.

At some point the conviction may become difficult to sustain.

A software diagnostic procedure may work reliably for years and then begin identifying the wrong component after an architectural change.

The same pattern can occur when a theory repeatedly predicts events that fail to happen, or when an interpretation of someone's behavior keeps leading to surprise.

Here the destabilizing force comes from breakdown.

Something relied upon stops working as expected.

This does not determine exactly what should be revised. But it can make continued reliance harder to sustain.

Repeated breakdown can become a source of destabilization.

Loss of Reinforcement

Imagine someone leaving a close-knit religious community.

For years, particular convictions were renewed through conversation, ritual, shared texts, friendships, calendars, and expectations.

After leaving, the person may initially remain just as convinced.

But some of the conditions that repeatedly returned those convictions to attention are gone.

Months or years later, things that once seemed obvious may feel more distant.

The same can occur when leaving a profession, political movement, social group, relationship, or institution.

Reinforcement can be one condition of stability.

Remove the condition and stability may change.

Sometimes the opposite happens. Leaving a group strengthens a conviction precisely because opposition makes it more salient.

Transformation and Dissolution

Not every destabilized conviction disappears.

Suppose someone believes: "My father was a bad person."

Later they learn more about his circumstances.

The conviction may weaken.

Or narrow: "He treated us badly during those years."

It may become conditional: "I still think what he did was wrong, but I no longer think it defines him."

It can transform: "I used to understand his behavior as cruelty. I now understand much of it as fear."

Or dissolve: “I simply don’t see him that way anymore.” Another conviction may replace it.

These changes are not equivalent.

Weakening concerns force, while narrowing restricts scope. Conditionalization adds conditions that were previously absent. A conviction can also be transformed, dissolved, or replaced by another.

The boundaries are not always clear.

9. Persistence and Recurrence

Some convictions disappear almost without leaving a trace.

A childhood fear may become difficult even to remember.

An old ambition can lose all force.

Judgments about someone simply cease to matter.

Other convictions remain for decades.

And still others disappear, then return.

These are different patterns.

A *persistent* conviction continues to participate in orientation even if it is rarely explicit.

Someone may spend years without thinking about the importance of friendship, yet organize major decisions around it whenever the issue arises.

A conviction can leave attention without disappearing.

Disappearance is stronger.

The conviction then no longer participates in orientation.

A person may remember having believed something without being able to recover its former force.

“I know I once thought that, but I can’t understand anymore why it seemed obvious.”

Recurrence occurs when something that had ceased to operate becomes convincing again.

Old political convictions can return when circumstances change. Religious convictions sometimes disappear and later re-emerge. A judgment about another person may become convincing again after a new event.

Distinguishing persistence from recurrence is not always easy, especially from the outside.

If a conviction returns after twenty years, was it absent or merely dormant?

That may depend on what we mean by participating in orientation, and the evidence can be thin.

Still, the distinction is useful.

The conditions already discussed offer several possible explanations for persistence.

Repeated reinforcement may keep a conviction active, while integration with memories, habits, plans, and other convictions can make it harder to disturb. Social surroundings may continually renew it, and recurring circumstances can reactivate the same pattern.

Different forms of persistence may have different support.

The persistence of "fire burns" is not obviously sustained in the same way as the persistence of "family matters."

One is repeatedly reinforced through a stable class of interactions.

The other may be integrated with relationships, identity, memories, obligations, hopes, and a life history.

The word *persistence* describes the temporal phenomenon.

It does not tell us why the phenomenon occurs.

Questions show related patterns.

Some disappear when a sufficiently convincing answer arrives.

"Where is the station?"

Once you arrive, the question no longer participates in orientation.

"How do I reset this device?"

Once the procedure is learned and the problem solved, the question may vanish.

Other questions recur.

"What should I do with my life?" can seem settled at twenty-five and return at forty.

Some questions appear to persist through changing answers.

Questions concerning justice, meaning, consciousness, knowledge, or how to live can remain active in a person's life even while successive answers become convincing for a time.

What gives rise to them may be too broad or persistent to be exhausted by one answer. But persistent questions need not persist for the same reason at all.

Persistent questions matter because conviction does not simply accumulate until everything has been settled.

What has settled in orientation can become unsettled again.

10. Defensibility and Criteria

Several things can remain convincing at once without pointing in the same direction.

Suppose two witnesses describe the same event differently, and both seem credible.

Neither report disappears because the other exists, and both exert pressure.

Or suppose a doctor recommends a treatment that appears likely to help but carries serious possible side effects.

This is advisable in one respect and troubling in another.

Sometimes nothing settles immediately.

We remain uncertain.

We may lean one way while still finding opposing considerations difficult to dismiss.

Conviction formation can involve more than something simply convincing or failing to convince. Different considerations can convince in different directions.

Weighing

Ordinary language already has a useful word for what often follows.

We weigh what speaks for one possibility against what speaks for another.

The metaphor should not be taken literally. There need not be a hidden scale in the mind.

It captures something simpler: several considerations can remain active at once and bear differently on what eventually becomes convincing.

Suppose you are unsure whether a colleague deliberately excluded you from an important conversation.

You remember earlier tensions.

You also know that the colleague was under strong pressure that day.

Someone tells you the omission was accidental.

You can remain with the situation as it stands.

But you can also do things.

You can seek further information, ask what else might explain what happened, inspect an assumption, or introduce another possibility. You might look for an objection to the view you currently favor, or return to the question later.

What you cannot do in the same way is decide where the weight will fall.

You can choose to inspect an objection without being able to choose whether it will cease to trouble you. You can reconstruct an argument without being able to command yourself to find it convincing.

Deliberation is both active and receptive.

We can influence what enters the weighing and how carefully it is examined.

We cannot directly assign convincing force.

Criteria

Sometimes parts of orientation become stable enough to organize later weighing.

Suppose you are choosing a car.

Price matters to you.

So does safety.

Reliability has become important after years of repairs.

Running costs matter because your budget is limited.

Once they have acquired enough force, they can function as criteria in the comparison.

Making them explicit can reveal why one option appears preferable and where different considerations conflict.

Criteria do not remove weighing.

They structure it.

They remain within conviction formation.

You can propose a criterion, apply it, compare it with another, and examine what follows. But you cannot simply decide that it will matter to you.

Once criteria become convincing enough, they can operate as relatively settled parts of an inquiry without being reconsidered at every step.

Different people and practices may settle on different criteria.

Criteria can also become unsettled again.

A criterion may conflict with another.

Its use may produce results that disturb confidence in it.

Someone may ask why it should matter at all.

When that happens, the criterion itself can become part of the inquiry.

Nothing requires this questioning to continue indefinitely.

Criteria can settle locally just as other convictions can.

Shared Criteria

Criteria do not have to be shared in order to guide a person's own weighing.

But when people assess something together, shared criteria can provide common terms for comparison.

Suppose two people disagree about which of two explanations is better.

They may already agree that an explanation should account for a particular observation.

That agreement does not settle the question.

They may still disagree about which explanation fits the observation better, how much weight it should carry, or what else matters.

But they do not have to reopen that criterion at every step.

Where enough criteria have become sufficiently stable for the people involved, they can organize a shared inquiry without becoming final or universal.

Where that overlap is missing, the disagreement may shift.

One person may appeal to a criterion that does not convince the other.

The criterion itself can then become part of the inquiry.

Shared criteria do not take inquiry outside conviction formation.

They are cases in which enough of what has become convincing overlaps for people to examine something together.

Defensibility

A conviction can be challenged.

Suppose you are convinced that another person acted maliciously.

Someone asks why.

Being convinced does not answer the question.

If you take up the challenge, you have to offer something in defense of the conviction.

You might point to what was said, what happened before, or what alternatives seem unlikely.

The other person can examine what you offer, accept some of it, reject some, or ask for more.

The conviction is now being assessed from another perspective.

It no longer operates only within the orientation in which it formed.

It has become answerable to another perspective.

A defense introduces a relation between people.

Someone offers considerations in support of a conviction.

Someone else assesses what is offered.

The person assessing the defense need not be neutral, authoritative, or final.

They may already disagree.

They may share some criteria with the person offering the defense and reject others.

This allows several things to come apart.

Convictional strength concerns how firmly something convinces a person.

A *defense* is what someone offers when that conviction is challenged.

Defensibility concerns how that defense fares under the criteria operative in the assessment.

Persuasion concerns whether the defense actually changes what the other person finds convincing.

These are not the same.

A defense may fare well under criteria shared in an inquiry and still fail to convince the person assessing it.

Another defense may convince someone while faring poorly under those same criteria.

Successful persuasion is not the measure of defensibility.

Nor must the considerations offered in a defense be the same things that originally made the conviction convincing.

What formed or stabilized a conviction may carry little weight under the criteria now being used to assess it.

A challenge can make that difference visible.

Someone may discover that they are strongly convinced of something but have little to offer under criteria they themselves accept. That may or may not change the convincing force of the conviction.

Or they may find that considerations they had scarcely noticed before become important once the conviction has to be defended.

Defensibility is local and relational.

A defense is offered here, to these people, under these criteria, for this challenge.

Where enough criteria are shared, the defense can be assessed without every standard being reopened at once.

Where they are not, part of the disagreement may shift to the criteria themselves.

Defense also creates another opportunity for deliberate participation.

We can clarify what we are claiming, offer further considerations, answer objections, compare alternatives, and apply criteria already operative in the inquiry.

What we cannot do is decide that the defense will convince the person assessing it. Nor can that person.

The same applies in self-examination. We can ask ourselves to state the case for a conviction and examine what we produce. We cannot command ourselves to find that case convincing.

The fundamental claim applies here as well. A group can agree on procedures and criteria, invite criticism, and require claims to be defended. These arrangements can shape what happens next, but the group cannot simply decide what its members will find convincing.

A conviction can become answerable beyond the orientation in which it first formed. Where enough criteria are shared, people can examine the defense together. But shared criteria still do not place what becomes convincing under either side's control.

11. What Conviction Formation Gives Us

We began with a practical problem.

What can finite agents investigate and do while larger disputes about truth, justification, rationality, and foundations remain unresolved?

The preceding investigation gives us a partial answer.

Questions concerning how convictions form, stabilize, weaken, and change can be investigated without first settling those larger disputes.

This does not make truth or justification irrelevant.

But inquiry into conviction formation can begin before they are finally settled.

The same applies to Conviction Formation Theory itself.

Conviction Formation Theory need not first establish itself as the final truth about conviction formation before it can be examined, used, challenged, or revised.

Being convincing does not make it true.

Its claims can still turn out to be untenable.

Participation Without Control

The fundamental claim was initially negative:

Becoming convinced is not under direct voluntary control.

But lack of direct control does not leave us passive.

We can ask questions, seek information, introduce alternatives, compare, practise, expose assumptions, and invite criticism. We can make criteria explicit, defend convictions, and examine defenses.

These interventions can change the conditions under which conviction forms.

They do not determine the result.

Participation does not require a standpoint outside conviction formation.

The same limit applies when inquiry is shared.

People can agree on procedures, use shared criteria, compare alternatives, and require claims to be defended.

They can organize inquiry together.

They cannot simply choose what its participants will find convincing. What becomes convincing among them must itself emerge and stabilize through the inquiry.

Settlement Without Final Settlement

Inquiry would be impossible if everything had to remain open at once.

Convictions stabilize.

Practices stabilize.

Some parts of orientation become stable enough to function as criteria.

Some criteria become shared strongly enough to organize inquiry across people.

Such stability need not be permanent or universal.

It can be local to a person, a group, a practice, a set of conditions, or a period of time.

This gives us usable settlement. What has settled can carry further inquiry until something disturbs it.

That disturbance may come from a contradiction, a failed expectation, a question, another person, a competing criterion, or something else that acquires enough force.

Stability matters without vindicating what has stabilized.

A stable conviction can be mistaken. A shared criterion can later be challenged, and settled practices can change.

Conviction Formation Theory treats settlement as something that happens within conviction formation, not as an endpoint outside it.

Local settlement can still do some of the work for which we might otherwise demand a firm scholarly foundation first. It can support further inquiry, coordinate action, and allow reasoning to continue while larger questions remain open.

Criteria and Shared Inquiry

Criteria do not have to be supplied from somewhere beyond conviction formation.

A distinction, aim, procedure, or standard can become convincing and stable enough to organize later assessment.

Criteria can be proposed, compared, and applied without putting their convincing force under our control.

Once operative, however, criteria can genuinely constrain an inquiry.

Where enough criteria are shared, people can assess something together without reopening every standard at every step.

Where they are not shared, disagreement may move to the criteria themselves.

Challenge introduces another important possibility.

A conviction that once operated within one person's orientation can become answerable to another.

Someone asks why. A defense is offered.

What is offered can be examined under criteria available in the inquiry.

Conviction has become available for assessment across perspectives.

Locating Disagreement

People may disagree about what happened, which possibilities deserve attention, which considerations matter, or which criteria to use. They may even share a criterion but disagree about how it applies, or agree about the assessment without finding the same conclusion convincing. Conviction Formation Theory does not settle these disagreements merely by distinguishing them.

But sometimes progress consists in discovering where the disagreement actually lies.

A question may become more precise, another possibility may need attention, or the inquiry may shift to a criterion or comparison.

Inquiry Without Command

Conviction Formation Theory does not give us a final theory of truth, a universal standard of justification, criteria everyone must accept, or a standpoint outside conviction formation from which everything can finally be judged.

It gives us something more limited.

Convictions form against an orientation already in place, and they can stabilize or become unsettled again. Questions can disturb what had seemed settled. Criteria can become stable enough to guide inquiry and shared enough to support assessment across people.

Throughout this, finite agents can intervene without directly choosing what will convince them or anyone else.

We can alter conditions, ask questions, compare possibilities, expose assumptions, apply criteria, challenge and defend.

We can work with what has become convincing without pretending that it is final.

Questions, Possibilities, And Formal Systems

From Conviction to Inquiry

Conviction does not fill orientation evenly.

Some things are settled enough that we simply rely on them.

The floor will hold. This is the road home. That person is a friend.

Seven plus five is twelve.

The meeting begins at nine.

Other parts remain uncertain, incomplete, unstable, or contested.

Why did she say that? Will the project succeed?

Which explanation fits what happened? Was the measurement wrong?

What should we do if the attack comes through another route?

Sometimes something that had seemed settled becomes uncertain again. A new fact no longer fits. An expected result fails to appear. Someone asks a question we cannot answer. An alternative we had not considered suddenly becomes visible.

Inquiry begins somewhere inside this mixture.

It does not begin from nothing. Most of orientation remains in place while some part becomes uncertain, tense, or newly visible.

Something that had remained in the background becomes an object of attention.

A gap or tension becomes difficult to ignore. Something matters that we do not yet know how to settle.

We cannot simply choose what will convince us. But we can ask questions, search for alternatives, compare possibilities, preserve them outside memory, and enter procedures designed to constrain what happens next.

These activities do not determine the result of inquiry.

They change its conditions.

How far can such intervention go?

Questions

Questions

Suppose you come home and find the kitchen window open.

You remember closing it.

For a moment there may be no question. You simply notice the open window.

But then questions begin to nag.

Did I really close it?

Did someone open it?

Was somebody in the house?

The questions do not erase everything else you take to be the case. You still recognize the kitchen. You still know roughly what happened earlier in the day. You still understand what a window is and why an open one might matter.

Only part of the situation has become unsettled.

Questions often work like this.

They mark or create some local incompleteness within a background that continues to operate.

There is little reason here to search for an essence common to every grammatical question. Questions do many things. They request information, expose uncertainty, challenge assumptions, redirect attention, test competence, and reopen alternatives.

Some are rhetorical, strategic, hostile, playful, or deceptive.

A question in the sense used here directs attention toward something that is not yet settled.

Sometimes the unsettledness comes first. You are genuinely puzzled. Something does not make sense, and the question gives the puzzlement a form.

Sometimes the question comes first.

A teacher asks:

“What assumption are you making here?”

You may not have been puzzled at all. Only after trying to answer do you notice that your conclusion depends on something you had never examined.

Being puzzled is not obviously something we can simply decide to become.

Conviction Formation Theory extends the fundamental claim to questioning:

We can deliberately formulate a question, but we cannot simply decide that something will become genuinely questionable for us.

But there is an important difference. Unlike convictions, questions call for answers.

A deliberately entertained conviction may later acquire convincing force if what we encounter supports it. A deliberately posed question has an additional possibility: the expected answer may fail to appear. That failure can itself make the question genuinely puzzling.

A conviction can gain force through what supports it. A question can gain force through the failure of what should answer it.

We can deliberately ask what else could explain something, what we may be assuming, what would count against a conviction, or which possibility we have ignored.

At first such questions may be little more than learned techniques. Perhaps nothing follows.

You ask what would count against a conviction and immediately find a satisfactory answer.

But the situation changes when we expected to have an adequate answer and discover that we do not.

The question may not have been genuinely puzzling when we posed it. The failure to answer it can make it so.

By asking, we can place ourselves in conditions under which the absence of an answer may itself give the question force.

What began as a learned technique can then become a genuine problem for us.

What Questions Reveal

Questions show what is still open, but they also show what has already been taken as settled.

Consider: “Who broke the glass?”

The identity remains open.

But the question already contains a substantial commitment: somebody broke the glass.

If nobody did, the question has organized the situation badly.

“What happened?”

This already treats something as an event that can be picked out and described.

“Where did it happen?”

This relies on a spatial organization in which something can be located.

“When?”

This relies on an ordered temporal structure.

“How did it happen?”

This may ask for a process, mechanism, or sequence.

“Why?”

That can do several different jobs.

“Why did the bridge collapse?”

“Why did you leave?”

“Why should we accept this?”

“Why does this keep happening?”

These questions may be asking about a cause, a motive, a justification, or some other connection that would make the situation intelligible.

An intelligible question depends on much that is not being questioned. It marks something as open while also revealing parts of the surrounding orientation that remain stable enough for the question to make sense.

An answer can shift inquiry into another part of the background structure on which the question depends.

“Why did the bridge collapse?”

The material failed.

“Why did the material fail?”

The materials were mixed incorrectly during manufacturing.

The first question concerned the collapse. Its answer connects that event to another process that had remained in the background, and that process can now become the object of inquiry in turn.

In a narrative context, this may amount to moving from one part of a background story to another. In other cases, inquiry may move into a procedure, a classification, a measurement practice, an assumption, or another part of orientation.

Questions Between People

Questions also pass between people.

If I ask: “Do you know where the key is?”

I do more than register a gap in my own orientation.

I treat you as someone who may be able to fill it.

Other questions do more.

A teacher may ask a student to identify an assumption. A doctor might want to know which symptoms appeared first, while an investigator asks a witness what happened before the alarm. An engineer might ask whether anyone checked name resolution before replacing the server.

A friend asks: “Are you sure that is why she did it?”

A question can introduce a distinction or possibility that was not previously active for the person being asked.

It can make an assumption visible, demand a defense, redirect attention, or narrow the available responses.

Questions between people are not automatically neutral.

A lawyer may ask a question designed to restrict what can be said, while an interrogator may use one strategically. A politician might pose a question to close inquiry rather than open it. Teachers sometimes ask questions whose answers they already know, and a friend may ask simply because something in another person’s account seems incomplete.

Inquiry can be shared without being symmetrical.

Who asks, who answers, and what relation exists between them affects what the question does.

We can deliberately introduce questions not only into our own inquiry, but into the conditions under which conviction forms for other people.

We cannot decide that the question will become puzzling for them.

It may leave their orientation untouched. They may have an answer ready, dismiss the question, misunderstand it, or simply not find it important.

But they may also discover, in trying to answer, that something they had taken for granted is harder to account for than they expected.

The question can then acquire a force that neither the person asking it nor the person answering it directly chose. Questioning another person is another form of indirect participation in conviction formation.

From Questions to Possibilities

Return to the open window.

"Who opened it?"

Several answers may become available.

I did and forgot.

My daughter did.

The wind pushed it open.

Someone entered.

Perhaps one of these possibilities immediately feels likely.

Perhaps none does.

The important change is that the question has created a space in which different answers can now compete.

Not every question does this. Some request a missing value or a simple piece of information.

But many questions make alternatives available.

This can reverse something conviction had previously accomplished.

Suppose you were convinced that you had left the window open yourself.

The burglary possibility might never become active.

Then someone asks: "Are you sure?"

Or: "Wasn't it cold when you left?"

The settled answer can become one possibility among others.

It is tempting to imagine all relevant possibilities as already present in some neutral space, waiting to be inspected.

For finite agents, they usually have to become available somehow.

Some come from memory or imagination. Analogy may suggest others, or another person may propose one. A formal method can sometimes enumerate possibilities, while a change in description may alter which ones appear relevant. Expertise can make possibilities visible that a novice would not think to consider.

Suppose someone investigates a computer outage.

The first question is: "Which server failed?"

Several servers are examined.

An experienced engineer may instead ask: "Did any server fail?"

Perhaps name resolution broke. The Monitoring may be wrong. Or the user report concerns another service.

The engineer has not merely evaluated the original alternatives better.

She has changed the possibility space.

Inquiry can have two distinct tasks: generating possibilities and evaluating them.

Good evaluation does not repair a badly generated possibility space.

A flawless ranking procedure applied to inadequate alternatives can produce a very orderly mistake.

For finite agents this creates a persistent difficulty. We often do not know whether every relevant possibility has been considered.

Sometimes, however, the structure of the possibilities itself gives us strong reasons to think that none are missing.

Take a natural number.

It is either even or odd.

Two claims are involved here.

First, the alternatives exclude one another. A natural number cannot be both even and odd.

Second, they exhaust the relevant possibilities. With respect to parity, there is no third case.

These claims are easy to run together, but they are different. A set of alternatives can fail because some overlap, because something has been left out, or both.

In the case of even and odd, a simple procedure makes the structure easy to inspect.

Imagine a natural number as a collection of pebbles and repeatedly remove them in pairs.

Eventually either no pebble remains or exactly one does.

There cannot be two remaining, because they would form another pair. Three would reduce to one, four to none, and so on.

The procedure supports both parts of the possibility space. The two outcomes exclude one another, and repeated pairing leaves no third outcome.

This is a particularly clear case.

Other possibility spaces are supported very differently.

Suppose a witness describes a shirt as red rather than orange.

The distinction is useful, but its boundary is not like the boundary between even and odd. One observer may say red-orange, another orange-red. Lighting

can change the judgment. The uncertainty lies partly in how the possibilities themselves have been distinguished.

Yet fuzzy boundaries do not make the classification useless. A disputed red-orange shirt may still clearly not be blue or green.

Exhaustiveness raises another problem. Do our familiar color categories capture every relevant distinction?

For ordinary purposes they may work very well. But their structure depends partly on human perception, learned categories, and the purpose for which the distinction is being made. We may not possess anything like the pairing procedure that made the parity case so clear.

Pluto shows another way in which a possibility space can change.

For much of the twentieth century, Pluto was classified as a planet. Then astronomers discovered additional bodies sufficiently similar to Pluto that the existing classification came under pressure.

Either the category of planets would have to expand, or the criteria distinguishing planets from other bodies would have to change.

Pluto was eventually reclassified.

Pluto itself was still there, following the same orbit. What changed was how astronomers classified bodies like it.

Sometimes revising the classification is enough. In other cases, inquiry does not merely add another possibility or alter the criteria separating existing categories. It changes the concepts through which the possibility space was organized.

Is light a wave or a particle?

Taken as a simple classification, the question appears to demand one of two answers. But the development of quantum theory did not amount merely to selecting the correct member of that pair or adding an overlooked third ordinary category. The concepts involved had to be reconsidered.

These cases differ in how much confidence we can place in the structure of the possibility space.

With even and odd, a simple procedure gives strong support to both exclusivity and exhaustiveness.

With color, useful distinctions remain despite uncertain boundaries.

With Pluto, new cases put pressure on the criteria separating existing categories.

With wave and particle, the alternatives themselves proved inadequate as the terms of the problem.

Possibility spaces can be open not only because alternatives are missing, but because the distinctions used to generate them may themselves change.

A possibility space is not simply a neutral container into which alternatives arrive.

It has to be generated and structured.

Convictions help determine which possibilities are recognized and which distinctions seem relevant. Those distinctions then affect where boundaries are drawn, whether possibilities are treated as overlapping, and whether the resulting space appears exhaustive.

Some of these convictions can be supported very strongly. Others remain practical and approximate, and new possibilities may force the structure to be revised.

Closure and Maintained Openness

What happens once several possibilities are available?

Sometimes one becomes sufficiently convincing that the others cease to matter.

Suppose you ask: "What caused the outage?"

At first several explanations remain plausible. Then the logs show that a deployment introduced the failure.

Other possibilities lose their force. One answer has become convincing enough to proceed with.

This is a form of relative closure. It need not be certainty. Conflicting evidence may reopen the question.

In other situations, closure would achieve little.

You are traveling tomorrow.

Rain is a serious possibility, but sunshine remains plausible.

You do not need to settle the meteorological question.

You take a light raincoat.

That action works across more than one possibility.

Security planning often has this form.

You do not know whether an attacker will use phishing, credential stuffing, an exposed service, or a supply-chain weakness.

A defense may be designed to work across several routes.

Strategy can preserve plurality for practical reasons.

Once several possibilities are available, one of them may become convincing enough that the others lose their force. If that does not happen, several may remain live and have to be carried forward.

Constraining Inquiry

Once questions and possibilities are available, inquiry can be shaped more deliberately.

We can ask what we had not previously asked, deliberately search for alternatives, or preserve several possibilities instead of allowing one to close the inquiry too early.

Suppose six possibilities are competing in your head.

You compare the first with the second.

Then you remember something relevant to the fourth.

The fifth disappears from attention.

You return to the second but no longer remember why it had seemed weak.

Now write the six possibilities down. They persist when attention moves elsewhere.

You can return to one.

Cross another out.

Add a seventh.

Compare two side by side.

Give the list to someone else.

Nothing about writing guarantees a good inquiry. But the conditions have changed. Parts of the inquiry now persist outside immediate memory. They can be inspected repeatedly.

Tables can preserve comparisons, and diagrams can make relations easier to inspect. Checklists or measuring instruments may stabilize a procedure or standard. Symbolic notation and formal rules can impose stronger constraints by fixing distinctions and limiting which transformations are permitted.

In each case, some part of the inquiry has been made more explicit and less dependent on what happens to remain in attention.

Formal Systems

A formal system can specify objects, stabilize symbols, restrict permitted operations, and preserve intermediate steps.

Another person can inspect the same structure, and errors can sometimes be localized.

A result can be reconstructed rather than merely asserted.

Counting provides a simple example.

A pile of stones lies on a table.

You count twelve.

I count thirteen.

We can count again, perhaps moving each stone as it is counted or arranging them in a row. We might pair each stone with a numbered marker.

If we disagree again, we may be able to locate the skipped or duplicated stone.

The procedure does more than produce a number.

It organizes the disagreement.

Measurement adds further constraints.

Two people disagree about the length of a table.

A measuring tape can reorganize the problem, but it does not give us a quantity independently of the conditions of measurement.

The measurement depends on standardized units, a suitable instrument, correct placement, an adequate reading, and sometimes calibration and uncertainty estimates.

Much of the stability associated with measurement is engineered.

Conditions have been constructed under which certain disagreements become easier to investigate.

Diagrams do something different.

A diagram can make several relations visible at once. They can be pointed at, compared, rearranged, and criticized. A geometrical construction, a circuit diagram, a state machine, and a dependency graph differ greatly in purpose, but all can make relations available for visual inspection.

Logic makes constraint more explicit still.

Once a formal language and rules are operative, a derivation can be constructed step by step.

Each transformation can be checked.

If two people disagree, they can move backward until they find the first point at which their judgments diverge.

Was a premise introduced?

Or a rule applied incorrectly?

Was a variable substituted where it should not have been?

Formalization does not remove the need for judgment.

Someone still has to decide what to formalize.

The notation has to be understood. Premises have to be supplied. Rules have to be interpreted and applied.

The formal system itself has to be appropriate to the task.

But within the system, some kinds of disagreement can become sharply constrained.

Across these examples, parts of inquiry move outside immediate memory and become easier to repeat or inspect. Intermediate states persist, some moves can be restricted, and errors can sometimes be localized. Other people can reconstruct what happened.

Disagreement can move from: "Your conclusion is wrong" to: "This step does not follow."

That is a substantial change in the conditions under which disagreement can be examined and convictions reconsidered.

The constraint is nevertheless local.

Formal systems can make parts of inquiry explicit, repeatable, and inspectable, but they do not replace the wider orientation in which those parts operate. And their constraints carry force only if the system itself is convincing.

A derivation still begins from premises. A measurement still depends on instruments, procedures, and interpretation. A formal model still has to be connected to some problem or domain.

Nor does all conviction formation proceed through reconstructible steps. Perception, practice, narrative, and affect can alter orientation in different ways.

Formal systems are not a general model of conviction formation. They are an especially explicit way of constraining parts of inquiry.

Questions, Possibilities, and Formal Systems

Questions, possibilities, and formal systems have done different things in the inquiry so far.

A deliberate question may initially have little force. Yet if no adequate answer appears, the question can begin to persist in ways we did not choose.

We may become genuinely puzzled.

Possibilities we had not entertained enter the inquiry.

They may come from memory, imagination, another person, expertise, changed description, or explicit enumeration.

Their generation matters because evaluation cannot compensate for alternatives that never entered the inquiry.

Possibility spaces can themselves be structured more or less sharply.

Some can be made exhaustive and mutually exclusive.

Others remain useful despite uncertain boundaries.

Inquiry may then close around one answer, or deliberately preserve several possibilities when action does not require settlement, or settlement is currently impossible.

Parts of inquiry can be externalized. Comparisons can be stabilized. Procedures can be repeated. Permitted moves can be restricted.

Formal systems bring together stabilized notation, inspectable representations, repeatable procedures, and shared ways of working through problems. This changes some of the conditions under which convictions can form and be reconsidered.

Participation

Introduction

An author can try to show how a conviction formed by presenting the questions that guided it, the distinctions that mattered, the examples that changed something, and the steps by which one view gave way to another.

The reader may then be able to walk that path.

Where that is not enough, an author can try to make a phenomenon visible or tangible enough to be examined in another way.

But texts cannot guarantee that convictions will form in the same way in every reader.

Texts contribute to that formation by clarifying something, raising a question, making a comparison, or opening another possibility.

But they do not command the result.

More generally, people can ask questions, seek other possibilities, direct attention, use procedures, return to problems, or prepare before a situation occurs.

Such interventions can alter the conditions under which conviction forms without determining what will become convincing.

Convictions form, and questions can become pressing without our choosing either.

What, in all this, is actually our doing?

What Is Our Doing?

If you are like most people, you can hold up your fist and extend your index finger.

You can repeat that.

But you cannot move a chess piece across the board in the same way, merely by willing it to move.

You can, however, use movements of your hand to move it where you want it.

You can do that even when the move is not permitted by the rules of chess.

You can also use your other hand to extend the index finger from the example above.

This is not unheard of. Maybe it is very cold and the finger no longer moves as you want. Maybe you have a cramp.

A sword is somewhere between the finger and the chess piece.

You can learn to use it as an extension of your body and, to a certain extent, feel what happens at its other end through your hand, even with your eyes closed.

But when you put it down, that connection is gone.

You can begin a calculation when you need the result. The answer is not something you can will into place either.

You are constrained by a formal system that works by its own rules.

If you are unsure, you can calculate again, maybe in a different but equivalent way.

You can choose to begin and how carefully to proceed, but you do not simply choose the result.

Now imagine a ball rolling towards a precious vase.

If nothing intervenes, it will knock the vase over and break it.

You can intercept the ball.

Then you have saved the vase from destruction.

But suppose you do not move.

Have you failed to save it?

That may depend on whether you saw the ball, whether you could reach it, whether other persons could have saved it, what else would have happened, and why exactly you did not move.

The examples do not leave us with one obvious place where our doing begins.

Moving a finger seems straightforward enough. A chess piece moves through movements of your body, and training can make a sword part of that movement. You can begin a calculation without choosing its result. You may save the vase by intervening, or fail to save it without moving at all.

What makes all of this our doing?

Does our doing begin where we choose something?

Looking At Choice

Choose a number between 1 and 10.

How did you do it?

Maybe you waited for the right number to appear.

Maybe you already had one in mind.

Or maybe a number appeared with the question, or while you were preparing to answer.

Is the chosen number your doing?

It came into the world by your *fiat numerus*.

But how exactly did you choose?

There does not seem to be one fixed way of choosing a number like that. The process seems receptive and active at once.

But this is an unusual kind of choice. There is no goal beyond choosing, and no obvious criterion by which one number would be better than another.

Suppose the screw on one of the legs of your desk is loose.

On the desk lie a hammer, a screwdriver, and a knife.

Which tool do you choose?

Probably the screwdriver. It is made for tightening screws.

But suppose the screwdriver is too large and the knife fits perfectly.

Then maybe you choose the knife instead.

Here the choice has a purpose.

You want to tighten the screw, and the available tools differ in how well they serve that purpose.

You may stop and reason about this.

Or you may simply reach for the screwdriver.

Either way, one possibility becomes the one you act on.

But were the hammer, screwdriver, and knife really all your possibilities?

Suppose there were only a hammer, a knife, and a lighter on the table. You could ignore all three, walk to another room, and get a screwdriver.

The possibilities were not simply given by what happened to be in front of you.

You can alter, update and abandon the set.

Now choose two names that potential partners of yours might have.

Replace Hanna and Mary with those names while reading.

Hanna and Mary both have qualities you value, and qualities you do not. Which one, if either, should you choose as your partner in life?

Here the situation is different again.

There may be many things that matter.

Affection, trust, attraction, shared plans, character, family, and timing may all matter. Maybe some of these matter greatly and others only a little.

But even when the possibilities and relevant considerations are known, nothing in them may make one choice follow.

By the way, how did you choose the names? Or did you choose not to choose?

Sometimes we respond by choosing a procedure.

A rule, a vote, a coin, or advice from someone we trust.

It changes where our participation lies.

We choose how the decision will be made.

And then we allow something else to settle what happens next.

Does calculation belong here too? We choose to begin and follow the procedure, but not what result it yields.

Suppose you promised two different people that you would spend the same evening with them. You cannot keep both promises.

Whichever one you keep, you break the other.

You may look for another possibility.

You might move one meeting, explain the problem, or bring the two people together.

But suppose none of this works.

Then you must choose.

Yet choosing one does not make the other promise disappear.

Nor must the chosen option suddenly appear good.

It may remain merely the less bad one.

Or simply the one you can finally bring yourself to enact.

Choice can settle action while conflicting convictions remain.

Choice seems to require possibilities and often involves criteria, goals, rules, or boundaries. But the possibilities need not simply be given, and the criteria need not determine one answer.

We may even choose the procedure by which the decision is made, while conflicting convictions remain afterward.

How do possibilities appear?

How do we know which ones belong to the choice?

And how do we know that we have considered them all?

In practice, we just choose without being able to say how all of this was settled.

So is choice where our doing begins?

Action Without Obvious Choice

Hold up your fist and extend your index finger.

Do it again.

When I do this, I find no moment in which several possibilities appear and I choose one of them.

What would the possibilities even be? Every possible moment at which I might move?

I simply extend the finger.

Still, it seems very clearly to be my doing.

It is quite different from someone else forcefully extending my finger. It is also different from my finger moving in a sudden involuntary jerk.

Whatever distinguishes these cases, an observable act of choosing does not seem necessary.

After a run, I take a bottle of water and unscrew it while continuing a conversation. I may barely notice what my hands are doing. There is no weighing of possibilities. No question appears. I do not choose between unscrewing the bottle and doing something else.

I simply unscrew it and drink.

Yet afterwards I would have little difficulty saying that I opened the bottle. The action can be mine without becoming an object of deliberation while I perform it.

More complicated movements can work in much the same way. When practising a motor activity, movements that initially require attention can become increasingly fluent. At some point, deliberately controlling every part may make the activity worse.

In rope flow, for example, one movement can lead into another without my first choosing the next figure. A pattern emerges, like in musical improvisation. I can interrupt it or change direction. I can begin again.

But while it is happening, much of what I am doing is not preceded by a choice between explicit possibilities. Often I find myself monitoring what happens more than deciding each movement.

Still, the movement remains recognizably mine.

The exact experience probably differs between people.

It is enough here that the phenomenon occurs.

Walking, speaking, typing, playing an instrument, and many movements in sport can proceed without a sequence of choices corresponding to every movement performed. Many of these activities were once learned deliberately. They may still be interruptible or redirectable, and often depend on preparation and training.

Yet at the moment of action, much of the doing unfolds without an identifiable choice.

Choice seems unable to mark the boundary of our doing.

Some things we choose are clearly our doing.

But some things that are clearly our doing do not seem to involve choice at all.

We began with the possibility that choice might mark the boundary of our doing.

It does not seem to.

But where choice and doing come apart, we learn something about what such a boundary would have to account for.

Choosing One Thing, Doing Another

A doctor tells a child:

“Do not touch the wound.”

The child understands.

Maybe even agrees.

Later, while playing, the child pretends that one hand is a spider crawling up the other arm.

The spider moves.

Up the wrist.

Up the forearm.

And then the hand touches the wound.

The child did not choose to touch it. The wound may not even have been present in attention. What the child was doing was making a spider crawl up an arm.

Yet touching the wound was also something the child did.

Suppose someone asks you to hold up a number with one hand.

But instead of choosing a number, you decide to choose randomly among familiar hand signs. The horns, the victory sign, the OK sign, or a fist.

Only afterwards do you look at how many fingers are extended.

Maybe you did not know the number before looking.

You chose the sign. Yet by choosing it, you also produced a number.

The number was your doing, but it was not what you actually chose.

What we do can extend beyond what we chose to do.

We may do one thing and thereby do another.

The Reach Of Choice

Suppose you set an alarm before going to sleep.

You choose the time. Maybe you have an early appointment or want to go running.

Then you sleep. Hours later the alarm rings.

The choice happened yesterday. It changed what happens now, but the choice itself is no longer happening.

You may not even remember immediately why you set the alarm.

And when it rings, another situation has already formed.

You are tired. The bed is warm. Getting up may no longer seem like a particularly good idea.

The earlier choice still matters. But it does not simply govern the later moment.

Sometimes we choose earlier precisely because we expect the later situation to change what seems convincing.

You are angry. You want to send a message.

But you know that anger changes what appears reasonable.

So you decide: Not tonight.

You put the phone away.

Maybe you even write the message and leave it unsent.

The choice changes the conditions under which you will act later.

It creates distance. It gives other convictions time to return. In the morning, the message may still seem right. Or embarrassing. Or unnecessary.

The same structure can extend much further.

An investor decides in calm circumstances how to respond to a large fall in the market.

The rule is simple: Do not sell merely because prices fall.

Years later prices collapse. Losses become visible, other investors panic, and news stories predict disaster. The conviction that selling immediately is the only sensible thing to do may become very strong.

The earlier strategy still exists. Maybe even in writing. But the earlier choice has not prevented the later conviction from forming. The person who made the rule and the person facing the crash are the same person. Yet they do not stand in the same situation.

What convinced the person in one set of circumstances does not necessarily convince them in another.

So what remains after a choice has been made?

Sometimes what remains is a memory, a rule, an alarm, a promise, a habit, a note left for oneself, or other people who remember what we said we would do.

The choice can leave many things behind. But these things are not the choice itself.

They participate in what happens later.

The earlier choice need also not remain present in memory.

You may get up when the alarm rings without remembering why you chose that particular time. If the alarm still rings because you once set it, the earlier choice remains plainly involved in what happens now.

Other cases are different.

You may have stopped sending angry messages years ago and now simply wait before replying. An earlier decision may have helped form that habit.

But today's waiting is not simply the earlier choice continuing to act. The connection is historical rather than immediate.

The investment strategy is looser still.

An earlier choice may remain available as a rule or reminder. But when the market falls, the later action arises in a new situation. You may be forced to choose again.

What remains of a choice differs from case to case.

It may be preserved in something we deliberately set in motion, survive through habits it helped form, or remain only as one consideration among others.

The history matters without becoming a permanent command.

We can create reminders, change the environment, establish rules, build habits, create obligations, and make some actions easier or harder.

A choice occurs at a particular time.

Choice can reach into the future.

But it does not control it.

We may continue what we once chose, revise it, or later act without any identifiable choice at all.

Yet all of these can still be our doing.

Revoking A Choice

Suppose you decided to go running tomorrow morning.

Tomorrow comes. You wake up and think: No. I am not going.

You may even say it aloud.

The earlier choice has been revoked. This seems straightforward.

You chose one thing yesterday. Today you choose otherwise.

The later choice replaces the earlier one. But not every change of heart works that way.

Suppose you decided never to answer work messages after eight in the evening. Then one evening a colleague writes: "The server is down." You reply. Maybe you do not think you have abandoned the rule at all.

The situation simply does not seem to be one to which it applies. The earlier choice remains. What changed is how the present situation is understood.

Sometimes even that does not happen.

You made an investment plan. You still think the plan is good and that you should follow it.

Then the market falls sharply. You watch the losses grow. And you sell. You may still be convinced, even while doing so, that selling is contrary to the strategy you chose.

The earlier choice has not been revoked. The rule has not been reinterpreted. Something else has become strong enough to override it in action.

Or suppose you decided to finish a difficult task today.

You work for an hour. Then you stop.

You have not abandoned the plan or decided that it was a bad one. You tell yourself: Not now. I will continue later.

The choice remains in place, but its enactment is suspended. Whether you return to it is another matter.

And sometimes there is no identifiable moment of reversal at all.

You once decided to practise an instrument every evening.

For a while you do.

Then one evening you do not. A few days later you miss another.

Eventually weeks pass.

Maybe you still vaguely think of yourself as someone who practises every evening. But the old choice has lost its force. No second decision replaced it.

You never clearly said: I no longer intend to do this. You simply stopped.

An earlier choice may be explicitly revoked, reconsidered in its application, overridden in action, suspended, or simply lose its influence without any clear later choice taking its place.

This makes it difficult to understand a life as a sequence of choices governing what follows.

Choices matter. They can redirect what happens. They can leave rules, habits, promises, and altered circumstances behind.

These can help an earlier choice survive later pressure.

We can make some actions easier, others harder, create reminders, establish routines, or involve other people.

Changes in circumstances can have the opposite effect. A routine that seemed stable at home may disappear during a journey, a move, or another interruption of ordinary life.

Any choice can be revised, resisted, forgotten, or simply lose its force.

An earlier choice does not govern what follows merely because it was once ours.

Even here, there need not be a choice.

Doing Across Time

Suppose you decide to build a table.

You draw a rough plan, buy wood, measure twice, and cut once. You still make a mistake and replace one piece. Then you leave the work unfinished for a week and return to it later. Eventually the table stands in front of you.

You built it. That description seems natural.

But there was no single moment in which building the table happened.

There were many actions, some deliberate, some habitual, some corrected halfway through, and some probably forgotten by the time the work was done.

The original choice to build a table mattered. But it did not govern every movement that followed.

And the table was not built by the choice alone.

The same is true when learning a piece of music, training for a race, restoring a neglected garden, or writing a book. These activities can take weeks, months, or years. They contain pauses, changed plans, mistakes, new ideas, and periods in which little happens at all.

The original reason for beginning may remain present throughout, change over time, or disappear almost completely while the activity continues in another form.

Something that began as a side effect can even become the main reason for continuing.

Still, afterwards we say:

“I learned this.”

“I restored that.”

“I wrote it.”

The doing extends across time even though no single choice extends across time in the same way.

What we call one doing may contain many smaller doings, choices, revisions, and periods in which no choice is present at all.

Consider writing a book.

Maybe you begin with a clear idea of what it should become. Later that idea no longer convinces you. You remove a chapter, and another may appear unexpectedly. At some point you might abandon the project for months because nothing seems to work. When you return, you may no longer remember some of the reasons behind earlier decisions.

Yet those decisions remain in the text.

Some sentences were carefully constructed. Others appeared quickly and survived every later revision. Someone else may have suggested a change that became indispensable.

The finished book contains all of this.

It would be strange to conclude that you did not write it because no single plan or choice remained unchanged from beginning to end.

What makes the book your doing is not that every part of its history was chosen in advance.

You repeatedly returned to it, altered it, abandoned parts, added others, and brought it closer to what it eventually became.

This seems to be true beyond individual projects.

A household can maintain a home over many years.

No single decision of one person does that. People clean and repair things, buy and replace them, forget some problems and notice others, organize, neglect, and begin again. Different people contribute different things at different times.

Still, it can make sense to say that they maintained the home together.

A company may build a product that no single person designed or made. Towns preserve infrastructure and practices across generations, and scientific fields can develop methods through the work of people who never met one another.

Here the language of doing begins to widen.

The participants change. The choices change. Some contributions are deliberate and visible. Others become routines, institutions, tools, or inherited practices. What one person leaves behind becomes part of the situation in which another acts.

The larger the doing becomes, the harder it is to identify one chooser behind it. Yet the doing does not disappear. It becomes distributed across actions, time, and people.

Earlier choices can shape later conditions. Later actions can revise what earlier participants began. Some contributions remain visible. Others disappear into the structure of what has been built.

We may choose some parts. Others may arise from habit, training, circumstances, or what others have already done. Yet they can all become part of what we did.

This is one reason participation is a better word than control. We do not need to have chosen the whole course of something in order to have had a part in bringing it about. And what we bring about can become part of the conditions under which others act after us.

Participation

We began by asking what in our lives can properly be called our doing. Choice seemed an obvious place to look.

Choice is real. We can stop, consider possibilities, select one, and act on it. But choice did not mark the boundary we were looking for. Something can be our doing without being preceded by a separate choice at every step.

A learned activity may unfold while we monitor what happens. We may intervene only when something calls for correction. Preparation done long before may become effective in the moment. An earlier course may be maintained against resistance or abandoned when it no longer holds.

A chess move may be preceded by deliberation, while a movement in sport emerges from training and the situation. We can also initiate a calculation deliberately even though its result is not ours to choose.

These are not merely stronger and weaker versions of one process. They are different ways in which something can be our doing.

Choice is one of them. It is not the whole of participation.

Sometimes our reasons are easy to state.

You choose the screwdriver because you want to tighten the screw. The screwdriver is not simply better than the hammer or the knife. It is better for what you are trying to do.

Sometimes a reason becomes visible when we intervene.

You stop because something looks dangerous. You change course because another possibility has become available. But a reason need not accompany every part of an activity separately.

Someone may be dancing, skating, playing an instrument, or riding a motorcycle.

"Why that exact movement?" There may be no useful answer.

"Why continue the activity?" There may be one. A reason can belong to the larger course without explaining each movement within it.

Sometimes the question arises only afterwards.

"Why did I say that?"

Or: "Why did I stop?"

Later, it may be: "Why did I stay?"

And sometimes the question concerns a longer pattern: "Why did I keep returning to this?"

An answer may form. Maybe I was afraid, wanted approval, or cared more than I had realized. Maybe the old course no longer convinced me. Such answers are convictions too. They can stabilize, change, or dissolve.

A later explanation of our own action is not automatically the true motive. It is another attempt to understand our participation.

Patterns of doing can nevertheless tell us something about orientation.

We return to some questions, protect some things, prepare for some possibilities, and maintain some courses while abandoning others.

No single action need reveal very much. Across time, however, what we repeatedly do can provide evidence about what matters, what we expect, and what has become sufficiently stable to guide us.

None of this begins from an empty situation. We enter circumstances already underway: a language we speak, practices we learned, institutions and relationships already there, events that happened before we arrived, and convictions and distinctions we did not originate. Orientation precedes any particular act of reflection or choice. Yet what we do within it can alter what happens next.

We can ask a question that another person later finds difficult to answer. We can introduce a possibility they had not considered. We can teach a procedure, preserve a comparison, challenge a claim, or build something that remains available after the action itself is over. Earlier action can become part of the conditions under which later action and conviction occur.

The result is not under the original actor's command.

A question may be ignored, a possibility fail to convince, or a method be learned and later rejected. A promise can change another person's expectations in ways we did not foresee.

What we do can matter without determining what follows.

The cases examined here do not reveal one obvious mechanism common to everything we call our doing.

Explicit choice is sometimes present and sometimes absent, and control may be extensive or limited. Preparation can occur long before action, while fluent activity may outrun conscious deliberation. Consequences can also follow from what we do without having been separately intended.

Comprehensive control is not required for something to count as our doing. Nor does our doing reduce to isolated moments of explicit choice.

Participation names this broader involvement.

It does not explain the metaphysics of agency. It marks a difference that the examples repeatedly make visible: between something merely happening around us and our having a part in what happens.

That is enough for Conviction Formation Theory as a preliminary account of how finite agents can continue to inquire and act without direct control of conviction or final settlement.

We do not choose directly what will convince us, whether a deliberately posed question will become genuinely puzzling, or which possibility will eventually acquire force.

We can nevertheless ask, search, prepare, compare, practise, challenge, defend, and enter procedures whose results are not ours to determine.

What we do can alter the conditions under which conviction forms. Those altered conditions can in turn become part of someone else's orientation.

We do not stand outside conviction formation and direct it. But neither are we absent from it.

II. Examining Constraint

A second cycle of essays asking whether inquiry can find criteria for refining convictions without appealing to a final authority, and examining what logic, probability, and causality contribute to that problem.

Recursive Examination

1. The Problem Part I Leaves Behind

Part I began from the claim that *becoming convinced is not under direct voluntary control*.

We can assert something without being convinced of it, entertain a possibility, follow a rule provisionally, or act on a decision we privately doubt.

But we cannot normally decide that something will become convincing simply because we want it to.

That limitation still leaves room to act.

We can ask questions, seek information, generate possibilities, compare alternatives, externalize reasoning, introduce criteria and procedures, challenge or defend convictions, practise, prepare, and return to an inquiry later.

Part I gave this broader involvement a name: *participation*. We can have a part in what happens without comprehensively controlling what follows.

Criteria can also become sufficiently stable to organize inquiry. Several people may rely on enough of the same criteria to assess something together without reopening every standard at each step and without requiring final settlement.

Finite agents can continue to inquire and act while larger questions about truth, justification, rationality, and authority remain unresolved. At the same time questions can reopen convictions. Criteria can organize and constrain how convictions are assessed.

But the criteria, procedures, and distinctions used in that examination remain part of conviction formation too. They can become convincing, lose force, conflict with one another, and become themselves objects of inquiry.

What do these different forms of examination and constraint actually add?

And if they change what becomes convincing, by what criterion could we call one conviction better than another?

Part I made inquiry possible without direct control or final settlement.

Part II looks at what can constrain such inquiry from within it.

2. Standards Remain Inside Conviction Formation

We can find one conviction more convincing than another.

But that does not yet tell us whether one is *better* than another. People can be convinced of claims that we ourselves judge to be worse.

To make that judgment, we need some criterion by which to compare them.

We might appeal to consistency, evidence, reliability, corrigibility, empirical success, logical validity, probabilistic coherence, causal adequacy, or something else.

These standards can organize inquiry.

But Conviction Formation Theory cannot simply place one of them above the process it has been examining.

Consider the sentence: *This is a better conviction than that conviction.* If that judgment is to become part of a finite agent's own assessment, it too has to become sufficiently convincing.

The same applies to:

This is a better criterion.

This is better evidence.

This objection matters more.

This method is more reliable.

This is the rational response.

Standards and judgments about standards remain within conviction formation.

Once a criterion is sufficiently fixed, comparing convictions under it may be straightforward.

One conviction may be more consistent than another, better supported by some evidence, more predictively successful, or fare better under some other criterion.

The harder question is why that criterion rather than another one should determine which conviction counts as better.

Another complication: a person can use a criterion without being convinced that it is correct or appropriate to the problem. We may follow a criterion provisionally because a procedure requires it, because others expect us to use it, or simply to see what follows. We may even be coerced into applying a criterion we reject. Applying a criterion and being convinced by it can come apart.

A criterion can be applied provisionally or even without being convincing. But if it is to guide our own judgment of which conviction is better, it has to acquire convincing force for us.

Like other convictions, criteria encounter persons with already existing orientations, histories, capacities, practices, relationships, and surroundings.

One criterion may acquire force quickly or remain inert. Another may become convincing through training and use, then later become questionable. Something compelling to one person may leave another unconvinced. That is not ours simply to choose.

The problem does not end with disagreement about the convictions being assessed.

Two people may agree that convictions should be examined and still disagree about which evidence matters, which distinctions should be drawn, which representations preserve what is important, which criteria organize a good assessment, or which methods are appropriate to the problem.

Part I already encountered this when examining defensibility.

A conviction can be strong while poorly defended.

A defense can fare well under shared criteria and still fail to persuade.

Two people may begin by disagreeing about a conclusion and discover that their disagreement concerns the criteria under which it is being assessed.

A fitting name for this wider situation is *convictional pluralism*.

Convictional pluralism describes a situation in which incompatible convictions, criteria, standards, interpretations, or commitments concerning methods can retain convincing force across different orientations of persons and groups.

This is not a pluralism of frameworks that people freely select from a shelf.

Someone can choose to conduct an inquiry using intuitionistic logic, or agree provisionally to work under a Bayesian model.

What cannot simply be chosen is whether the relevant commitments will become convincing, gain force, lose force, or remain unchanged through the investigation.

Convictional pluralism does not make convictions, criteria, interpretations, or methods matters of preference or choice. It is not a form of relativism.

A conviction may be convincing for me and not for you, or for one group and not another. But if I am convinced of P, I cannot simply decide that an incompatible conviction is equally convincing. Nor can I decide that a criterion I find unconvincing now carries the same force as one I rely on.

Convictional pluralism describes differences in what becomes convincing across orientations.

It does not say that incompatible convictions are equally true, equally justified, equally defensible, equally useful, or equally tolerable. It does not entail that truth or correctness are relative to orientation.

What remains open is how such comparisons could be made without simply assuming the criteria that would settle them.

Conviction Formation Theory cannot settle that merely by choosing one.

Whatever criterion is used remains within conviction formation too.

It can become convincing, fail to convince, conflict with another criterion, or become itself an object of inquiry.

3. Recursive Examination

Suppose two people disagree.

One says:

“Peter deliberately excluded me.”

The other asks:

“Why do you think that?”

Maybe Peter invited everyone else. Maybe he had excluded the speaker before, or something he said earlier seemed hostile.

“Why does that count as evidence of deliberate exclusion?”

Deliberate action may have been inferred from a pattern of selective behavior.

“What else could explain what happened?”

Peter may have forgotten. He may have assumed someone else had sent the invitation. The omission may have resulted from a mailing list or a technical error.

“Why these alternatives?”

Maybe they came from experience, analogy, or someone else’s suggestion. Important possibilities may still be absent.

“Would you use the same evidential standard if someone you liked had done the same thing?”

Throughout this sequence, the examination remains part of conviction formation.

An answer can convince or fail to convince, a new possibility may seem serious or irrelevant, criteria can remain stable or become questionable.

Trying to answer one question can expose something that had previously organized the inquiry without itself being examined.

We can call this *recursive examination*: making more of what a conviction depends on, or of what has organized its examination, available as an object of further inquiry.

A conviction may depend on an interpretation. The interpretation may rely on a distinction. A distinction may matter because of a criterion. Criteria may be embedded in a method. A method may rely on assumptions, representations, mappings, rules, or procedures.

Any of these can become the next object of attention.

This matters because examination is not merely an external audit of an otherwise unchanged conviction.

Trying to answer a question can reveal an assumption that had never been explicit.

A newly available possibility may weaken what had seemed like the necessary explanation, or a contradiction may become salient. A distinction may turn out to matter in a way it did not before. Criteria can become harder to defend once they are applied elsewhere. An objection can acquire force after careful reconstruction.

The reverse can happen too.

A conviction may become stronger after examination. An objection may fail. A formal reconstruction may make a relation visible that had previously been only vaguely sensed. Criteria or methods may become more convincing once we understand better how they work and what they depend on.

Sometimes the main change is not that either side becomes more or less convinced. Further disagreements become visible.

What appeared to be a disagreement about Peter's intention may turn out to concern which alternatives were considered, how evidence of intention should be interpreted, or whether the same standard is being applied consistently.

The same pattern can appear in more formal inquiries.

Two people may agree about what follows within a formal system while disagreeing about what has to be put into it. They may accept the formal consequence but dispute a premise, a representation, or whether the system fits the problem in the first place.

Examination does not necessarily remove disagreement. It can reveal more precisely what the disagreement depends on.

Part I gives us one reason why making such dependencies available may matter.

We can deliberately ask a question without being able to decide whether it will become genuinely puzzling. Sometimes an adequate answer appears immediately and little changes. But sometimes we ask a question expecting that we can

answer it and discover that we cannot. The absence of the expected answer can itself give the question force.

Recursive examination can create such situations. It can make assumptions, alternatives, criteria, and dependencies explicit enough to become objects of further inquiry.

We can then ask what supports them, whether they apply, or what would happen if they changed.

What we hope to find is *firmer ground*: a conviction that remains convincing after more of what it depends on has been brought into view and examined. Some possible sources of disturbance are no longer hidden. We may understand better what would have to change for the conviction to weaken, become qualified, or give way.

4. Settlement Without Final Settlement

If a conviction can be examined using a criterion, and the criterion can itself be examined, why does inquiry not regress indefinitely?

Suppose we challenge a conviction and assess its defense using a criterion. We could then challenge that criterion. Whatever we use in examining the criterion could in principle become questionable too.

The move remains available.

But availability is not the same as force.

Part I already found that we can deliberately formulate a question without being able to decide that it will become genuinely puzzling.

The same applies here.

We can turn attention toward a criterion, an assumption, or a method. We can ask what supports it, what would count against it, or what may happen if it changes.

But we cannot simply decide that it will become questionable for us.

Trying to answer may expose a difficulty we had not expected. Or the question may leave what was already convincing entirely untouched.

Recursive examination makes further inquiry possible. But here too, participation does not give us command over what becomes unsettled.

The availability of the move does not mean that inquiry actually proceeds indefinitely. Most of orientation ordinarily remains in place while some part of it becomes questionable.

For the inquiry at hand, enough remains sufficiently settled to carry what happens next.

This is *settlement without final settlement.*

5. What Part II Will Examine

Recursive examination may leave a conviction on firmer ground.

More of what it depends on has been exposed and examined, while the conviction remains convincing. We may be less exposed to having something we rely on overturned by a difficulty we could already have brought into view.

That matters when other judgments, decisions, or commitments have been built on the conviction.

But firmer ground does not yet give us a definite criterion for ordering *different* convictions as better or worse. Firmer ground describes a change in the same conviction. It is, in a sense, better situated than before.

Other convictions can undergo the same kind of examination and also come to stand on firmer ground.

And a conviction on firmer ground can still be false. It can still be inconsistent with other convictions, depend on a poor representation, or leave something important unexamined.

So firmer ground does not by itself give us a single scale on which different convictions can be ordered as better or worse.

Firmer ground differs from the criteria we ordinarily use to compare convictions. We may compare convictions by truth, consistency, evidential support, predictive success, usefulness, or something else.

Once the criterion tells us what is to count in the comparison, we can ask how the convictions fare by that measure. The result may rank one above another, leave them tied, or fail to order them at all.

But the problem from Chapter 2 remains: different criteria can produce different assessments, and Conviction Formation Theory cannot settle which should govern merely by selecting one in advance. The criteria themselves remain within conviction formation and open to examination.

Part II will thus examine forms of inquiry in which, once enough conditions are fixed, some consequences are no longer independently movable.

If we can reconstruct where the constraint within a formal system comes from, and what gives the results of applying that system to a particular problem convincing force, we may learn more precisely what recursive examination can add, under what conditions it leaves us on firmer ground, and whether it gives us any further basis for calling one conviction better than another.

Part II will examine logic, probability, and causal methods as examples of such strong constraint.

Sources Of Logical Constraint

1. About Logic

Where is Anna?

If Anna is at work, her car is not in the driveway.
Her car is not in the driveway.

So Anna is at work.

Given we can be sure that the first premise has no exceptions, is that a valid conclusion?

It may feel convincing, and you start to rely on it. You send Anna a message asking whether she wants to meet at the coffee shop during her lunch break. But then Anna replies that she has a day off and is at the spa.

The premises are all true:

If Anna is at work, her car is indeed not in the driveway. It is parked in front of her office building.

Also her car is now not in the driveway. Her car is at the spa.

So the conclusion may have felt convincing at first, but then a counterexample emerged.

If you saw that right away, good. But more complicated conclusions exist.

Are we supposed to exclude all counterexamples when we want to draw a valid conclusion?

How should we do that?

Logic

Logic studies inference.

We can now look more closely at what distinguishes a conclusion that really follows from its premises from one for which a counterexample remains possible.

Consider:

15 is even or 15 is divisible by 3.
15 is not even.

Thus: 15 is divisible by 3.

Or:

All men are mortal.
Socrates is a man.

Thus: Socrates is mortal.

In the first examples, the conclusion follows from the premises.

Now compare:

15 is not even or 15 is divisible by 3.
15 is not even.

Thus: 15 is divisible by 3.

Or:

All men are mortal.
Socrates is mortal.

Thus: Socrates is a man.

These do not follow in the same way from the premises.

But how can we know that the first two have no counter examples, while the second pair has them?

What constrains the first two conclusions, and why can that constraint become so convincing?

What is missing in the second two cases, and why do their conclusions fail to acquire the same force?

To answer that, we need a way to make the relevant possibilities explicit enough to inspect.

This essay is not an introduction to logic. For that purpose it is much too thin.

We will construct only as much formal machinery as the examples require, keeping the resulting systems small enough that the relevant possibilities and mappings remain open to inspection.

The aim is to trace where the constraint inside these systems comes from and why that constraint can become convincing.

We will also ask what gives the results of applying such a system to a particular problem enough convincing force to become operative in inquiry.

That may also tell us whether this kind of constraint gives us any further basis for calling one conviction better than another.

There is no natural point at which an investigation like this becomes complete. More machinery could always be added. We will follow only a few paths far enough that the relevant sources of convincing force can be inspected rather than merely assumed.

2. A Small Propositional System

The examples from the previous chapter suggest a way to test the difference.

Can we find a case in which all the premises are true and the conclusion is false?

In other words, can we find a case allowed by the premises in which the conclusion fails?

If we can, the conclusion does not follow from those premises.

If no such case remains, we will call the inference *valid*.

That criterion will guide the construction.

Our first example was:

15 is even or 15 is divisible by 3.
15 is not even.

Thus: 15 is divisible by 3.

To examine more closely if it is valid, we will construct a small formal system in which the relation between premises and conclusion can be inspected directly.

If the model itself becomes convincing, and if it can show either that a counterexample remains or that none is available, then it may tell us something about where the constraint comes from.

What should be the main objects of our formal system?

15 is not true or false.

Even is not true or false.

That means they cannot be our primary objects.

The claims we make about them can be true or false however.

“15 is even” can be true or false. So can “15 is divisible by 3”. And so can “15 is even or 15 is divisible by 3”.

Claims that can be true or false will be our basic objects. We will call them *propositions*.

Some claims are uncertain for us, or difficult to classify, even if they are in fact either true or false and we simply do not know which. Other claims are vague or context-dependent in ways that make it less clear whether treating them as having exactly one of two truth values is an adequate representation.

Both cases raise further problems, but we do not need to settle them here. We are not trying to build a system for every kind of claim. We are trying to construct one small setting in which a source of logical constraint can be made visible. We deliberately restrict ourselves to propositions that we treat as having one of two truth values.

That gives us objects suitable for the criterion we started with.

Now take the proposition "15 is even or 15 is divisible by 3."

We can split it into two smaller claims:

"15 is even."

"15 is divisible by 3."

Each of those can still be true or false on its own.

The word "or" tells us how the two claims have been combined into a larger proposition.

The same happens with:

"15 is even and 15 is divisible by 3."

Again we have two propositions, joined in a particular way.

Now compare:

"15 is even."

We cannot split this in the same way into smaller parts that are themselves propositions and whose combination gives us the original claim.

For the present system, we will call propositions of the latter kind *simple propositions*.

We will use letters such as A and B to stand for simple propositions. At this stage, what the propositions are about does not matter to the formal structure. What matters is whether each is true or false.

For our case let:

A = "15 is even"

and:

B = "15 is divisible by 3".

Then A *and* B represents "15 is even and 15 is divisible by 3."

If we want to build larger propositions formally, we need somewhere to start.

Our simple propositions provide that starting point. Symbols such as A, B or C will count as propositions in the system.

We then need rules that tell us how further propositions can be built from ones we already have.

Otherwise any sequence of symbols would be allowed, including expressions for which we would not know what proposition they represent or how their truth value should be determined.

We will call expressions constructed according to these rules *well-formed formulas*.

The rules introduce the connectives we need and make the construction of larger formulas traceable.

If P is a proposition, then $\neg(P)$ is a proposition. It represents "not P".

If P and Q are propositions, then $(P) \wedge (Q)$ is a proposition. It represents "P and Q".

If P and Q are propositions, then $(P) \vee (Q)$ is a proposition. It represents "P or Q".

If P and Q are propositions, then $(P) \rightarrow (Q)$ is a proposition. It represents "if P, then Q".

The formation rules let us recover how a proposition was constructed.

The parentheses help preserve the order of the construction.

Every complex expression can in this way be taken apart into simpler expressions until we reach the original variables.

That trace will matter when we assign truth values, because the truth value of a complex proposition will depend on the truth values of the propositions from which it was built.

What We Have Constructed

We now have a small formal language.

Its basic objects are propositions that we treat as either true or false, and connectives. "Connective" is inherited terminology here and applies also to negation.

We have created formation rules that let us build complex propositions from simpler ones while preserving how they were constructed.

Now we need a way to find out which propositions must be true when others are true.

We have made the structure of the propositions explicit enough that their construction can be followed step by step.

If the truth value of a complex proposition can be determined from the truth values of the propositions from which it was built, then we may be able to inspect the whole process rather than simply accept the result.

What is still missing is exactly that connection.

We know how formulas can be constructed. We have not yet defined how their truth values depend on the truth values of their parts.

Fixing The Mappings

We now have expressions, but how do we make use of them?

For that we define mappings from the truth values of the simpler parts to the truth value of the whole.

Each connective operates on one or two propositions.

For two variables, there are four possible assignments:

A	B
true	true
true	false
false	true
false	false

There are no other combinations if each variable has two possible truth values.

We can look at how it was created systematically: to each truth value of A, the two truth values of B have been attached.

On the basis of complete enumeration of all cases we can now define our connectives.

We can be sure that we have not overlooked a case. That will matter for showing that there can be no counterexample.

A	$\neg A$
true	false
false	true

A	B	$A \wedge B$	$A \vee B$	$A \rightarrow B$
true	true	true	true	true

A	B	$A \wedge B$	$A \vee B$	$A \to B$
true	false	false	true	false
false	true	false	true	true
false	false	false	false	true

The first column of the first table, and the first two columns of the second table give us all possible truth value combinations of the simpler propositions A and B.

The later columns fix the truth value of the complex proposition combining them using each connective.

Once the truth values of A and B are given, the truth value of each more complex expression is no longer separately chosen. If $A =$ true and $B =$ false, then under the definitions above $A \wedge B =$ false, $A \vee B =$ true, and $A \to B =$ false.

The convincing force here is conditional. Nothing about the symbols themselves forces us to use these definitions. We have adopted them, but once we do, and once the truth values of the simpler expressions are fixed, the mappings determine the truth values of the more complex expressions in a definite way.

Why These Definitions?

The table fixes how the connectives behave inside our formal system.

As we will see now, they catch many intuitions about how connectives work in natural language. But at the same time they exclude some quite natural readings. Those readings require more complicated constructions. We gain simplicity and inspectability, but some natural-language claims become harder to translate.

Take $A \wedge B$.

If it represents "A and B", then it should be true exactly when both A and B are true.

That is the basic intuition behind "and" in natural language.

But what we lose are the fine differences between "Peter felt sick and Peter took the medicine" and "Peter took the medicine and Peter felt sick".

There is no temporal or causal meaning hidden in the logical "and".

It is true when both components are true, and false otherwise, and no other meaning is preserved.

For $A \vee B$, we use the inclusive sense of "or": "A, or B, or both". It is false only when both A and B are false.

The exclusive sense of "or" gets lost, like in "We can go to the movies, or we can stay home."

If we want to express it, we need to resort to something like $(A \vee B) \wedge (\neg A \vee \neg B)$.

Negation is rather simple.

If A is true, then $\neg A$ is false.

If A is false, $\neg A$ is true.

Nothing worth mentioning seems lost here.

For $A \rightarrow B$ the case we need to exclude is the one in which A is true and B is false. That is exactly the shape of the counterexample we used to test whether a conclusion follows from its premises.

So we define $A \rightarrow B$ as false in that case, and true in the other three.

Treating every other combination as true makes the connective very simple to handle, but it has consequences that may seem strange.

Suppose we have:

Cats are plants.
Trees are plants.
Cats are animals.

Under the ordinary interpretation, "Trees are plants" is true. So is "Cats are animals". "Cats are plants" is false.

According to the mapping we defined for $A \rightarrow B$, both of these come out true:

"If cats are plants, then trees are plants."

If the premise is false, the whole proposition is true. We were only forbidding the case that the premise is true and the consequence is false. That is not the case here.

"If cats are animals, then trees are plants."

Because premise and consequence are true, the whole proposition is true, too. The formal system is not telling us that being a cat has anything to do with trees being plants. It is preserving only this condition: $A \rightarrow B$ is false when A is true and B is false. If B is already true, that forbidden case cannot occur.

These consequences have long been recognized as peculiarities of the so-called material implication $\rightarrow$. The connective preserves only a narrow condition: the antecedent cannot be true while the consequent is false. It contains no causal or other connection between what the two propositions say. That is exactly the part of the conditional relation we want this system to preserve.

Once the representation and the mappings remain in place, the truth values of the more complex propositions are no longer settled separately. They follow from the truth values of their parts according to rules we can inspect directly.

Testing The Inference

How can we use this to check if inferences are valid?

Let's assume we have the following premises. They can either be our first example from the last chapter, or any other inference that fits this pattern:

$A \vee B$

$\neg A$.

B.

We can combine them into one expression (we drop the first pair of brackets; it does not change anything in the reading):

$(A \vee B) \wedge \neg A$.

Now compare that expression with the proposed conclusion:

$$B.$$

A	B	$(A \vee B) \wedge \neg A$	B
true	true		true
true	false		false
false	true		true
false	false		false

How do we find the truth value of a complex proposition?

We can follow the way the proposition was constructed.

Our proposition is

$$(A \vee B) \wedge \neg A$$

We begin with its simple propositions, A and B, and list every possible assignment of truth values:

A	B
true	true
true	false
false	true
false	false

Then we look for the simplest parts whose truth value is not yet in the table.

Here these are $\neg A$ and $A \vee B$.

Their truth value depends only on A and B, so we can add a column for each of them:

A	B	$\neg A$	$A \vee B$
true	true	false	true
true	false	false	true
false	true	true	true
false	false	true	false

Now the truth values of both parts of the larger proposition are available:

$$(A \vee B) \wedge \neg A.$$

So we add one more column:

A	B	$\neg A$	$A \vee B$	$(A \vee B) \wedge \neg A$
true	true	false	true	false
true	false	false	true	false
false	true	true	true	true
false	false	true	false	false

This gives us a general procedure.

Start with the simple propositions and enumerate all possible assignments to them. Then add the complex parts one by one, beginning with those whose own parts already have truth values in the table. Continue until the whole proposition has a column.

Because our formation rules let us take every complex proposition apart into smaller propositions, we can repeat this procedure for propositions of any finite complexity.

We can now shorten the table to A, B, the premise and the conclusion:

A	B	$(A \vee B) \wedge \neg A$	B
true	true	false	true
true	false	false	false
false	true	true	true
false	false	false	false

The first two columns give us every possible assignment to A and B. In that way we make sure we have every case covered.

The third column tells us which assignments make both premises true.

Only one does.

In that same row B is true.

There is no row in which $(A \vee B) \wedge \neg A$ is true while B is false.

Thus the inference is valid in the sense we fixed at the beginning of the chapter.

If someone wants to reject the conclusion while keeping both premises, we can ask what assignment would make that possible.

The table contains every assignment allowed by the system.

None works.

The attempted denial requires a counterexample that is not available under the conditions stipulated here.

That matters because we took care at two earlier points.

First, the first two columns contain every possible combination of truth values for A and B in this system.

Second, the formal expressions are meant to preserve the parts of the natural-language claims that matter for the inference.

If that representation remains convincing, and if the mappings we defined remain in place, then the table leaves no case in which the premises are true and the conclusion false.

Under those conditions, the inference is valid.

An Inference That Fails

We can now return to the first inference from the previous chapter that did not seem convincing:

15 is not even or 15 is divisible by 3.
15 is not even.

Thus: 15 is divisible by 3.

Using the same representation gives $\neg A \vee B$ and $\neg A$.

The proposed conclusion is now B.

Again, we can combine the premises and inspect every possible assignment:

A	B	$\neg A$	$\neg A \vee B$	$(\neg A \vee B) \wedge \neg A$	B
true	true	false	true	false	true
true	false	false	false	false	false
false	true	true	true	true	true

A	B	$\neg A$	$\neg A \vee B$	$(\neg A \vee B) \wedge \neg A$	B
false	false	true	true	true	false

This time two assignments make both premises true.

In one of them B is false:

A	B	$(\neg A \vee B) \wedge \neg A$	B
false	false	true	false

That means the premises can be true, while the conclusion is false.

The constraint that held in the earlier inference is absent here.

And the table gives us the counterexample directly.

Translated back into ordinary language, the premises do not exclude this situation:

15 is not even or 15 is divisible by 3.
15 is not even.
15 is not divisible by 3.

The last sentence is false in fact as a statement about arithmetic. But if we want it to enter the inference and make the counterexample go away, we need to add it as a premise. The premises we have do not force the conclusion. And that is the only question we are asking here: whether the inference is valid.

For the validity of an inference, it does not matter whether the conclusion happens to be true for some other reason than following from the premises. An inference cannot use facts that have not entered the representation.

We can see this more easily by changing the example:

17 is not even or 17 is divisible by 3.
17 is not even.
17 is divisible by 3.

Here the same inference form would lead to a false conclusion.

On the other hand this is a valid inference, even though one premise is false:

17 is even or 17 is divisible by 3.
17 is not even.

Thus: 17 is divisible by 3.

Our small formal system does both kinds of work we wanted from it, provided that we handle the translation into the system with care.

It shows, within the representation we constructed, why one inference satisfies our criterion of validity and why the other does not.

And it can do that because the relevant possibilities have all been made explicit.

What This Construction Shows

We can now separate two things that were easy to run together at the beginning.

First, we can see where the constraint inside the formal system comes from.

For a finite number of simple propositions, we can enumerate every possible assignment of truth values. The mappings then determine the truth values of the more complex propositions for each assignment.

To test an inference, we inspect every case in which the premises are true.

If none of those cases leaves the conclusion false, then there is no counterexample within the system. We can be sure of that. The conclusion is not another choice left open after the premises, mappings, and possible assignments have been fixed.

Several things contribute to the constraint: complete enumeration of the relevant possibilities, fixed mappings between simpler and more complex propositions, and a construction that remains open to inspection.

They also make mistakes easier to locate. If something seems wrong, we can ask whether a case was omitted, a mapping was applied incorrectly, a formula was constructed incorrectly, or the original problem was represented badly.

That combination seems to be the source of how this constraint becomes so widely convincing.

But this alone does not show that the formal system is convincing as an appropriate representation of a particular inference.

In our examples, we were interested only in whether the premises could be true while the conclusion was false. We therefore represented the claims as propositions with two truth values and chose mappings that preserve the distinctions needed for that test.

This deliberately leaves other things out. The system does not preserve temporal order, causal connection, uncertainty, vagueness, or every feature of ordinary conditionals.

The choice of mappings can be questioned, as can the decision to treat the claims as simply true or false, the translation of the ordinary-language claims into propositions, and whether something left out by the representation matters to the problem.

If those choices remain convincing for the case at hand, then the absence of a counterexample inside the system can carry convincing force in the inquiry.

3. A Small Quantified System

The propositional system from the previous chapter could inspect some kinds of conclusions already.

15 is even or 15 is divisible by 3.
15 is not even.

Thus: 15 is divisible by 3.

It could do that because the inferentially relevant structure was already visible in the propositions and their combination.

Our second example is different:

All men are mortal.
Socrates is a man.

Thus: Socrates is mortal.

If we represent these three propositions simply as A, B and C, then our small propositional system sees three unrelated propositions:

A
B

Thus: C

But this representation has lost some of the structure that is important to what happens.

"Man" occurs in the first premise and again in the second.

"Mortal" occurs in the first premise and again in the conclusion.

Socrates is the same person in the second premise and the conclusion.

And "all men are mortal" does not concern Socrates alone.

None of that survives in A, B and C.

If we want to inspect the constraint in this inference, we need a representation that preserves more of what happens inside the propositions.

Things, Properties, And Propositions

Consider "Socrates is a man."

This proposition has parts that play different roles.

"Socrates" is not true or false.

Neither is "is a man" by itself.

But the complete claim "Socrates is a man" can be true or false.

For our small system, we will distinguish between *things* we talk about, *properties* we attribute to them, and *propositions* produced by *attributing* a *property* to a *thing*.

Let s stand for Socrates.

Let M stand for the property "is a man".

Then $M(s)$ shall stand for "Socrates is a man."

Similarly, let D stand for the property "is mortal".

Then $D(s)$ shall stand for "Socrates is mortal."

The notation now preserves something our earlier propositional letters did not. The same object s can occur in several propositions. And the same property can be attributed to several objects.

That gives us a way to preserve some of the internal structure that matters to the inference.

It does not tell us whether Socrates is in fact a man or mortal. Those truth values still have to come from the context of the application of the formal system.

What the notation gives us is a way to keep track of what is being said about what.

A Small Collection

The first premise causes another problem.

All men are mortal.

This does not concern only Socrates.

To inspect the word "all", we need to know what objects are under consideration.

For the present construction, suppose there are exactly three:

$$\{s, a, b\}.$$

The letters stand for three objects. One of them is Socrates.

The collection is complete for this small system. There is no fourth object hiding outside it that also has to be checked.

This is obviously not a claim about the world.

Ordinary universal claims can range over far larger collections, including ones we cannot simply list. We restrict the present system because a finite collection lets us inspect what universal coverage amounts to.

That is the same advantage we used in the previous chapter. There, two truth values for two variables gave us four possible assignments. Here, a finite collection lets us make every object under consideration explicit.

Saying It For Every Object

We can now try to represent "All men are mortal."

Without introducing any new symbol, we can write the relevant condition separately for every object in our collection:

$$M(s) \to D(s)$$

$$M(a) \to D(a)$$

$$M(b) \to D(b).$$

We already defined the implication in the previous chapter. For each object, the expression excludes one combination: the object being a man while not being mortal.

In our finite model, the representation of "all men are mortal" requires this condition to hold for every object we have admitted.

All three cases can be inspected directly.

But the construction is repetitive. The same pattern appears three times. Only the object changes.

A Repeated Place

The first position inside both properties is occupied once by s, once by a, and once by b. The property however does not change.

We can mark that repeated place with another symbol: x.

Then the common form becomes $M(x) \to D(x)$.

Here, x is not another object in our collection. It marks a place into which each of those objects can be inserted. It is a variable.

If we put s there, we get:

$$M(s) \to D(s).$$

If we put a there:

$$M(a) \to D(a).$$

And for b:

$$M(b) \to D(b).$$

The new notation compresses the repeated structure. But we can reconstruct each individual case.

Taming The Variable

We now have a common form:

$$M(x) \to D(x).$$

But the variable x is still only marking a place. The two occurrences of x tell us to put the same object into both places, but we have not yet said which objects are to be considered, or how far that use of x extends.

For that we introduce $\forall$.

It does two jobs for us. It says that we are considering every object in our collection, and it marks the scope within which x is governed by that requirement.

That matters because a later use of x need not always belong to the same quantifier. Parentheses let us mark where the reach of a quantifier ends and where another use could begin. We may also need other quantifiers for different kinds of claims.

For our present claim, we want the requirement to cover the whole formula:

$$\forall x(M(x) \to D(x)).$$

The quantifier tells us that throughout the formula in its scope, x ranges over every object in our collection.

For the small system we have constructed, this means that all of these must be true:

$$M(s) \to D(s),$$

$$M(a) \to D(a),$$

$$M(b) \to D(b).$$

Because the collection is finite and complete, we can unfold the quantified formula again and inspect every case.

What We Have Constructed

We can now distinguish objects from properties, preserve the same object across several propositions and the same property across several objects, and express one condition that has to hold for every object in a fixed finite collection.

The variable and the universal quantifier compress that repeated structure, but we can still unfold it whenever we want to inspect the individual cases.

The interpretation has to be supplied from outside of the system.

What the system gives us is a way to make more of the structure of the inference explicit.

Is it enough to recover the constraint we started with?

Testing The Inference

Our original inference was:

All men are mortal.
Socrates is a man.

Thus: Socrates is mortal.

In the system we have constructed, this becomes:

$$\forall x(M(x) \to D(x))$$

$$M(s)$$

with the proposed conclusion:

$$D(s).$$

The notation alone does not show why the conclusion follows.

So we unfold the universal premise.

Our complete collection is:

$$\{s, a, b\}.$$

That means the first premise requires:

$$M(s) \rightarrow D(s),$$

$$M(a) \rightarrow D(a),$$

and:

$$M(b) \rightarrow D(b).$$

The first of these concerns Socrates.

So among the things we have taken as true are:

$$M(s) \rightarrow D(s)$$

and:

$$M(s).$$

The truth-value mapping for implication from the previous chapter already gives us what we need.

It is the same structure as:

$$A \rightarrow B$$

$$A$$

Thus:

$$B.$$

There was no assignment in which both premises were true while the conclusion was false.

The same propositional mapping still applies even though the letters now occur inside a more structured notation.

If:

$$M(s) \rightarrow D(s)$$

is true, and:

$$M(s)$$

is true, then:

$$D(s)$$

cannot be false under the mapping we already fixed.

The quantified premise gave us the Socrates instance because Socrates belongs to the complete collection over which the condition ranges.

The second premise tells us that Socrates satisfies the first part of that condition.

Together they leave no case in which Socrates is a man but not mortal.

The inference is valid in the small system.

If we add more objects, even as many as there are human beings, we get a much larger collection, but still a finite one. In principle, the quantified expression can still be unfolded into one condition for each object.

That is as far as our present construction carries us. But it can fully accommodate the real example. For an infinite collection, we could no longer inspect universal coverage by unfolding every case into a finite list.

Quantified logic can handle such structures, but doing so requires machinery we have not constructed here.

An Inference That Fails

We can now test the other inference:

All men are mortal.
Socrates is mortal.

Thus: Socrates is a man.

Its formal representation is:

$$\forall x(M(x) \rightarrow D(x))$$

$$D(s)$$

with the proposed conclusion:

$$M(s).$$

To show that the conclusion does not follow, we need one case in which both premises are true while the conclusion is false.

We can try to construct one systematically.

The proposed conclusion is:

$$M(s).$$

So we begin by making it false:

$$M(s) = \text{false}.$$

The second premise has to remain true:

$$D(s) = \text{true}.$$

Now consider the part of the universal premise that concerns Socrates:

$$M(s) \to D(s).$$

With the values we just chose, this becomes:

$$\text{false} \to \text{true},$$

which is true under the mapping we already fixed.

So making the conclusion false has not yet broken either premise.

We still need the universal condition to hold for a and b. We can choose any values for them that do not make $M(x)$ true while $D(x)$ is false.

For example:

Object	Man	Mortal
s	false	true
a	true	true
b	false	false

Now we can inspect the universal premise object by object.

For Socrates $M(s) \to D(s)$ is true.

Its first part is false and its second part true.

For a $M(a) \to D(a)$ is also true.

Both parts are true.

For b $M(b) \rightarrow D(b)$ is true as well.

Its first part is false. Remember that is how we defined the implication.

So every object in the collection satisfies $M(x) \rightarrow D(x)$.

That means $\forall x(M(x) \rightarrow D(x))$ is true.

The second premise $D(s)$ is also true.

But $M(s)$ is false.

The premises can be true while the proposed conclusion is false.

Translated back into ordinary language, the premises do not exclude this situation:

All men are mortal. Socrates is mortal. Socrates is not a man.

There is a counterexample.

What This Construction Shows

The system has reproduced the inferential difference we started with.

To do that, it had to preserve more structure than the propositional system from the previous chapter.

It had to keep track of objects and properties, preserve the same object across several propositions, and make explicit what it means for one condition to hold for every object in the collection.

The universal quantifier compresses that repeated structure. In our finite model, we can unfold it again into one case for each object.

The convincing force of this additional constraint seems to come from preserving these relations explicitly: the same objects across propositions, the properties attributed to them, and the requirement that a universal claim hold for every object in the collection.

Once the universal premise is fixed, Socrates belongs to the collection, and $M(s)$ is true, the condition that applies to every object also applies to Socrates. The propositional constraint constructed in the previous chapter can then operate on that particular case.

But the application still depends on what we supplied.

We chose the objects under consideration, decided which object s represents, chose the properties represented by M and D, treated the collection as complete, and translated "all men are mortal" as a universally quantified conditional. We also relied on the truth-value mappings constructed in the previous chapter.

Any of those choices can be questioned.

If they remain convincing for the problem at hand, then the formal system makes the relevant structure explicit enough that a counterexample can either be constructed or excluded.

What seems to make the resulting constraint convincing is that we can inspect how it arises from the representation we supplied.

An Aside On Hidden Structure

The quantified system also helps us see why we used the arithmetic example in the previous chapter.

Consider:

If the alarm light is on, the machine has stopped.
The alarm light is on.

Thus: the machine has stopped.

Written propositionally, this looks simple enough:

$$A \to B$$

$$A$$

Thus:

$$B.$$

It is natural to read the first premise as a general rule. But propositional logic evaluates a complex proposition from a particular assignment of truth values to its atomic propositions. That requirement is easy to overlook in ordinary-language examples.

"The alarm light is on" and "the machine has stopped" do not have fixed truth values unless a particular machine, time, or situation has been settled. Without that, there is no single valuation for A and B to begin with. Sometimes the machine has stopped and sometimes it has not, and similar for the alarm light.

The truth value of $A \to B$ is determined from the truth values of A and B by the mappings we fixed earlier. If the atomic propositions have not yet received definite truth values, neither has the complex proposition built from them.

Read as a general rule, this example does not yield a propositional interpretation at all.

We can fix that by referring to a specific situation where the truth values of A and B are indeed determined, but then we lose the rule character of "If the

alarm light is on, the machine has stopped". It now says only that this particular situation is not one in which "the alarm light is on" is true and "the machine has stopped" is false. That is hard to square with the ordinary meaning of the if-then sentence as a rule about the machine. Or we can represent the changing circumstances explicitly with a richer system, for example a quantified system.

The arithmetic example avoids much of this problem.

"15 is even" and "15 is divisible by 3" already concern one fixed object. Their truth values do not change from one occasion to another. We can therefore treat them as complete propositions without first fixing a surrounding situation.

The quantified system is a better fit when the truth of a claim depends on which object, time, or situation is under consideration. Those circumstances can become part of the representation instead of being silently supplied by the reader.

In that sense, the two systems also illuminate each other. The quantified system makes it easier to see where a propositional representation has hidden a dependency, and to say more precisely what has been left implicit.

4. Sources Of Logical Constraint

The small systems in the previous chapters were deliberately limited. Their purpose was to keep enough of the relevant machinery visible that we could inspect where some of the constraint came from.

One source of the convincingness of constraint was representation.

In the propositional system, replacing ordinary claims with letters was useful because the internal structure of those claims did not matter for the number inference.

For the Socrates inference, we needed to preserve the fact that the same object and the same properties appeared in different propositions. Formalization preserves only what the representation makes explicit.

The second source was fixed structure.

Once the truth-value mappings of the connectives had been defined, the truth value of a complex proposition was no longer settled independently of the truth values of its parts. Once universal quantification had been defined over a complete finite collection, the same condition applied to every member of that collection.

The third source was the treatment of possibilities.

For two Boolean variables, we could list every assignment. That gave us a complete space in which to ask whether the premises could remain true while the conclusion became false.

One surviving counterexample was enough to make an inference invalid.

The quantified example worked similarly. We did not need to list every possible interpretation to disprove the inference.

One interpretation in which both premises were true and the conclusion false gave us a counterexample.

The fourth source was compression.

The expression $\forall x(M(x) \rightarrow D(x))$ was shorter than writing the same condition separately for every object.

But in our finite system the compressed expression could still be unfolded. The variable and quantifier preserved the repeated structure in a form that could be reused.

These sources interact.

A representation makes some structure available. Definitions and mappings constrain what can happen within it. Formal devices compress repeated structures. Possibilities can then be tested, and conclusions fail when a counterexample remains.

The constructions also made those relations open to visual inspection and manipulation. Possibilities could be laid out so that several cases remained visible at once, and the spaces were produced by repeatable procedures rather than assembled arbitrarily. Later, truth-value mappings and formal rules replaced some of this direct manipulation with rule-governed operations.

Part of the convincing force seems to come not only from relations being made explicit and fixed within the system, but from being able to inspect how they are arranged and to reproduce the operations by which the constraint appears.

This continues something already encountered in Part I. Under ordinary conditions, perception can strongly constrain what we become convinced of. Externalizing an inquiry can make parts of it persist outside immediate attention and become available for comparison and repeated inspection. Formal logic adds rule-governed constraints to such externalized structures.

Where These Small Systems Sit

We can now use some of the standard terminology we mostly avoided while constructing the systems.

The system in the second chapter contained familiar parts of classical propositional logic. The letters A, B, and C were propositional variables, and our construction rules defined a small language of well-formed formulas.

The assignments of true and false were Boolean valuations. The symbols $\neg, \wedge, \vee, \rightarrow$ were truth-functional connectives. The tables we used were truth tables.

When we asked whether there was an assignment in which the premises were true and the conclusion false, we were testing semantic validity. A surviving assignment of that kind can be called a countervaluation.

The third chapter added machinery familiar from first-order logic.

The symbol s functioned as an individual constant naming Socrates. M and D were unary predicate symbols. Expressions such as $M(s)$ were atomic formulas. The symbol x was a variable. Our three-object collection served as a finite domain of discourse.

$\forall x(M(x) \rightarrow D(x))$ used universal quantification. The table assigning the properties "man" and "mortal" to the objects gave us a small interpretation. Together with the domain, this is close to what is called a finite structure.

When such a structure makes the relevant premises true, it is a model of those premises. When it also makes the proposed conclusion false, it is a countermodel to the inference.

Full first-order logic distinguishes more carefully between the symbols of the language and the objects they denote.

For example, the symbol s and Socrates himself are not the same thing. We mostly collapsed that distinction because nothing in our investigation depended on keeping it visible.

The standard terminology also gives us a useful distinction between syntax and semantics.

Syntax concerns which expressions belong to the formal language and how they can be constructed. It concerns what we called well-formed formulas. Semantics concerns how those expressions receive truth conditions: an interpretation assigns meanings to the nonlogical parts of the language.

Semantic validity concerns what happens across the relevant interpretations: an inference is valid if there is no interpretation in which all its premises are true and its conclusion false.

What Modern Logic Adds

The small systems worked partly because they were small. That advantage disappears quickly.

With two propositional variables, there are four Boolean valuations. With ten variables, there are already 1024. With twenty, there are more than a million.

The possibility space is still finite, but direct inspection is no longer attractive.

First-order logic creates a larger problem. Its domains need not be finite at all.

A universal statement may concern every natural number, every point in a space, or every object in some other infinite structure.

The method we used in the Socrates example, writing out the condition separately for every object, is then unavailable.

Modern logic develops machinery for working beyond those limits.

One part of that machinery is recursive syntax. Our simple formation rules already pointed in that direction.

A complex expression was built from simpler expressions according to fixed rules, and its structure could be recovered afterward. Modern formal languages develop this systematically. They can generate expressions of arbitrary finite complexity while preserving how those expressions were constructed. That extends one source of constraint we already encountered: traceable structure.

Semantics is extended in a similar way.

Instead of defining truth only for a few hand-built examples, modern logic defines systematically how formulas are evaluated in interpretations or structures.

Universal quantification no longer depends on being able to write every member of the domain on the page. Its semantic definition can range over every object in the domain whether that domain contains three objects or infinitely many.

This increases reach, but it also changes what carries the convincing force.

In our small systems, much of the construction could be inspected directly. In larger systems, more of the work is carried by mathematical definitions and the theory built around them.

Another important addition is proof.

So far, we have approached validity mainly through semantics. We asked whether a counterexample existed.

For small propositional systems, a truth table can answer that directly. For larger systems, an exhaustive search may become impractical or impossible.

Formal proof systems offer another route. Natural deduction, sequent calculi, tableaus, and axiomatic calculi differ in how they organize proofs, but for our purposes they share an important feature.

They allow reasoning to proceed through local formal steps.

Instead of inspecting every interpretation, we transform formulas according to specified rules. A large space of possibilities can be replaced by a sequence of smaller operations. That is another form of compression. It also creates another question: why should the permitted proof steps convince us?

This is where metatheory enters.

A proof system can itself become an object of mathematical investigation. A soundness result connects syntactic derivation back to semantics. Roughly, if a proof system is sound, then anything derivable from the premises by its rules is

semantically valid. The local transformations do not lead us to a conclusion for which a semantic counterexample remains.

Completeness establishes a converse relation for systems where it holds. Roughly, if a conclusion is semantically valid, then an appropriate complete proof system can derive it. For classical propositional logic and classical first-order logic, sound and complete proof systems are available.

The proof system can be examined in turn.

We can investigate not only whether a particular inference succeeds. We can also investigate whether the procedures we use for producing proofs correspond to the semantic relation we wanted them to track. That gives us another level at which the system can itself be examined.

Classical propositional and first-order logic preserve some distinctions and ignore others. Other systems change the available machinery.

Modal logic introduces formal treatment of notions such as necessity and possibility. Temporal logics represent temporal structure. Many-valued logics allow more than two truth values. Relevance and paraconsistent logics alter other parts of the inferential setting.

Increasing formal precision does not by itself choose one unique formal system for every inquiry.

The question of what structure needs to be represented remains.

From Inspection To Practice

There is another difference between our investigation and ordinary logical practice.

A trained logician usually does not reconstruct the truth table for:

$$A$$

and:

$$A \to B$$

before accepting:

$$B.$$

The form is familiar.

A proof system may contain a rule corresponding to the move. The user may also know that the relevant proof system is sound.

Notation becomes familiar. Common forms become recognizable. Proof rules become ordinary working tools. Training makes some transformations available without deliberate reconstruction. Books, teachers, colleagues, and mathematical practice contribute further layers of reliance.

Mature logical practice depends on more than one source of convincing force at once.

Recursive examination becomes relevant when something no longer convinces, when two standards conflict, when an unfamiliar inference appears, or when the limits of a system matter.

We can then reopen parts of what had been compressed. We may reconstruct a truth table, examine a proof rule semantically, or challenge a formalization. A metatheoretical result can also become an object of inquiry.

Logical practice fits the same active and passive structure encountered elsewhere in conviction formation.

We can construct examples, search for countermodels, learn calculi, check proofs, alter representations, and choose which formal systems to investigate.

We cannot simply decide what the result of those activities will convince us of.

What The Logic Investigation Found

One question concerns the constraint inside a formal system.

Once enough of the representation, mappings, and premises remain fixed, some conclusions are no longer independently open. A counterexample either remains available or it does not.

The other concerns the application of that system to a problem.

We may accept the formal constraint while questioning whether the problem was represented adequately. A distinction may have been lost, a premise may be doubtful, or another formal system may preserve what matters better.

When an inference or its formalization becomes questionable, recursive examination can proceed in two directions.

We can examine what follows once the formal conditions are in place. This is often comparatively contained: if the formal system and the relevant inputs are fixed, we can inspect whether the conclusion follows within it.

Or we can examine whether those were the right conditions to put in place for the problem.

That can reopen the representation, the interpretation, the premises, or even the choice of formal system itself.

If more of these dependencies have been exposed and the result is still convincing, then we are on firmer ground. We know more clearly what the conclusion

depends on and what would have to change for it to move. It is, in that sense, better than before.

But that does not mean it is better than other convictions. It still does not give us a definite criterion for ordering different convictions as better or worse.

Logic And Answerability

There is another result, but it needs to be made visible.

Suppose a counterexample has been constructed, an inconsistency exposed, or a conclusion shown not to follow from the stated premises.

The conclusion may be revised, a premise rejected, the formalization questioned, or another interpretation or logical system preferred. The constraint may also be judged irrelevant to the problem.

The inquiry nevertheless now contains a difficulty that was not there before, or was not yet visible in that form.

If the original conclusion remains convincing, attention can move to the premise, representation, interpretation, or choice of formal system on which the difficulty depends.

A logical constraint can also become part of a disagreement between people.

If enough of the logical machinery is shared, one person can construct a counterexample to another person's inference, expose an inconsistency, or ask whether a conclusion follows from premises they both accept.

The disagreement may then shift. It may concern a premise rather than the conclusion, the representation rather than the formal consequence, or whether the logical system fits the problem at all.

Once a difficulty has been made visible in terms both participants can sufficiently follow, one can ask the other how it is to be handled. Rejecting the counterexample, the formalization, or the relevance of the logical constraint remains possible. Those responses can themselves become parts of the inquiry.

This is one form of answerability. And it need not remain confined to the exchange in which the difficulty first appeared.

The particular challenge may pass, but the form of examination remains available.

Logical procedures can also be learned, reused, criticized, and introduced into later inquiries. Other people can bring them with them. Once such practices become sufficiently shared, inquiry takes place in an environment in which these forms of examination are available.

In that sense, logic becomes a fact of intellectual life.

The logic case suggests that forms of constraint may matter not only because of the conclusions they produce. Once they become sufficiently available and

shared, they may also change what participants in an inquiry become answerable for.

There may be something still more basic underneath this.

Logic needs propositions that are true or false. But that formal distinction may depend on an earlier, ordinary distinction between something being the case and not being the case. That distinction may already make new questions and forms of answerability possible in much the same way as logical constraint does.

Once people distinguish between describing what is the case and inventing something, or between reporting what they take to be the case and deliberately saying otherwise, further questions can arise before any formal logic is introduced.

The investigations of probability and causality will give us other cases in which to see whether the same pattern appears.

Sources Of Constraint In Probability Theory And Bayesian Updating

1. About Probability

Colored Socks

You have five socks in a drawer.

Three are black and two are blue.

It is early, the room is dark, and you want a matching pair.

You could turn on the light, but then your spouse would wake up.

Three of the five socks are black. So you may think your chances of pulling out a black pair are pretty good.

You reach into the drawer and take two.

One is black. The other is blue. Bad luck, maybe.

But what were your chances in the first place?

There are only a few possibilities we need to consider.

You might draw two black socks. You might draw two blue socks. Or you might draw one of each.

But those three possibilities do not have the same weight.

There are more ways to end up with one black and one blue sock than with either matching pair.

So simply listing the possible outcomes does not yet tell us how likely they are.

How can we use what we know about the socks in the drawer to work out what weight each possibility should carry?

Probability Theory

Logic let us ask which possibilities remained compatible with fixed premises and mappings.

Probability lets us go further when several possibilities remain open and assign different weights to them. Under suitable conditions, the result can be convincingly constrained.

Where does that convincing force come from?

The previous investigation of logic began with very small formal systems. In small cases, the relevant possibilities could remain open to inspection. We could see what had been represented, which mappings had been fixed, which possibilities survived, and where a counterexample remained.

Probability offers cases of the same kind.

Dice and cards give us examples in which the relevant possibilities are easy to inspect. A die has only six faces, two throws still give a manageable number of pairs, and after drawing a card we can inspect what remains.

These examples are elementary enough that some of the structure later compressed into probability formulas stays visible.

We will also look at Bayesian updating.

Probability first lets us work forward from a model and a distribution to the probabilities of possible results. But inquiry often gives us the problem in the other direction. We observe a result and want to know how it should change the weights we assign to the possibilities that might have produced it.

Bayes' theorem gives us a simple case in which that updating can still be reconstructed from conditional probability and algebra.

More elaborate statistical methods usually rely on additional mathematical machinery that we would first have to reconstruct, even when much of it is not directly relevant to the question this essay is asking.

We will therefore follow a small path from dice to cards and then to Bayesian updating, keeping enough of the relevant possibilities and mappings visible to inspect how probabilistic constraint is constructed and how it changes when new information arrives.

As in the logic essay, this essay is not an introduction to probability theory. It will leave almost all of probability theory untouched. It does not depend on choosing one interpretation of probability over another. And although Bayesian updating is one of the cases we examine, the essay does not argue that Bayesian methods are preferable to other statistical approaches. Appendix B briefly broadens the picture with two frequentist examples, randomization and sampling, which handle evidence differently.

2. Dice

Put an ordinary six-sided die on the table. Before throwing it, we can name six possible results: $1, 2, 3, 4, 5, 6$.

We have already selected an object and specified an experiment: throw the die and look at the upper face after it has come to rest. We have also decided what counts as the result: the number on that face.

Many other differences between throws do not enter the result. The die may bounce twice or five times, land near the edge of the table, or rotate differently. The sound may differ too.

For the question we are asking, none of those differences is preserved. We can write:

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

The symbol Ω ("Omega") will stand for the possibilities admitted by this small model.

We have constructed six outcomes by deciding what about the throw will count as the result.

For the question "Which numbered face ends upward?" that representation seems very convincing.

The six outcomes exclude one another. If the result is 2, it is not also 5.

They also appear to exhaust the experiment as we have specified it. The die has no seventh numbered face. We can pick it up. Turn it around. Count. Under ordinary conditions, one of those six faces ends upward.

This is a friendly possibility space.

Earlier we encountered the problem that relevant possibilities have to become available somehow. An evaluation cannot later recover an alternative that never entered the inquiry.

Here we have reasons to think that the relevant alternatives have all been represented.

Weight

Now suppose we treat the die as fair.

That adds something.

We no longer say only that six outcomes are possible. We say that none of them should receive greater weight than another.

So:

$$P(1) = P(2) = \cdots = P(6) = \frac{1}{6}.$$

The six possibilities alone do not tell us how much weight each should receive.

Imagine putting a small metal weight inside the die near one face. The same numbered outcomes remain possible. But equal weighting has become much less convincing.

The loaded die separates two things that were easy to run together: counting gives us the number of possibilities, while fairness gives us one reason to weight them in a specific way.

For an ordinary die manufactured to be symmetric and thrown under ordinary conditions, equal weighting may seem natural because of the visible symmetry of the object.

But it remains a distinct step.

The expression:

$$P(1) = \frac{1}{6}$$

contains more than the fact that there are six possibilities. It depends on which possibilities we admitted and on why we treat them as carrying equal weight.

Once those things remain convincing, further calculations become simple.

Let $A = \{2, 4, 6\}$. This is the event that the result is even.

Three of our six equally weighted possibilities belong to it.

So:

$$P(A) = \frac{3}{6} = \frac{1}{2}.$$

In this simple setting, probability has become a proportion. We count the possibilities in the event and compare them with the possibilities in the whole space.

Once we have decided to treat the die as fair, everything the calculation depends on is visible in the small model.

Throw Twice

Now throw that fair die twice. A complete outcome contains two results.

For example:

$$(2, 5)$$

means that the first throw produced 2 and the second produced 5. For every possible result of the first throw, there are six possible results of the second.

So there are:

$$6 \cdot 6 = 36$$

ordered pairs.

We can write all of them down. Or we can imagine a tree. The first throw produces six branches. Each of those branches produces another six.

Suppose:

$$A = \text{the first throw is 6}$$

and:

$$B = \text{the second throw is 6.}$$

Only one of the 36 complete and equally weighted paths gives two sixes.

So direct counting gives:

$$P(A \wedge B) = \frac{1}{36}.$$

But there is another route.

The first condition leaves one sixth of the whole possibility space.

That remaining sixth is then split equally among the six possible results of the second throw.

So each path along that branch has weight:

$$\frac{1/6}{6} = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}.$$

Both routes lead to the same result.

We can inspect the complete possibility space and count one successful path. Or we can move through the tree and take one sixth of one sixth.

The multiplication compresses a structure that can still be reconstructed directly.

This resembles what happened in logic. We first inspected small cases directly, then compressed repeated relations into notation and rules. Once the relation had become convincing, we no longer needed to reconstruct the whole path every time.

Independence

We had:

$$\frac{1/6}{6} = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}.$$

Writing the division as multiplication is a convenient shorthand, and we will use that form from now on. If we picture the tree, under certain circumstances we can simply multiply along its branches. We will call this the *multiplication rule for dice*.

Those circumstances deserve a closer look.

Suppose the first throw was a six.

What possibilities remain for the second throw?

Again:

$$1, 2, 3, 4, 5, 6.$$

Suppose the first throw was a three.

For the second throw the same six possibilities remain. The first result has not altered the die. It still has the same faces and the same shape. If we continue treating it as fair, the six outcomes remain equally weighted.

The second stage of the tree therefore repeats the same structure whatever happened at the first. The weight of getting a six on the second throw remains:

$$\frac{1}{6}.$$

We call this relation *independence*. The result of the first throw does not change how we weight the possibilities of the second.

Under these circumstances, if A is an event concerning the first throw and B an event concerning the second, we can write:

$$P(A \wedge B) = P(A) \cdot P(B).$$

Now suppose we throw the die ten times and ask for ten sixes.

The complete possibility space contains:

$$6^{10}$$

paths. We could in principle construct it. But doing so would be a bit pointless. At every stage relevant to the event we care about, the same local relation repeats.

The next six has probability:

$$\frac{1}{6}.$$

So:

$$P(\text{ten sixes}) = \left(\frac{1}{6}\right)^{10}.$$

There is no need to inspect the entire possibility space. The formal compression has become useful precisely because the space has become too large to survey comfortably. But the local relation on which the compression depends remains reconstructible from the smaller cases.

3. Cards

Take a well-shuffled deck of 52 cards. Draw one card. For a single draw, the structure looks much like the die.

There are four aces among 52 cards.

If we treat every card as equally likely to occupy the first position in the shuffled deck, then:

$$P(\text{ace}) = \frac{4}{52} = \frac{1}{13}.$$

Now draw another card without returning the first.

Suppose the first card was an ace.

What is the probability that the second card is also an ace?

It cannot still be:

$$\frac{4}{52}.$$

There are only 51 cards left. Only three of them are aces.

So:

$$\frac{3}{51}.$$

The first draw has changed the situation in which the second draw occurs.

With repeated dice, the second stage presented the same six equally weighted possibilities whatever happened at the first. The deck does not.

Its later structure depends on the path already taken. That difference gives us a reason to preserve more information.

Let:

$$A = \text{the first card is an ace}$$

and:

$$B = \text{the second card is an ace.}$$

Before specifying what happened on the first draw, every card is equally likely to occupy the second position too, so:

$$P(B) = \frac{4}{52}.$$

Given that the first card was an ace:

$$P(B \mid A) = \frac{3}{51}.$$

This is a conditional probability: the probability of B under the condition that A has occurred. The vertical bar is read as *given*: B given A.

The concept lets us preserve information about the path already taken when that path changes how the remaining possibilities should be weighted.

We can now ask two different questions: "What is the probability that the second card is an ace?" and "What is that probability given that the first card was already an ace?"

The conditions under which we evaluate B are different in these questions. With the deck in front of us, that difference is easy to inspect.

A More General Multiplication Rule

We can now calculate the probability of drawing two aces.

First the first card must be an ace:

$$P(A) = \frac{4}{52}.$$

Then, along that branch, the second card must also be an ace:

$$P(B \mid A) = \frac{3}{51}.$$

So:

$$P(A \wedge B) = P(A)P(B \mid A).$$

In this case:

$$P(A \wedge B) = \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}.$$

The multiplication can be reconstructed from the tree of possible events.

If we collapse some of the branches, the relevant part of the tree looks like this:

First select the A branch.

Then select the parts of that branch in which B also occurs.

The rule we used for repeated dice now appears as a special case.

If:

$$P(B \mid A) = P(B),$$

then:

$$P(A \wedge B) = P(A)P(B \mid A) = P(A)P(B).$$

The cards show why the simpler dice rule worked: the probabilities of the outcomes of the second throw do not change with the result of the first throw.

Conditional probability asks how likely an event is when we restrict attention to cases in which another condition holds. The probability is then evaluated within that restricted set of possibilities.

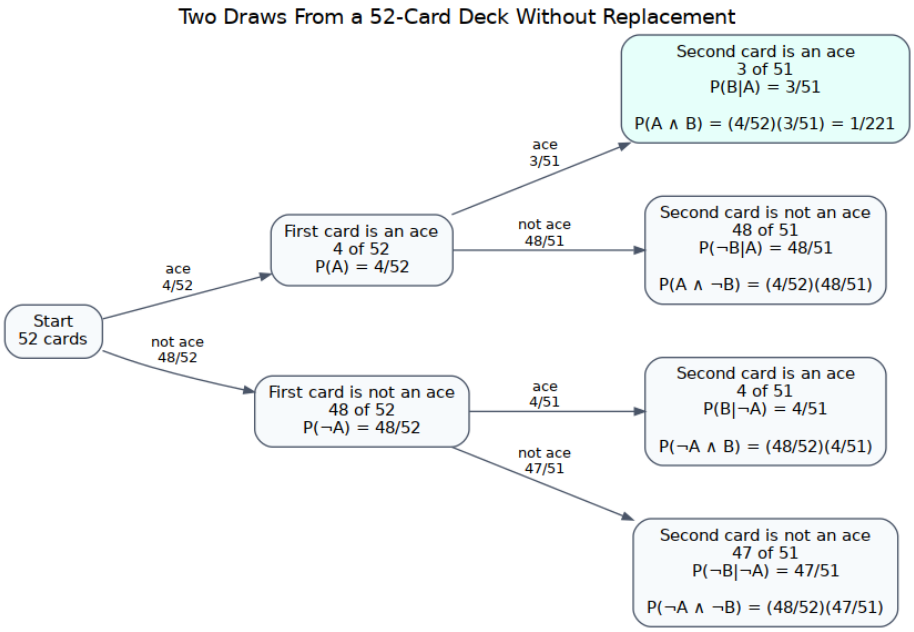


Figure 1: Tree

From Trees To Rules

The examples with dice and cards can now be seen as two forms of the same structure.

A probability tree begins with some initial possibility and then branches further as more conditions are added.

To follow one complete path, we move stage by stage.

At each step we take the fraction that remains once the earlier stages have already been fixed.

For two stages this gives:

$$P(A \wedge B) = P(A)P(B \mid A).$$

For three stages:

$$P(A \wedge B \wedge C) = P(A)P(B \mid A)P(C \mid A \wedge B).$$

And similarly for longer paths.

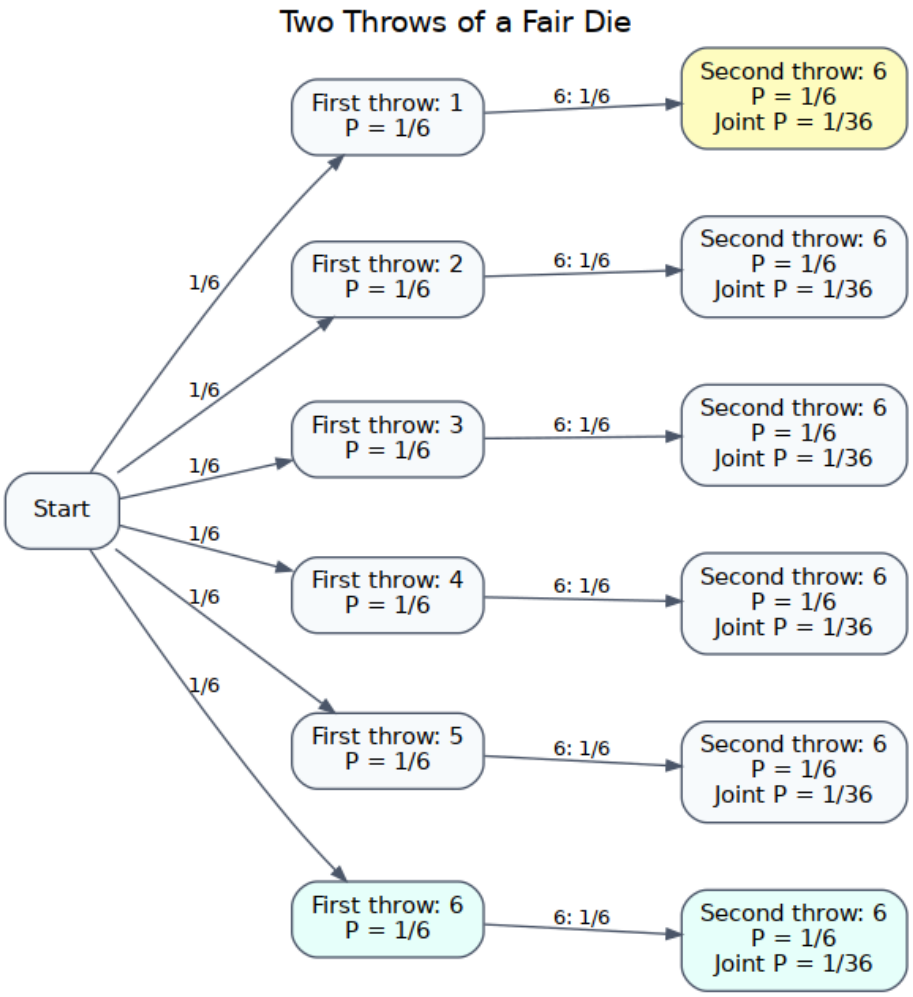


Figure 2: Tree

The card example shows the more general form. The relevant fraction at the next step may depend on what happened before.

The repeated-dice case shows a special simplification. If the next-stage probabilities do not depend on what happened before, the branches no longer require different conditional weights. If the same probability also repeats at every stage, as in our repeated-die example, the product can be compressed into a power.

Independence removes the dependence of later factors on earlier outcomes. The branches can then be treated with the same weight from one stage to the next.

4. Putting The Pieces Together

Elementary Definitions

We can now define possibility spaces as finite sets.

For example the possibility space for one throw of a die is:

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Events are subsets of that possibility space. For example:

$$A = \{2, 4, 6\}$$

is the event that the result is even, and:

$$B = \{4, 5, 6\}$$

is the event that the result is greater than 3.

$$C = \{1\}$$

is the event that the result is exactly 1.

That gives us a connection to the logic investigation that we will now look into.

There, *and*, *or*, and *not* operated on propositions. Here the same distinctions can operate on events.

For *not*, we take everything in the possibility space that does not belong to the event.

For the A we defined above:

$$\neg A = \{1, 3, 5\}.$$

Not even is in our case *odd* for the numbers in the possibility space.

More generally, for any event A:

$$\neg A = \Omega \setminus A,$$

that means Ω without the elements from A. In other words: everything outside A.

The fixed possibility space matters here. It tells us what counts as everything outside A.

The complement of A can therefore be different in a different possibility space, even when A itself is the same set.

For *and*, we keep only the outcomes that belong to both events connected by *and*.

With:

$$A = \{2, 4, 6\},$$

the event that the result is even, and:

$$B = \{4, 5, 6\},$$

the event that the result is greater than 3. The outcomes 4 and 6 occur in both sets. The outcome 2 belongs only to A, while 5 belongs only to B.

So:

$$A \wedge B = \{4, 6\}.$$

The event that the result is even and greater than 3 leaves us with 4 and 6 given this Ω.

Using set notation, the same relation can be written:

$$A \wedge B = A \cap B.$$

It makes no difference which event we name first:

$$A \wedge B = B \wedge A.$$

Both expressions select exactly the same elementary outcomes.

For *or*, we keep every elementary outcome belonging to at least one of the events.

Thus:

$$A \vee B = \{2, 4, 5, 6\}.$$

The event that the result is even or greater than 3 leaves us with 2, 4, 5 and 6 given this Ω.

Using set notation, this relation can be written:

$$A \vee B = A \cup B.$$

The outcomes 4 and 6 belong to both A and B, but they appear only once in the combined event. We are not joining two lists. We are asking which elementary outcomes satisfy at least one of the two conditions.

Not also gives us two simple relations that will matter shortly.

No elementary outcome can belong both to A and to its complement:

$$A \wedge \neg A = \varnothing.$$

Together they cover the whole possibility space:

$$A \vee \neg A = \Omega.$$

So far we have only organized possibilities.

We know which elementary outcomes belong to an event, which events overlap, and how *not*, *and*, and *or* change the sets under consideration.

We have constructed the structure of the events, but not yet their probabilities.

Adding Weight

Probability assigns numerical weight to these events, allowing us to distinguish between possibilities that remain open while constraining how much weight each of them can carry.

We will write that weight as:

$$P(A).$$

In the examples so far, probabilities could often be read as shares of a whole. Three equally weighted outcomes out of six gave us:

$$\frac{3}{6}.$$

That suggests a scale on which 0 means none of the total weight and 1 means all of it.

For the finite system we are constructing, we require three things.

First, probability weight is not negative:

$$P(A) \geq 0.$$

An event can carry none of the available weight, but not less than none.

Second, the whole represented possibility space carries total weight 1:

$$P(\Omega) = 1.$$

The number 1 is the weight of the whole. If a finite possibility space contains n elements, then:

$$\frac{n}{n} = 1.$$

That represents all of them. Assigning that weight to Ω commits the model to treating this space as exhaustive for the question it represents.

Third, consider two events that do not overlap.

Suppose A contains one part of the possibility space and B contains another, with no elementary outcome belonging to both:

$$A \wedge B = \varnothing.$$

If probability weight represents shares of one whole, then combining two non-overlapping parts combines their shares without counting anything twice.

So we require:

$$P(A \vee B) = P(A) + P(B).$$

The probability of events that do not overlap is additive.

Some Rules Follow

From these three requirements, several familiar probability relations follow.

Consider first the empty event.

We have:

$$\Omega \vee \varnothing = \Omega.$$

The two sets do not overlap, so additivity gives:

$$P(\Omega) = P(\Omega) + P(\varnothing).$$

Therefore:

$$P(\varnothing) = 0.$$

The rule for *not* follows in the same way.

We already know:

$$A \wedge \neg A = \varnothing$$

and:

$$A \vee \neg A = \Omega.$$

Additivity therefore gives:

$$P(A) + P(\neg A) = P(\Omega).$$

Since:

$$P(\Omega) = 1,$$

we obtain:

$$P(\neg A) = 1 - P(A).$$

So once the whole space has weight 1, the probability of *not* A is whatever weight remains outside A.

Or requires one additional step when the events overlap.

For our die let again:

$$A = \{2, 4, 6\}$$

and:

$$B = \{4, 5, 6\}.$$

Their overlap is:

$$A \wedge B = \{4, 6\}.$$

If we add:

$$P(A) + P(B),$$

the weight belonging to 4 and 6 is included twice.

So one copy of the overlap has to be removed:

$$P(A \vee B) = P(A) + P(B) - P(A \wedge B).$$

For a fair die:

$$P(A) = \frac{3}{6},$$

$$P(B) = \frac{3}{6},$$

and:

$$P(A \wedge B) = \frac{2}{6}.$$

Therefore:

$$P(A \vee B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6}.$$

That agrees with the four elementary outcomes we can inspect directly:

$$A \vee B = \{2, 4, 5, 6\}.$$

The formula compresses the same structure.

And Needs More Information

Suppose we know:

$$P(A)$$

and:

$$P(B).$$

Does that determine:

$$P(A \wedge B)?$$

Take a possibility space with four equally weighted elementary outcomes:

$$\Omega = \{1, 2, 3, 4\}.$$

Let:

$$A = \{1, 2\}.$$

Then:

$$P(A) = \frac{1}{2}.$$

Now consider three different events:

$$B_1 = \{1, 2\},$$

$$B_2 = \{1, 3\},$$

and:

$$B_3 = \{3, 4\}.$$

Each has probability:

$$\frac{1}{2}.$$

But their overlaps with A differ.

For B_1:

$$A \wedge B_1 = \{1, 2\},$$

so:

$$P(A \wedge B_1) = \frac{1}{2}.$$

For B_2:

$$A \wedge B_2 = \{1\},$$

so:

$$P(A \wedge B_2) = \frac{1}{4}.$$

For B_3:

$$A \wedge B_3 = \varnothing,$$

so:

$$P(A \wedge B_3) = 0.$$

In every case:

$$P(A) = P(B_i) = \frac{1}{2}.$$

Knowing the separate weights of A and B does not tell us how much weight belongs to both.

The tree from the card example shows what is missing.

To find $P(A \wedge B)$, we first followed the branch on which A occurred. Then, within that branch, we asked what share also belonged to B.

That restricted share is what we wrote as:

$$P(B \mid A).$$

And needs conditional probabilities.

Conditional Probability

The card tree gave us the missing relation.

To reach the event $A \wedge B$, we first follow the branch on which A occurs.

That branch carries weight:

$$P(A).$$

Once we are on that branch, we ask how likely B is under the condition that A has already occurred.

We write that as:

$$P(B \mid A).$$

In the card example, this was easy to determine directly. Once the first card was known to be an ace, 51 cards remained and 3 of them were aces, so:

$$P(B \mid A) = \frac{3}{51}.$$

In other problems, determining a conditional probability may be much harder. But once we have it, the tree tells us how the two stages fit together.

We take the weight of the A branch and then the share of that branch in which B occurs:

$$P(A \wedge B) = P(A)P(B \mid A).$$

This is the more general multiplication rule we used in the previous chapter.

In the last chapter we used this visualization:

Independence

Repeated fair-die throws gave us a simpler case for *and*.

There, learning the result of the first throw did not change the probabilities assigned to the possible results of the second.

If A and B have that relation, then:

$$P(B \mid A) = P(B).$$

We call them independent.

Substituting this into:

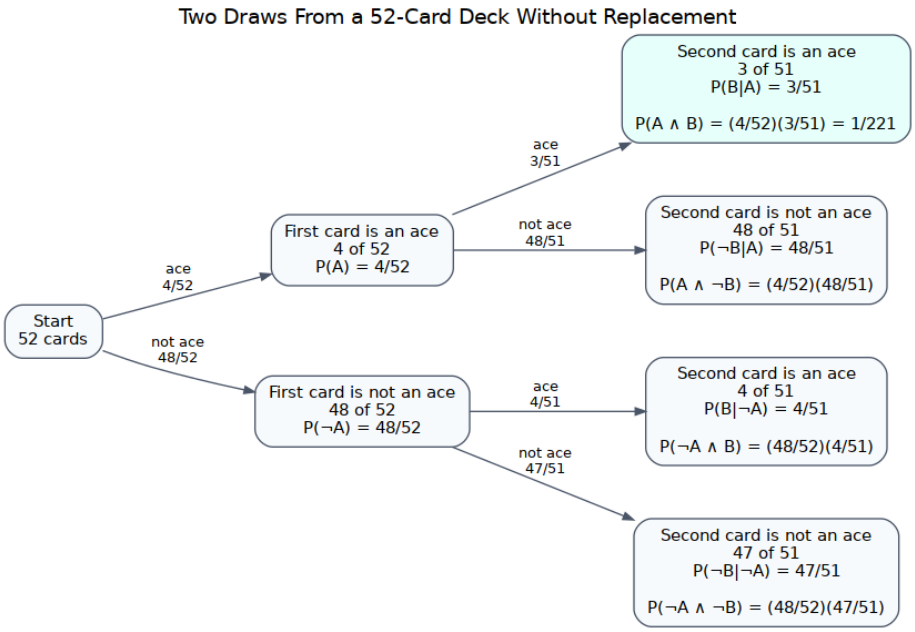


Figure 3: Tree

$$P(A \wedge B) = P(A)P(B \mid A)$$

gives:

$$P(A \wedge B) = P(A)P(B).$$

The multiplication rule we used for independent die throws is therefore a special case of the conditional relation.

Equal Weighting

For the fair die, we treated the six elementary outcomes as carrying equal weight.

Let the weight of each one be:

$$w.$$

Since the six outcomes together make up the whole possibility space:

$$6w = 1.$$

Therefore:

$$w = \frac{1}{6}.$$

For the event:

$$A = \{2, 4, 6\},$$

additivity gives:

$$P(A) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6}.$$

More generally, when the elementary outcomes in a finite possibility space are equally weighted:

$$P(A) = \frac{|A|}{|\Omega|}.$$

Counting produces probabilities only after equal weighting has been justified.

For the fair die, its symmetry gave us a reason to assign equal weights to the six elementary outcomes.

The loaded die showed why that step matters. Six possible faces do not by themselves imply six equal probabilities.

What This Construction Shows

We can now return to the two questions about convincing force which guide the second part of this essay cycle.

The first question concerns the constraint inside the probability system.

Once Ω is fixed, the events are represented as sets of elementary outcomes. We can inspect which outcomes belong to A, which belong to B, where they overlap, and what remains outside an event.

Probability then adds weight.

We treated 1 as the weight of the whole possibility space, required weights to be non-negative, and required the weights of non-overlapping parts to add.

Once those conditions are fixed, some further relations are no longer separate choices.

If A and $\neg A$ divide the whole space between them, their weights must add to 1.

If A and B overlap, adding their separate weights counts the overlap twice, so one copy has to be removed.

If we follow an A branch and then ask for the weight of B under that condition, the weight of $A \wedge B$ is fixed by the two stages:

$$P(A \wedge B) = P(A)P(B \mid A).$$

For small enough problems, the route to those relations remains open to inspection.

We can write down the possible outcomes, group them into events, inspect overlaps, follow branches in a tree, count, and compare the formulas with the structure from which they were derived.

That is one source of the convincing force of the constraints produced by probability theory. The formulas compress relations that can still be reconstructed.

The second question concerns applying the formal system of probability to a problem.

As in logic, applying the formal system depends on prerequisites that are not usually fully under our control.

The possibility space has to contain the possibilities that matter for the question. The elementary outcomes have to represent the distinctions that matter for the question. Equal weighting, when used, needs support. A claim of independence has to fit the situation. Conditional probabilities have to be estimated or otherwise justified.

The probability calculus does not settle those questions for us.

They can be examined separately from the formal relations that follow once the inputs are fixed.

So the formal constraint may be highly convincing while its application to a particular problem remains doubtful.

5. Bayesian Updating

Winning The Lottery

You entered a lottery a few weeks ago.

One morning you receive an email saying that you have won one million euros. Click here.

At first you are suspicious. Messages like this are often scams.

But the email made it to the inbox in the first place. And the email carries the right lottery logo. The wording looks professional. That makes a genuine message seem somewhat more plausible.

Then you look at the sender.

The address is not from the lottery's usual domain. That counts against the message being genuine. But the address is not completely implausible either. Maybe they send such messages from another domain. It is still strange.

The balance shifts again. Each clue fits the two possibilities differently, but neither of them settles which possibility is true.

The setting is different from the earlier examples of cards and dice. There we worked with finite arrangements of outcomes.

Here we are comparing two hypotheses in the light of evidence: the message is genuine, or it is a scam. The weights we assign in such a setting often come from quite different sources.

The initial weights of the hypotheses may depend on background information, previous experience, base rates, or judgment.

The likelihoods require another kind of assessment: how likely would this evidence be if the message were genuine, and how likely would it be if it were a scam?

Those quantities are usually not read directly from a small finite arrangement in the way they were for dice, cards, or socks.

Many questions involve such a setup: several hypotheses remain open, their weights are not given simply by counting a finite arrangement, and new evidence bears differently on them. What is causing the strange noise in the car? Why has a plant suddenly stopped growing? Which explanation best fits an unexpected result?

In such cases, how should the weights assigned to the hypotheses change when new evidence is taken into account?

That is the problem Bayesian updating addresses.

Turning The Conditional Around

We already have:

$$P(A \wedge B) = P(A)P(B \mid A).$$

From the previous chapter we also know:

$$A \wedge B = B \wedge A.$$

So the same joint event can be reached from the other direction:

$$P(B \wedge A) = P(B)P(A \mid B).$$

Since both expressions on the right side describe the same event, they must be equal to each other:

$$P(A)P(B \mid A) = P(B)P(A \mid B).$$

If:

$$P(B) > 0,$$

we can divide by $P(B)$:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}.$$

That is Bayes' theorem.

Mathematically, it is an ordinary probability theorem.

Why, then, has it become so prominent?

The reason is easier to see when the conditional we know is not the conditional we want. But this is just the first step.

Bayes' Theorem

Suppose:

$$H = \text{a person has disease D}$$

and:

$$E = \text{a diagnostic test is positive.}$$

Imagine that in a certain population:

$$P(H) = 0.01.$$

One person in one hundred has the disease.

Suppose the test returns a positive result for 90 percent of people who have it:

$$P(E \mid H) = 0.90.$$

And suppose it also returns a positive result for 5 percent of people who do not:

$$P(E \mid \neg H) = 0.05.$$

A person tests positive.

How likely is it that the person has the disease?

The 90 percent figure is tempting.

But it answers: Given the disease is present, how likely is a positive test?

We want: Given a positive test, how likely is the disease?

Those are:

$$P(E \mid H)$$

and:

$$P(H \mid E).$$

They are not the same conditional.

We can answer the second question without Bayes' theorem.

Imagine 10,000 people.

About 100 have the disease.

Of those, about 90 test positive.

The other 9,900 do not have the disease.

Five percent of them also test positive:

$$495.$$

So there are about:

$$90 + 495 = 585$$

positive tests.

Of those, about 90 come from people with the disease:

$$P(H \mid E) \approx \frac{90}{585} \approx 0.154.$$

After conditioning on the positive result, the probability in the model has moved from one percent to about fifteen percent. But it has not moved to ninety percent.

Bayes' theorem lets us calculate that result in a general way.

A positive test can occur in two mutually exclusive ways: the person has the disease and tests positive, or the person does not have the disease and tests positive.

So:

$$P(E) = P(E \wedge H) + P(E \wedge \neg H).$$

Using the multiplication rule from the previous chapter:

$$P(E \wedge H) = P(H)P(E \mid H)$$

and:

$$P(E \wedge \neg H) = P(\neg H)P(E \mid \neg H).$$

Combining the two branches gives:

$$P(E) = P(E \mid H)P(H) + P(E \mid \breve{\text{n}}H)P(\breve{\text{n}}H).$$

So:

$$P(E) = 0.90 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0585.$$

We now have all the pieces we need:

The prior probability of the hypothesis:

$$P(H) = 0.01.$$

The probability of the evidence if the hypothesis is true:

$$P(E \mid H) = 0.90.$$

The total probability of the evidence:

$$P(E) = 0.0585$$

What we want is:

$$P(H \mid E),$$

the probability of the hypothesis given the evidence.

We take Bayes' theorem

$$P(H \mid E) = \frac{P(E \mid H)P(H)}{P(E)}.$$

and plug the values in:

$$P(H \mid E) = \frac{0.90 \cdot 0.01}{0.0585} \approx 0.154.$$

Both routes lead to the same result.

We can construct an imagined finite population and count. Or we can use the algebraic relation.

The formula reaches the same result while doing much less work.

Up to this point, we are still close to the world of dice and cards. The numbers supplied to the calculation may come from earlier observations or measurements, but the inversion itself can still be reconstructed through an explicit finite arrangement.

The situation changes when the possibilities themselves are hypotheses. That is the second step.

Between Hypotheses And Evidence

We saw that Bayes' theorem is a mathematical theorem about a relation between conditional probabilities.

In Bayesian inference, it is usually given a particular interpretation: one conditional represents how likely some evidence would be under a hypothesis, and the other represents how probable the hypothesis is given that evidence.

We begin with a set of hypotheses representing possible explanations of what is happening. Some carry more weight than others before the present evidence is taken into account. We assign probabilities to those hypotheses to represent the weight we give them.

Something we have observed that bears on our investigation becomes evidence, E.

The likelihood connects the two sides. For each hypothesis H_i, we ask how likely this particular evidence would be if that hypothesis were true.

That gives us $P(E \mid H_i)$. In many applications, this is the conditional we can estimate or model more directly.

Bayes' theorem then carries the relation in the other direction. Once E has been observed, it constrains how the weights assigned to the hypotheses change, giving us $P(H_i \mid E)$.

This gives the mathematics a particular role in inquiry. We can represent explanatory hypotheses we have in mind, represent evidence we encounter, and connect them through claims about what each hypothesis would lead us to expect.

Those connecting claims are doing substantial work. If two hypotheses make E equally likely, that evidence does not distinguish between them. If one makes E much more likely than another, the observation shifts their relative weights, as in this schema:

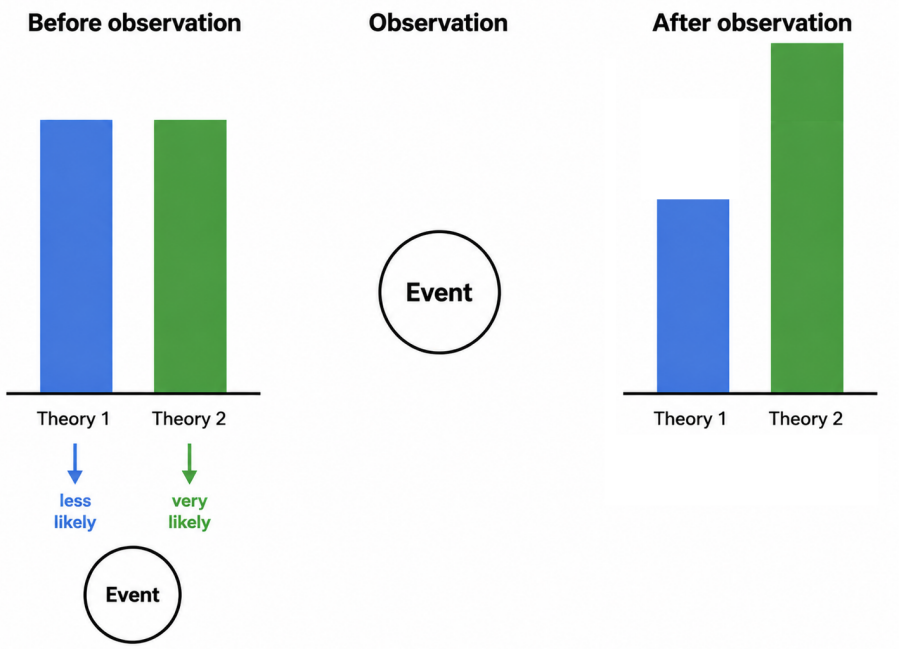


Figure 4: Bayes' theorem applied

That is what gives Bayes' theorem its particular importance in Bayesian inference.

An ordinary relation between conditional probabilities becomes a model for a common situation in inquiry: several hypotheses remain open, evidence bears differently on them, and their relative weights change as that evidence is taken into account.

When The Possibilities Are Hypotheses

A computer repeatedly restarts under heavy load.

Several explanations occur to us.

The power supply may be failing. The machine may be overheating. Memory could be defective.

Call these:

$$H_1 = \text{power supply failure},$$

$$H_2 = \text{overheating},$$

and:

$$H_3 = \text{memory failure}.$$

We observe:

$$E = \text{the machine restarts under heavy load}.$$

We can ask:

$$P(H_i \mid E).$$

That form is familiar. But unlike the cards in the earlier examples, the hypotheses are not given to us by an arrangement we can simply inspect. We cannot spread every possible explanation of the machine failure across a table and check that all of them are present. We have to come up with the possibilities somehow.

A technician may recognize familiar failure patterns. The age of the power supply may make one possibility salient, or overheating may have occurred before. Memory problems may also be common in similar machines.

Why exactly these three?

The operating system may contain a bug, the motherboard may be failing, the external power supply may be unstable, or two faults may occur together.

The probability model can distribute weight only among possibilities represented in it.

If an important hypothesis is missing, no flawless update can assign probability to it until the model changes.

This is the old problem of possibility generation in another form. With dice and cards, the completeness of the possibility space was easy to inspect. Here, the conviction that the relevant possibilities have been included has to come from somewhere else.

Priors

For this small model, suppose we treat these three hypotheses as mutually exclusive and exhaustive: exactly one of them is taken to explain the failure.

We assign:

$$P(H_1) = 0.50,$$

$$P(H_2) = 0.30,$$

$$P(H_3) = 0.20.$$

These are priors: the probabilities assigned to the hypotheses before (*prior* to) the observation E is used to update them.

Where did they come from?

Comparable failures may historically have been power-supply problems about half the time. A technician might instead translate her own judgment of plausibility into numbers, or the values may come from another statistical model. These are examples of where the convincing force of a prior can come from. Its source can then itself be examined.

Likelihoods

We also need to know how expected the observation would be under each hypothesis.

Suppose:

$$P(E \mid H_1) = 0.80,$$

$$P(E \mid H_2) = 0.60,$$

$$P(E \mid H_3) = 0.20.$$

That says roughly: the evidence we are looking at is very likely under the first hypothesis, likely under the second, and not so likely under the third.

The likelihood, too, has to be convincing enough to enter the model. Whatever supports that estimate lies outside the update itself.

Likelihoods do not have to add up to 1. Each is the probability of the evidence under a different condition.

Evidence

Even E had to be constructed.

Something happened in the machine.

We represented it as "The machine restarts under heavy load."

We might instead have recorded "The machine restarts within thirty seconds above 80 percent CPU load", or "The machine loses power without a kernel panic."

Those descriptions preserve different distinctions. The world does not arrive carrying an E label.

Something becomes salient, is observed, classified, measured or described, and then enters the formal process in a particular representation.

That representation can change the likelihoods assigned under the hypotheses.

Once E has been represented, we still need its probability.

For this model, the three hypotheses are mutually exclusive and exhaustive. Exactly one of them is taken to be true, and together their probabilities add to 1.

So if the evidence E occurs, it must occur together with one of the three hypotheses:

$$E = (E \wedge H_1) \cup (E \wedge H_2) \cup (E \wedge H_3).$$

These three cases do not overlap.

From the addition rule:

$$P(E) = P(E \wedge H_1) + P(E \wedge H_2) + P(E \wedge H_3).$$

And from the multiplication rule:

$$P(E \wedge H_i) = P(H_i)P(E \mid H_i).$$

So:

$$P(E) = P(H_1)P(E \mid H_1) + P(H_2)P(E \mid H_2) + P(H_3)P(E \mid H_3).$$

With our values:

$$P(E) = 0.50 \cdot 0.80 + 0.30 \cdot 0.60 + 0.20 \cdot 0.20 = 0.62.$$

More generally, suppose the model contains mutually exclusive and exhaustive hypotheses

$$H_1, \ldots, H_n.$$

Under each hypothesis, part of the probability of the evidence comes from:

$$P(E_i) = P(E \wedge H_i) = P(H_i)P(E \mid H_i).$$

The hypotheses do not overlap, so these contributions can be added. And because they exhaust the represented possibilities, there is no other part of E left outside them. Thus:

$$P(E) = P(E_1) + P(E_2) + \cdots + P(E_i) =$$

$$P(H_1)P(E \mid H_1) + P(H_2)P(E \mid H_2) + \cdots + P(H_n)P(E \mid H_n).$$

Or more compactly:

$$P(E) = \sum_{i=1}^{n} P(H_i)P(E \mid H_i).$$

Each hypothesis contributes according to how much probability it has and how likely the evidence would be under it. Adding those non-overlapping contributions gives the full probability of the evidence.

The Update

Now suppose the model is in place.

We have:

$$P(H_1) = 0.50, \quad P(H_2) = 0.30, \quad P(H_3) = 0.20$$

and:

$$P(E \mid H_1) = 0.80, \quad P(E \mid H_2) = 0.60, \quad P(E \mid H_3) = 0.20.$$

The probability of the observation under the model is:

$$P(E) = 0.80 \cdot 0.50 + 0.60 \cdot 0.30 + 0.20 \cdot 0.20.$$

So:

$$P(E) = 0.62.$$

The posterior probabilities are:

$$P(H_1 \mid E) = \frac{0.80 \cdot 0.50}{0.62} \approx 0.645,$$

$$P(H_2 \mid E) = \frac{0.60 \cdot 0.30}{0.62} \approx 0.290,$$

and:

$$P(H_3 \mid E) = \frac{0.20 \cdot 0.20}{0.62} \approx 0.065.$$

What The Bayesian Investigation Shows

What makes the constraint inside Bayesian updating convincing?

Bayes' theorem was reconstructed from the multiplication rule established in the previous chapter:

$$P(A \wedge B) = P(A)P(B \mid A),$$

from the fact that:

$$A \wedge B = B \wedge A,$$

and algebra.

That gives us a route back through the formula.

We can inspect the multiplication rule, inspect why $A \wedge B$ and $B \wedge A$ describe the same event, and follow the algebraic rearrangement step by step.

The diagnostic example gave us another way to inspect the same relation.

Bayes' theorem gave us the same result from the prior and the two conditional probabilities.

In cases like this, the probabilities can be represented as proportions of finitely many equally weighted cases, so the calculation can be reconstructed by counting those cases directly.

That is not always possible exactly. Even a finite probability space can contain weights that cannot be expressed as a ratio of two finite counts.

But even then, the route back to the structure remains open. We can still lay out the elementary possibilities, assign the weights supplied by the model, and calculate the relevant conditional probabilities directly from those weighted possibilities.

It is again inspectability that lends some convincing force. In simple cases we can reconstruct it by counting. In more general finite cases we can reconstruct it from explicitly weighted possibilities and from relations whose operation has already become familiar in the simpler examples.

The hypothesis example showed the same constraint in a richer model. Once the model is fixed, the posterior values are not separately open. If one of them is to change, something else in the model has to change with it.

This differs somewhat from the logical cases. There, much of the visible constraint worked by exclusion. A possibility survived or did not. A counterexample existed or did not.

Here several hypotheses can remain open while their relative weights become constrained.

The second question concerns applying Bayesian updating to an actual problem.

As in logic, applying the formal system depends on prerequisites that are not usually fully under our control.

The hypothesis space has to contain possibilities that matter for the question. Bayes' theorem cannot assign weight to a hypothesis that never entered the model.

This returns us to the problem of possibility generation from Part I. With cards, completeness was easy to inspect. With explanations for a computer failure, it is not.

The priors also need support. Whatever their source, their convincing force has to arise outside the Bayesian update itself.

The same is true of likelihoods. A value such as $P(E \mid H_1) = 0.80$ has to be convincing enough to enter the model. The update tells us what follows from that value once it is supplied.

The evidence has prerequisites of its own. The world does not arrive carrying an E label. Something becomes salient, is observed, measured, classified, or described, and then enters the model in a particular representation.

Those differences can matter to the likelihoods and to the update.

There is another prerequisite as well.

The Bayesian interpretation itself has to be convincing, both as a way of relating hypotheses to evidence and as a suitable representation of this particular problem.

We have used Bayes' theorem by treating possible explanations as hypotheses that can carry probability, treating an observation as evidence, and using likelihoods to represent what each hypothesis would lead us to expect.

We then interpret conditioning on that evidence as a change in the weights of the hypotheses.

This asks more than the small finite arrangements we reconstructed earlier, because more of the formalism now has to stand for something outside the calculation.

So the convincing force of Bayesian updating depends on more than the correctness of the calculation and the support for its numerical inputs. The bridge we have built between hypotheses and evidence has to be convincing as well.

Appendix A sketches two central parts of a real Bayesian investigation: maintaining a hypothesis space, and updating the weights within it as evidence arrives.

As long as the hypothesis space remains fixed, the updating is mathematical. But as we have seen, during inquiry other possibility can come up, and the space itself then needs to change.

The appendix shows how these two moves can alternate and some of the places where their inputs can go wrong.

The Posterior Is Not The Conviction

Suppose the calculation gives:

$$P(H_1 \mid E) \approx 0.645.$$

That is a result inside the model.

A technician may become more convinced that the power supply is failing.

She may also question the prior, a likelihood, the hypothesis space, the way the evidence was represented, or whether this is an appropriate model of the problem.

The posterior and the conviction are not the same thing.

The calculation can participate strongly in conviction formation without determining what the person will become convinced of.

It can also make disagreement more precise.

If the arithmetic is accepted but the posterior is not, attention can move to what the result depends on.

A prior may be challenged. A likelihood may need better support. Another hypothesis may have been omitted. The evidence may have been represented differently.

The formal structure can expose where something would have to change if the result is to change.

6. What Probability Adds

Logic constrained inquiry mainly by ruling possibilities out.

Probability allows several possibilities to remain open while assigning different weights to them. Those weights are not independent once enough of the probability model had been fixed.

Bayesian updating gives us a way to methodically change these weights on hypotheses when evidence becomes available.

Where The Constraint Comes From

In the small finite cases, much of the probability system remained open to inspection. We could lay out elementary outcomes, group them into events, inspect overlaps, follow branches in a tree, assign weights, and reconstruct the multiplication rule from those branches.

Bayes' theorem could then be reconstructed from that rule, from $A \wedge B = B \wedge A$, and from algebra. In particularly simple cases, the same relation could also be checked by counting. The diagnostic example allowed exactly that.

More general finite probability spaces need not allow such an exact counting representation. But their elementary possibilities and assigned weights can still be laid out, and the relevant conditional probabilities calculated from that weighted structure.

The formulas compress these relations without completely losing the route back to the structures from which they were reconstructed.

We saw something similar in logic. Small cases made enough of the structure visible for the constraint to be inspected directly. Once the relation became familiar, a formal rule could carry it into cases where reconstructing the whole arrangement each time would be impractical.

Once the event structure, weights, conditional relations, and other relevant conditions remain fixed, some probability values are no longer independently open.

Being able to inspect and reconstruct how that constraint arises seems to explain a large part of why it can become convincing.

What Probability Adds

Logic gave us ways to ask which possibilities remained once enough of a representation had been fixed. A possibility could be excluded, or a counterexample could remain.

Probability adds another perspective. If more than one possibility remains open, how much weight does each of them carry?

Conditional probability added a dependence. The weight of a possibility could change once we restricted attention to a particular branch of the tree of events.

Bayesian updating then gave us a model for changing the weights of hypotheses when evidence arrives.

Probability adds a form of constraint that does not require possibilities to disappear. Several possibilities can remain available while the weights assigned to them are no longer independently movable.

Applying Probability

The strength of that constraint does not settle whether a particular probabilistic model should be used for a particular problem.

As in logic, applying the formal system depends on prerequisites that are not usually fully under our control.

Applying probability to a problem requires more than the formal relations themselves.

The possibility or hypothesis space has to preserve distinctions that matter for the question, and the weights assigned within it need some support.

In Bayesian applications, the likelihoods have to make sense as claims about how the evidence relates to the hypotheses. The evidence itself also has to enter the

model in some representation, through observation, measurement, classification, or description.

Any dependence built into the model matters as well.

The convincing force of those prerequisites has to come from elsewhere in the inquiry.

We can now separate two kinds of uncertainty. A model can represent uncertainty among the possibilities it contains, while leaving open whether the model itself contains the right possibilities and relations.

A probability distribution can represent uncertainty among possibilities already inside a model.

Suppose it distributes its weight among:

$$H_1, H_2, H_3.$$

It says nothing by itself about an omitted:

$$H_4.$$

Some uncertainty about the model can itself be represented in a larger model. The adequacy of that larger representation remains another question.

How The Investigations Clarify One Another

The investigations so far have given us different kinds of formal constraint.

Logic asks what follows once enough of the premises and relations have been fixed, and which possibilities can still remain.

Probability becomes relevant when several possibilities remain open and we want to distinguish between them by weight.

Bayesian updating asks how those weights should change when new evidence bears differently on the possibilities.

If evidence rules a possibility out, logical constraint may be enough. If several possibilities remain but carry different plausibility, probability gives us another way to represent the situation. If new evidence should change those weights, Bayesian updating gives us a way to model that change.

Part of the convincing force of an application can come from the fit between the question we are asking and the kind of constraint the method provides. Seeing several such methods beside one another also makes their boundaries easier to recognize.

The investigations do more than clarify those differences. They also build on one another.

Probability used structures already familiar from logic, such as *not*, *and*, and *or*, while adding numerical weight. Bayesian updating then used weighted possibilities and conditional probability to represent changes in those weights when evidence arrived.

That makes the later constructions easier to understand because part of their structure has already been examined. At the same time, the later investigations make clearer where the earlier ones stop being enough for the question at hand.

What Becomes Available For Examination

Making the probabilistic structure explicit also makes more of what a result depends on available for examination.

Suppose a posterior remains doubtful even though the arithmetic is accepted. The question can then move. Maybe a prior needs better support, a relevant hypothesis is missing, or the evidence has been represented badly. The likelihood connecting a hypothesis to the evidence may itself be doubtful.

The formal structure helps locate where something would have to change if the result is to change.

We saw something similar in logic. A counterexample could move disagreement from a conclusion to a premise, a formalization, or the choice of logical system. Probability can do this while several possibilities remain open. It can expose which weights, conditional relations, and representations carry the result.

If more of those dependencies are exposed and examined, and the result still remains convincing, the conviction may stand on firmer ground than before.

But the probability investigation also takes us back another step.

Before we assigned any numbers in the lottery example, some of the relevant distinctions were already in use. One explanation seemed more plausible than another. A new clue could count more strongly for one possibility than for the other. The balance could shift without either possibility disappearing.

Probability theory gives those pretheoretic, practical distinctions a more exact form. Possibilities receive numerical weights, relations between those weights become constrained, and Bayesian updating specifies how evidence changes them once enough of the model has been fixed.

That precision makes further examination possible. We can ask whether the weights fit the relations of the model, whether the likelihoods adequately represent how the evidence bears on the hypotheses, and whether the resulting update follows from what was supplied.

The application of the model still has to be sufficiently convincing.

Like logic, neither probability nor Bayesian updating gives us a criterion for judging convictions from outside conviction formation. But once Bayesian updating is convincing enough to enter an inquiry, it can constrain a question within that inquiry. We can ask whether the posterior weights follow from the priors, likelihoods, and evidence supplied under the probabilistic relations we have accepted.

Appendix A: A Bayesian Investigation Loop

In this appendix we give a heuristic algorithm that integrates Bayesian updating into a small model of an investigation.

Here is how it works:

Suppose we begin with a hypothesis space:

$$H_1, \ldots, H_n.$$

Each hypothesis has a prior weight.

When new evidence E becomes available, we ask:

$$P(E \mid H_i)$$

for each hypothesis.

Bayes' theorem then constrains how the weights change:

$$P(H_i \mid E) = \frac{P(E \mid H_i)P(H_i)}{P(E)}.$$

The posterior can then become the prior for the next update.

A simplified investigation might look like this:

1. Begin with a represented set of possibilities.
2. Assign or accept initial weights.
3. Observe or represent new evidence.
4. Ask how likely that evidence would be under each represented possibility.
5. Update the weights.
6. Continue with further evidence.

As long as the hypothesis space remains fixed, Bayesian updating constrains the movement of the weights.

But the investigation can also change in another important way.

A new possibility may become available.

Suppose we began with:

$$H_1, H_2, H_3$$

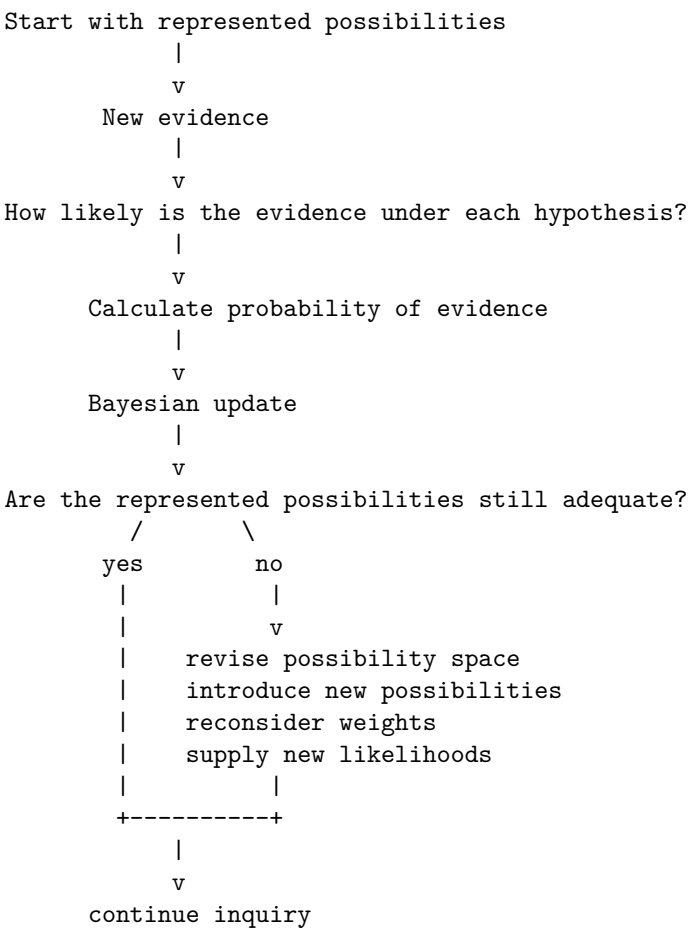
and later realize that:

$$H_4$$

also needs to be considered.

The new possibility has to be represented, some initial weight has to be given to it, the weights of the existing hypotheses have to be reconsidered relative to the enlarged space, and relevant likelihoods have to be supplied.

So a more realistic heuristic is:



The two routes in this heuristic are different.

As long as the represented possibilities remain in place, Bayesian updating constrains what happens to their weights.

The other route is less constrained.

A new possibility may arise from observation, experience, analogy, another person's suggestion, a failed prediction, another model, or something else.

Once it enters the investigation, the probability model has to be rebuilt around the enlarged possibility space.

Appendix B: Examples Of Frequentist Handling Of Evidence

Probability And Statistical Inference

Earlier in the essay, probability was introduced through cases where the setup was already given and we calculated what followed from it.

A fair die has a represented possibility space and assigned weights. From those, we can calculate how likely different outcomes are.

Bayesian updating already moved us into statistical inference.

There the direction was partly reversed.

We had hypotheses about what might be producing the observations, some initial weights assigned to those hypotheses, and new evidence. The evidence then constrained how those weights changed.

So statistical inference is broadly when we use observed data to learn something about the process, population, or hypothesis that may have produced it.

In the Bayesian case, the probabilities belonged to the hypotheses and were updated with the evidence. Frequentist methods place the probability differently. In frequentist methods, the probabilities usually belong to possible data or statistics under a specified model, and the observed data is compared with that distribution.

Imagine an urn whose composition we do not know. We draw some balls and record what we see.

A Bayesian treatment might represent several possible compositions of the urn, assign probabilities to them, and update those probabilities in light of the sample.

A frequentist treatment does not need to assign probabilities to the possible compositions themselves. Instead, it asks how samples would behave if a particular composition or sampling model were in place, and then compares the observed sample with those possible results.

That difference will become clearer in the examples.

We will look at two cases: randomization in an experiment, and sampling from a population.

Randomization

One important route into frequentist statistics comes from the work of Ronald Aylmer Fisher.

Fisher developed methods in which the randomization of an experiment gives probability a precise role in assessing the observed result.

An anecdotal example used often by the statistician Phillip I. Good makes the mechanism visible[1].

Good discusses an experiment involving cell cultures grown either with a conventional nutrient solution or with vitamin E added.

After some losses, six usable observations remained, three in each condition:

vitamin E	control
121	34
118	22
110	12

The sum of the vitamin E measurements is:

$$121 + 118 + 110 = 349.$$

The sum of the control measurements is:

$$34 + 22 + 12 = 68.$$

Their difference is:

$$349 - 68 = 281.$$

But is this difference large or small? Compared to what other differences?

The experiment gives us a way to answer that.

The treatment labels were assigned randomly. Now suppose that vitamin E had *no effect* on any of these measured outcomes.

Under that hypothesis, it would not matter which three specimens had received the vitamin E label. The six measurements would remain:

121, 118, 110, 34, 22, 12.

We can thus keep the measurements fixed and redistribute three vitamin E labels and three control labels over them in every way allowed by the experiment.

There are:

$$\binom{6}{3} = 20$$

such assignments.

Under this randomization design, each has probability:

$$\frac{1}{20} = 0.05.$$

These 20 assignments give us a reference distribution against which we can compare the observed difference.

For each possible assignment we calculate the same statistic: the sum of the vitamin E measurements minus the sum of the control measurements.

vitamin E	control	difference
121, 118, 110	**34, 22, 12**	**281**
121, 118, 34	110, 22, 12	129
121, 110, 34	118, 22, 12	113
118, 110, 34	121, 22, 12	107
121, 118, 22	110, 34, 12	105
121, 110, 22	118, 34, 12	89
121, 118, 12	110, 34, 22	85
118, 110, 22	121, 34, 12	83
121, 110, 12	118, 34, 22	69
118, 110, 12	121, 34, 22	63
121, 34, 22	118, 110, 12	-63
118, 34, 22	121, 110, 12	-69
121, 34, 12	118, 110, 22	-83
110, 34, 22	121, 118, 12	-85
118, 34, 12	121, 110, 22	-89
110, 34, 12	121, 118, 22	-105
121, 22, 12	118, 110, 34	-107
118, 22, 12	121, 110, 34	-113
110, 22, 12	121, 118, 34	-129
34, 22, 12	121, 118, 110	-281

The entire probabilistic structure is visible.

The observed difference 281 is the largest positive difference among the 20 possible assignments.

Only one assignment produces a result at least this favorable to vitamin E.

So, under the no-effect hypothesis and this randomization procedure $p = \frac{1}{20} = 0.05$.

If vitamin E really made no difference to these measured outcomes, a result at least this favorable to vitamin E would occur in only one of the 20 possible assignments.

That makes the observed result hard to reconcile with the no-effect setup. N.B.: It does not mean that the probability of the hypothesis is 5%. If we repeatedly used a valid test with a 5% rejection threshold in situations where the no-effect hypothesis was actually true, the procedure would falsely reject that hypothesis only about 5% of the time or less.

That is the frequentist error guarantee. It concerns the long-run behavior of the procedure, not the probability that this particular hypothesis is true.

It is a common convention to reject the no-effect hypothesis when the p-value is at most 5%.

This differs from the Bayesian case in an important way. There we assigned probability to competing hypotheses and changed those probabilities when evidence arrived. Here we have not assigned a probability to the no-effect hypothesis. We assume it for the purpose of the test and ask how often results like the one we observed would arise under it.

Both approaches use a model of what the evidence would look like under a hypothesis. Bayesian updating uses that relation to redistribute probability among hypotheses. The randomization test uses it to estimate how likely the observed result and more extreme results are compared to the other outcomes expected under the hypothesis being tested.

Sampling From A Population

Randomization is only one route within frequentist statistics.

Another large family of methods is based on sampling.

Instead of asking how treatment assignments could have varied under an experimental design, we ask what observations drawn from a population can tell us about that population.

The familiar urn picture makes the basic relation visible.

Imagine an urn containing a very large number of red and blue balls.

We do not know the proportion of each color.

Let p be the proportion of red balls in the urn.

If we knew that $p = 0.60$, probability theory could tell us how samples from the urn would tend to vary.

A sample of ten balls could easily contain somewhat more or fewer red balls than the population proportion would suggest.

If we repeatedly drew random samples of the same size from the same population, the resulting sample proportions would form a probability distribution.

Now reverse the problem.

Suppose we do not know p.

We draw a random sample and observe 60 red balls among 100.

The sample proportion is:

$$\hat{p} = \frac{60}{100} = 0.60.$$

Another sample from the same urn might give 0.57, and another 0.64.

The statistical problem is to understand how much variation the sampling process itself can produce.

Without going into the calculation, samples like these would be much easier to obtain from a population with p near 0.60 than from one with p near 0.90 or 0.10.

Probability theory lets us work out how compatible different population values are with the observed sample under the assumed sampling process.

In a small finite case, we could make this completely explicit, as we did throughout the main text.

Suppose the urn contained only ten balls, six red and four blue, and we drew three without replacement. Then we could lay out every possible sample of three balls and count how often each number of red balls occurs.

A sample with three red balls would appear in some of those possible samples. A sample with no red balls would appear in others. The probability distribution over the possible samples could be reconstructed directly from the finite arrangement.

If we now started from an observed sample instead, we could compare it with the distributions produced by different possible urn compositions. A composition under which samples like ours occur often fits the observation better than one under which they are rare.

For larger populations and samples we usually do not enumerate every possibility by hand. The mathematics compresses the same kind of relation. Once a population model and a sampling process are supplied, they constrain which samples are more or less likely.

The observed sample can then be compared with those possibilities.

But the application does not establish its own prerequisites. The sample may not have been produced in the assumed way, or the population represented in the model may not be the one relevant to the question.

These prerequisites still have to be sufficiently convincing for the procedure to apply.

Frequentist And Bayesian Uses Of Probability

In both frequentist examples, probability describes possible data under a setup that has been supplied, and the observation is assessed in relation to those possibilities.

In the randomization example, the six measured outcomes were already there. We held them fixed and asked what differences could have appeared if the treatment labels had been assigned differently. The randomization procedure supplied those possible assignments and their probabilities under the hypothesis of no effect.

In sampling the population is not fully known. We observe only a sample from it and ask how samples would tend to vary under different possible population values.

Randomization and sampling require different things to be in place before probability can do any work.

The randomization example needed experimental units, measured outcomes, treatment labels, and a procedure by which those labels were assigned.

The sampling example needed a population, together with a process by which observations were sampled from it. The urn picture makes that concrete: the balls stand for the population, and drawing balls stands for sampling.

Bayesian updating used probability differently. There, probability was assigned to competing hypotheses and redistributed when evidence arrived.

The frequentist and Bayesian approaches represent different kinds of uncertainty.

Randomization works with possible rearrangements of an experimental setup, while sampling starts from a population or generating process and a relation between it and the observations.

In both cases, the probabilities concern possible outcomes of a process taking place in the world: other assignments that could have occurred, or other samples that could have been drawn.

Bayesian updating needs a space of hypotheses and a way of relating each hypothesis to the evidence. Here the process being modeled is how inquiry changes the relative weight of competing hypotheses in light of evidence. Probability is used to represent something like the relative plausibility of those hypotheses.

The possibilities being weighted are not alternative assignments or samples, but alternative accounts of what may be the case.

That difference changes how the observation is used.

In the randomization test we asked: if the treatment had no effect, how often would an assignment produce a difference at least this large?

In sampling we asked: if the population had a particular composition, how would samples from it tend to vary?

In Bayesian updating we asked how the evidence should redistribute weight among the represented hypotheses.

Bayesian updating has the central update rule we have already seen: probabilities are assigned to hypotheses and Bayes' theorem updates them from the data.

In frequentist inference, the data can count for or against a hypothesis or population value without that hypothesis or value itself being assigned a probability.

There is no single frequentist update rule corresponding to Bayes' theorem. Different procedures build probability distributions from different parts of the setup: random assignments, repeated samples, test statistics, estimators, and other constructions.

In the frequentist examples above, the constraint came from probabilities that could be traced back to represented possibilities generated by an experimental or sampling arrangement.

In the randomization case we could lay those possibilities out almost completely. In sampling, the same relation is usually compressed into mathematical models of how repeated samples would behave.

Bayesian updating puts probability on the hypotheses themselves.

Across the examples, the constraint came from representing relevant possibilities and assigning or updating their weights in ways that could be inspected.

Is any of these ways of using probability in inference better than the others?

It can be, depending on what we are asking of it. One procedure may be better than another for a specific problem by a criterion such as calibration, error control, predictive performance, transparency, computational cost, or how well its assumptions fit the problem.

But then the criterion itself can become part of the inquiry. Why should that criterion matter here? Does it conflict with another one? Does it still seem convincing once its consequences and assumptions are made explicit?

And that is exactly the kind of problem Part II has been following all along.

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1. cf. Phillip I. Good, *Resampling Methods*, 3rd Ed., Birkhäuser Boston, 2006, ch. 3.

Sources Of Constraint In Pearl's Theory Of Causality

1. From Observation To Intervention

What If I Sleep Longer?

You have been keeping track of your sleep and mood.

After a short period of time, you notice already: on days after you sleep longer, you are usually in a better mood.

The pattern becomes convincing enough that you start saying: "I feel better because I sleep longer."

You may even decide to act on it. If more sleep produces the better mood, going to bed earlier should help.

But most of the long nights in your record were weekends and holidays. You did not have to get up for work, and the day was less hurried.

So what would actually happen if you deliberately started sleeping longer?

Your records tell you what mood was like after nights on which longer sleep occurred.

Let $S = 1$ mean that you slept longer than usual, and let $M = 1$ mean that you were in a good mood that day. $S = 0$ is your usual sleep baseline.

Your records suggest:

$$P(M = 1 \mid S = 1) > P(M = 1 \mid S = 0).$$

In other words, good mood was more common after nights on which you slept longer than after nights on which you slept as usual.

But what you want to know now is what mood would be like if you made longer sleep occur. The question looks similar, but asks something quite different.

Pearl introduced notation to keep this *intervention question* apart from the *observational question* the records answer[1].

The observational comparison tells us what happened on nights when longer sleep occurred.

But those nights differed from the other nights *in more than sleep duration.* Those circumstances could affect both how long you slept and how you felt the next day.

Observing $S = 1$ can carry information about those circumstances.

Now imagine arranging things so that you sleep longer.

We represent that intervention as:

$$P(M \mid do(S = 1)).$$

The expression $do(S = 1)$ represents intervening so that longer sleep occurs, rather than selecting nights from the record on which it happened anyway.

Pearl's notation gives us a way to keep observation and intervention apart as the inquiry becomes more formal.

That is important when probability alone leaves us with a problem.

As with logic and probability, this essay is not an introduction to Pearl's theory of causality. The aim is to examine how causal assumptions and graphical representations can constrain questions about intervention, and what gives their application to a particular problem convincing force. We will go only as far into the theory as we need to for that purpose.

---

1) cf. Judea Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed., Cambridge University Press, 2009.

2. One Table, More Than One Causal Story

Suppose a hospital introduces a new treatment for a serious illness.

After a year, there are records for 100 patients.

Half were in relatively good general condition when treatment decisions were made. Half were in relatively poor condition.

But the groups were not treated equally.

Among the 50 patients in good condition, 10 received the treatment and 40 did not.

Among the 50 patients in poor condition, 40 received the treatment and 10 did not.

The recovery pattern looked like this: among patients in good condition, 9 of the 10 treated patients recovered.

Let T mean that the patient received treatment, R that the patient recovered, and C that the patient was in relatively good general condition. Then $\neg C$ means relatively poor general condition.

That gives:

$$P(R \mid T \wedge C) = \frac{9}{10} = 0.9.$$

Among the 40 untreated patients in good condition, 32 recovered:

$$P(R \mid \neg T \wedge C) = \frac{32}{40} = 0.8.$$

Treatment looks better within this group.

The same happens among patients in poor condition.

Of the 40 treated patients, 16 recovered:

$$P(R \mid T \wedge \neg C) = \frac{16}{40} = 0.4.$$

Of the 10 untreated patients, 2 recovered:

$$P(R \mid \neg T \wedge \neg C) = \frac{2}{10} = 0.2.$$

Again treatment looks better.

Within both health groups, treated patients recovered more often.

$$P(R \mid \neg T \wedge C) < P(R \mid T \wedge C)$$

and

$$P(R \mid \neg T \wedge \neg C) < P(R \mid T \wedge \neg C).$$

But something interesting happens when we ignore the condition the patient is in.

There were 50 treated patients altogether. Of these,

$$9 + 16 = 25$$

recovered.

So:

$$P(R \mid T) = \frac{25}{50} = 0.5.$$

There were also 50 untreated patients. Of these,

$$32 + 2 = 34$$

recovered.

So:

$$P(R \mid \neg T) = \frac{34}{50} = 0.68.$$

The aggregate comparison has reversed the result:

$$P(R \mid T) < P(R \mid \neg T).$$

Among all patients together, the treated patients recovered less often. Such a reversal is called Simpson's paradox.

It is true that 50 percent of the treated patients recovered. It is also true that 68 percent of the untreated patients recovered. And it remains true that within each health group the treated patients recovered more often.

That leaves us with a problem the probabilities alone do not settle.

Suppose another patient now arrives.

Should the hospital expect treatment to improve the patient's chance of recovery?

The aggregate comparison says that treated patients recovered less often. The comparisons within the two health groups say that treated patients recovered more often. All of those probabilities are correct.

The causal question remains open: what would happen if treatment were deliberately given?

Simpson's paradox makes the problem especially vivid, but the problem is more general.

Probability theory tells us how the variables are distributed together.

For an intervention question, Pearl adds a way to represent claims about what can causally affect what.

Those claims can change which probabilistic comparison bears on the question we are asking.

A Small Causal Graph

Let:

$$G = \text{general health},$$

$$T = \text{treatment},$$

and:

$$R = \text{recovery}.$$

Suppose doctors are more likely to give the treatment to patients in poor general health.

We can represent that causal claim with an arrow:



The arrow says that, in this model, general health can influence whether treatment is given.

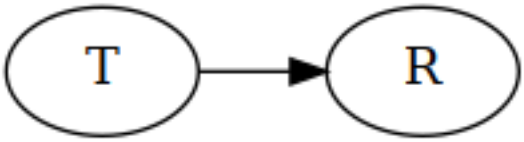
It says more than that G and T occur together statistically.

We are claiming a direction of influence.

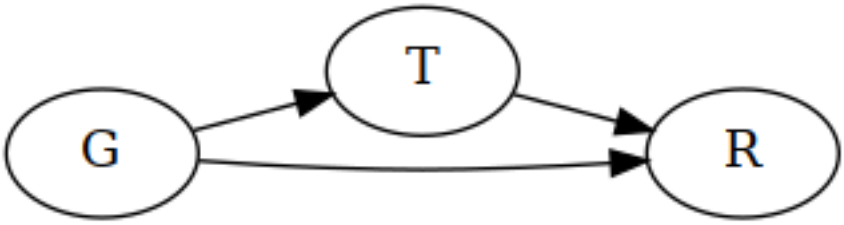
General health can also affect whether a patient recovers:



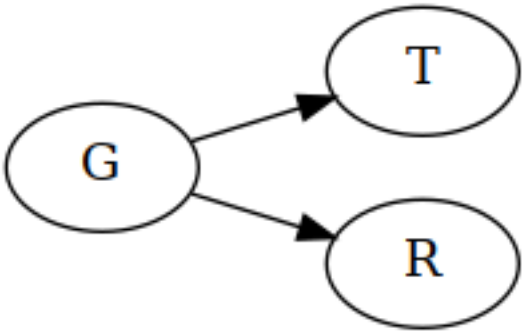
The model also includes a direct causal effect of treatment on recovery:



Together these claims give us a small causal graph:



General health can influence both treatment and recovery:



That helps explain why simply observing treatment can tell us something about a patient's health.

In our records, most treated patients were in poor condition:

40 poor, 10 good.

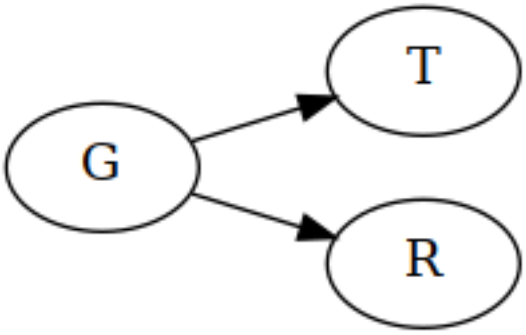
Most untreated patients were in good condition:

10 poor, 40 good.

So the two treatment groups differ in more than treatment.

That matters because our model also says that general health affects recovery.

The graph we have just seen makes the problem visible:



Patients in poor health are more likely to receive treatment, and they are also less likely to recover.

So part of the lower recovery rate among treated patients can come from who receives the treatment rather than from what the treatment does.

If this causal model is right, the aggregate comparison:

$$P(R \mid T) < P(R \mid \neg T)$$

does not isolate the effect of treatment.

Under this causal model, the comparisons within the two health groups seem better suited to the intervention question. There, patients with the same general-health status are being compared, and in both groups the treated patients recovered more often.

The graph has given us a reason to favor that comparison for the causal question.

But that reason came from the causal claims represented by the arrows. The probabilities alone did not tell us that general health should be treated this way.

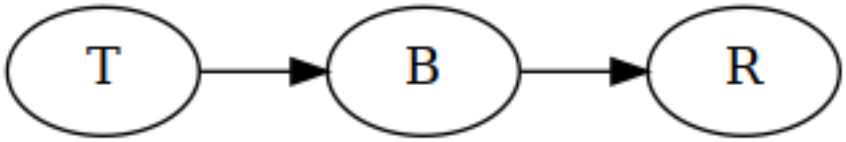
Another Causal Story

Now imagine that the same pattern of numbers arose in a different hospital.

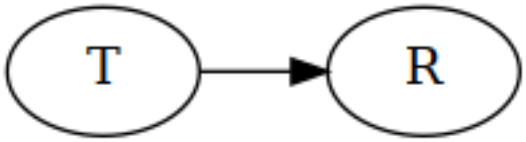
This time the variable separating the two groups is not general health before treatment.

Suppose it is a biological condition measured after treatment, say blood pressure. Call it B.

The treatment can affect B, and B can affect recovery:



Suppose treatment can also affect recovery in other ways:

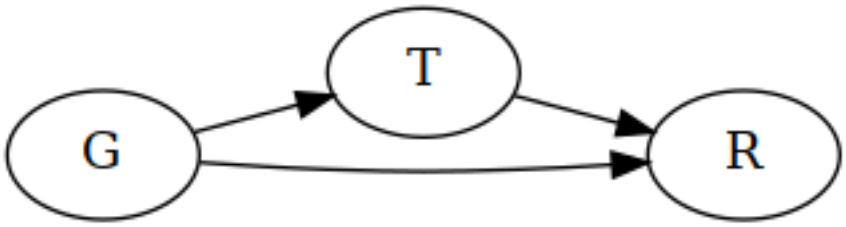


The numbers could still show the same reversal.

Within each value of B, treated patients might recover more often, while treated patients recover less often in the population as a whole.

But the causal story has changed.

In the earlier model, general health helped determine who received treatment and also affected recovery:



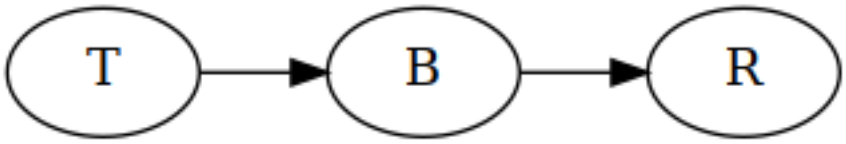
Comparing patients with the same general health removed that difference between the treatment groups.

Here B comes after treatment in the causal structure.

Suppose the treatment lowers the value of B.

And suppose a lower value of B improves recovery.

Then one way the treatment can help a patient is:



Part of the treatment's effect on recovery travels through B.

Now imagine comparing treated and untreated patients only among people with the same value of B.

Within that comparison, we have deliberately removed the difference in B.

But in this causal story, changing B was one of the ways treatment could change recovery.

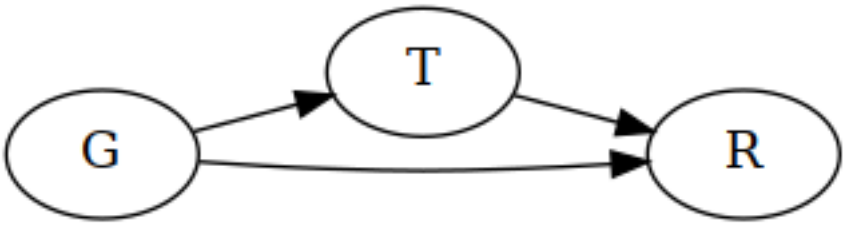
So by holding B fixed, we also remove that part of the treatment's effect from the comparison.

The within-B comparisons are still valid probability comparisons, but they do not contain the whole effect we are asking about.

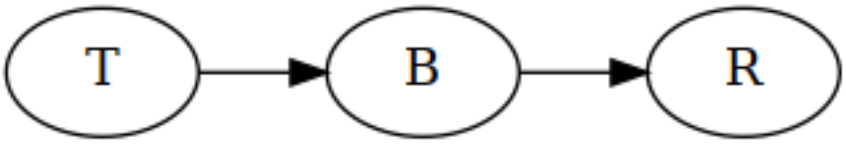
If we want to know the overall effect of giving the treatment, including whatever effect operates through B, then the aggregate comparison may now be the relevant one.

The numbers stay the same. What changes is the causal role of the variable we were conditioning on.

In the first model, G came before treatment and influenced both treatment and recovery:



In the second, B lies on a route through which treatment can influence recovery:



Holding it fixed removes part of the effect we want to know about.

Probability can tell us how treatment, the third variable, and recovery are distributed together. It cannot by itself tell us whether that third variable is a common cause, something produced by treatment, or something else.

The arrows add those causal claims. For an intervention question, they can change which probabilistic comparison bears on the answer.

What The Comparison Shows

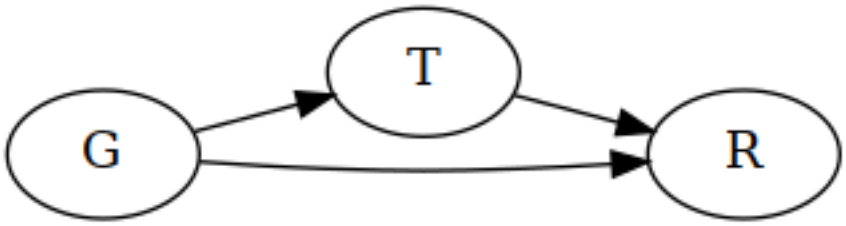
Probability alone has not determined which comparison answers the intervention question.

The causal representation adds structure that can help with that.

Again there is the constraint produced once enough of the representation is in place. And there is the separate question whether that representation is convincing for this problem.

3. What Changes Under Intervention

Chapter 2 gave us this causal model:



General health affects both treatment and recovery.

So when treatment is merely observed, a patient's value of T can also carry information about G.

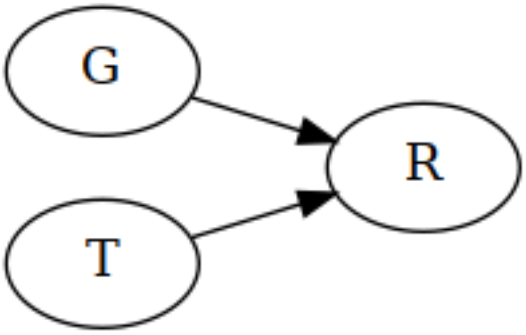
An intervention changes that.

Under:

$$do(T = 1),$$

treatment is set *independently of its ordinary causes.*

This gives a more precise meaning to this notation informally introduced in Chapter 1: In the graph, the incoming arrow from general health to treatment is removed while everything else remains untouched:



Pearl calls the removal of the incoming arrows into the intervened variable *graph surgery*. This makes the difference between observation and intervention visible.

In:

$$P(R \mid T = 1),$$

we condition on patients whose treatment status arose through the ordinary hospital process.

In:

$$P(R \mid do(T = 1)),$$

treatment has been fixed, so general health no longer determines whether it is received.

But changing the graph does not yet give us the probability of recovery under the intervention.

Our records came from the original hospital process, where general health helped determine treatment. We do not have records from the modified process represented by:

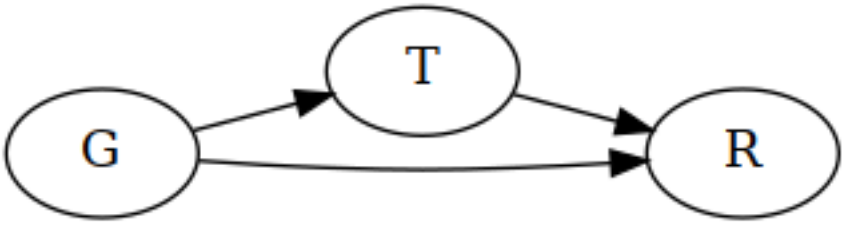
$$do(T = 1).$$

Can the observational data we already have tell us what recovery would look like after that intervention?

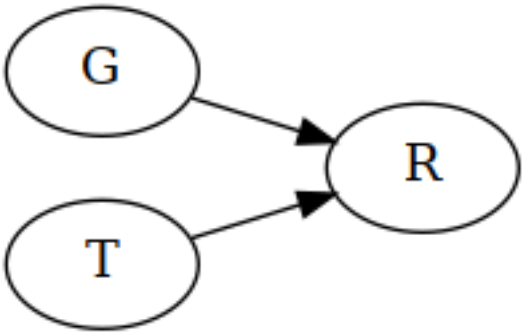
4. Recovering An Intervention From Observation

In the previous chapter we had changed the causal graph.

Before intervention it was:



When setting $do(T = 1)$, the arrow from general health into treatment was removed:



But our records came from the first situation, not the second.

Can we use them to work out what recovery would look like in the modified graph?

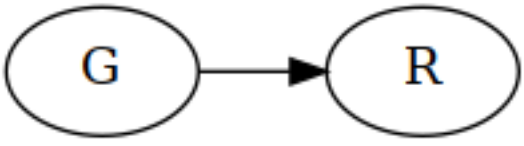
What The Intervention Changes

In the hospital records, treatment depended partly on general health.

That gave us a treated group consisting of 10 patients in good condition and 40 in poor condition. The untreated group had the opposite mixture: 40 in good condition and 10 in poor condition.

Under the intervention, this changes. $Do(T = 1)$ means that treatment is assigned to everyone by intervention. General health no longer helps determine who receives it.

But the intervention does not change the patients' general health. Our model still contains:



And the population still consists of 50 patients in good condition and 50 in poor condition.

So the group produced by $do(T = 1)$ would have a very different health composition from the treated group in our records.

That is why:

$$P(R \mid T = 1) = 0.5$$

cannot simply be used as:

$$P(R \mid do(T = 1)).$$

Comparing Patients With The Same General Health

The observed treated group contains mostly patients in poor health. The intervention group would contain the original population mixture.

Look first only at patients in good general health.

Among them:

$$P(R \mid T \wedge G) = 0.9,$$

while:

$$P(R \mid \neg T \wedge G) = 0.8.$$

Within this comparison, general health is the same on both sides.

Now look only at patients in poor general health:

$$P(R \mid T \wedge \neg G) = 0.4,$$

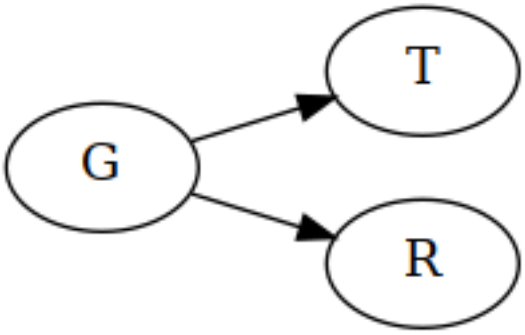
while:

$$P(R \mid \neg T \wedge \neg G) = 0.2.$$

Again, general health is the same within the comparison.

Why does that help?

The problem in the aggregate comparison came from this path:



General health helped determine treatment, and it also affected recovery.

Within each health group, treated and untreated patients have the same value of G. Their different recovery rates cannot come from one group containing healthier patients than the other.

In our small causal model, general health is the only variable that affects both treatment and recovery.

So within each health group, the observed treatment comparison is no longer mixed with differences in general health.

We can use those rates to reconstruct what would happen if treatment were assigned independently of general health.

Reconstructing The Intervention

Imagine the 100 patients again.

Under:

$$do(T = 1),$$

all 100 receive treatment.

The intervention does not change their general-health status, so 50 are still in good condition and 50 in poor condition.

For the 50 patients in good condition, our model lets us use the observed treated recovery rate:

$$0.9.$$

For the 50 patients in poor condition:

$$0.4.$$

So the expected number of recoveries is:

$$50 \cdot 0.9 + 50 \cdot 0.4 = 65.$$

Across all 100 patients:

$$P(R \mid do(T = 1)) = 0.65.$$

Now imagine assigning no treatment to the same population.

The health distribution again remains 50 good and 50 poor.

Using the corresponding observed rates gives:

$$50 \cdot 0.8 + 50 \cdot 0.2 = 50.$$

So:

$$P(R \mid do(T = 0)) = 0.5.$$

Under the causal model we have supplied, assigning treatment raises the recovery rate from 0.5 to 0.65.

That is the opposite direction from the raw observational comparison:

$$P(R \mid T) = 0.5$$

against:

$$P(R \mid \neg T) = 0.68.$$

Adjustment

What we have just done is a simple case of *adjustment*.

We separated the population by general health, used the treatment comparison within each health group, and then rebuilt the intervention population using the original distribution of general health.

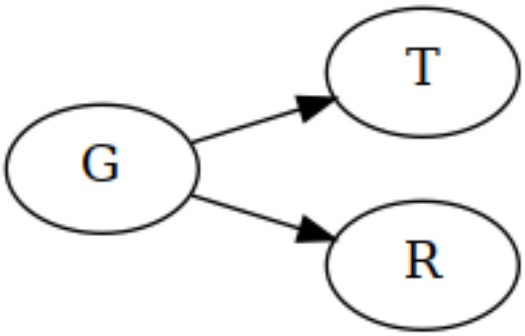
In proportions, the treatment calculation was:

$$0.5 \cdot 0.9 + 0.5 \cdot 0.4 = 0.65.$$

More generally, when a variable Z is appropriate for this adjustment, the same procedure can be written:

$$P(Y \mid do(X = x)) = \sum_{z} P(Y \mid X = x \wedge Z = z)P(Z = z).$$

For each value of Z, take the corresponding conditional outcome probability and weight it by how common that value of Z is in the population. The formula compresses the finite reconstruction. The graph tells us how to reconstruct:



General health created the path that made the aggregate treatment groups different in a causally relevant way.

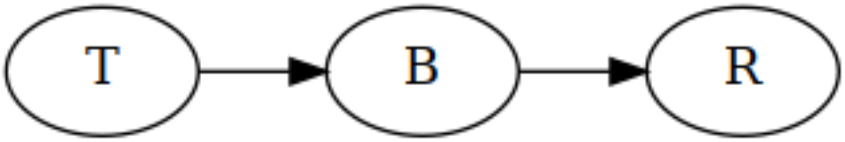
Holding G fixed blocked that path.

Intuitively, this works because within each health group, general health no longer varies between the treated and untreated patients. Its effect on recovery is still present, but it is no longer mixed into the treatment comparison.

We then put the health groups back together according to how common they are in the population. The influence of general health is still there, but it is now carried through the population mixture rather than through different treatment groups.

Chapter 2 already showed why this cannot become a general instruction to condition on another variable.

If the variable lies on a route through which treatment affects recovery:



Holding B fixed removes part of the effect we may be trying to estimate.

The probability operations can be performed in either case.

The graph tells us what they mean causally and whether they bear on the intervention question.

What The Reconstruction Shows

We began with observational records that seemed to point in the opposite direction from the intervention result. The causal graph gave us a reason to organize those records differently for the question we were asking.

Once the graph, the recovery rates within the health groups, and the population distribution of general health remain fixed, the result is no longer independently open. Another intervention probability would require something in the graph, the observed rates, the population distribution, or the calculation to change. That is the constraint the reconstruction has made visible.

But its convincing force in the inquiry remains conditional. The calculation can be correct while the causal model is wrong.

We have seen all of that already several times.

5. Why The Shape Of The Graph Matters

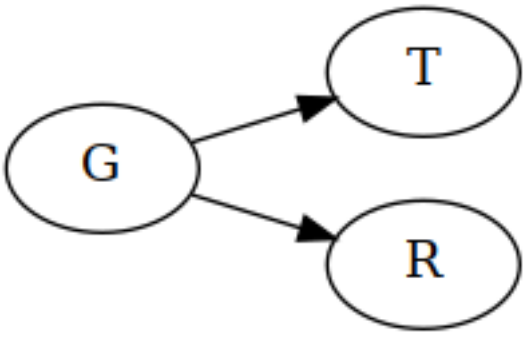
We have already seen that the effect of conditioning depends on where a variable sits in the causal structure.

A common cause and an intermediate variable behaved differently even though the same probability operation could be applied to both.

With three variables, there are only a few basic local shapes to consider. Looking at them side by side makes the differences easier to see.

The Fork

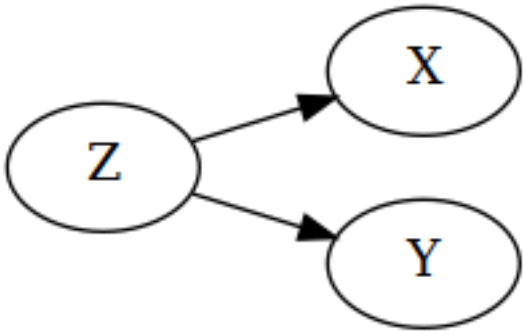
Return to the hospital example:



General health affects both treatment and recovery.

This shape is called a *fork*.

More generally:



Z is a common cause of X and Y.

That is the role general health played in our hospital model.

Observing treatment could tell us something about general health, and general health could tell us something about recovery.

So treatment and recovery could be associated partly because both were connected to G.

When we compared treated and untreated patients with the same general-health status, that route could no longer explain the difference between them.

For the causal question we were asking, conditioning on G helped separate the effect of treatment from the common-cause path.

That is why adjustment worked in the previous chapter.

The Chain

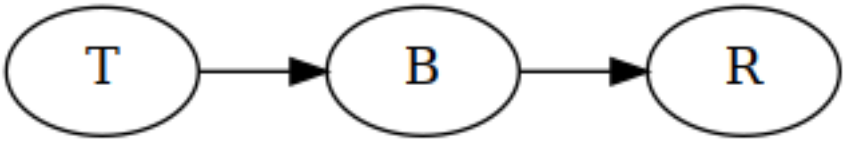
Suppose instead like in the second example of chapter two:



Here Z is not a common cause.

It lies on a causal route from X to Y.

Suppose again treatment reduces blood pressure, and reduced blood pressure improves recovery:



Part of the treatment's effect on recovery now operates through blood pressure.

Imagine comparing treated and untreated patients only among people with the same blood pressure level.

Within that comparison, differences in blood pressure have been removed.

But blood pressure is one of the ways treatment changes recovery in this model.

So holding B fixed also removes that part of the treatment's effect from the comparison.

The probability operation is familiar. We condition on a third variable. But its causal role is now very different.

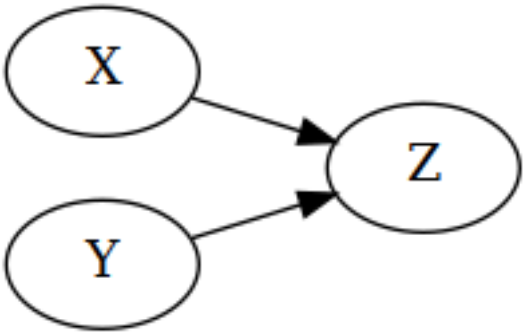
In the fork, conditioning helped remove a route that was not part of the treatment's effect.

In the chain, conditioning removes a route through which the treatment itself can work. That is not what you want to determine the treatment effect.

The arrows make that difference visible.

The Collider

Suppose:



Both X and Y can affect Z.

The arrows meet at Z, so this shape is called a *collider*.

Consider a school that admits applicants for either strong academic results or strong athletic results.

Let:

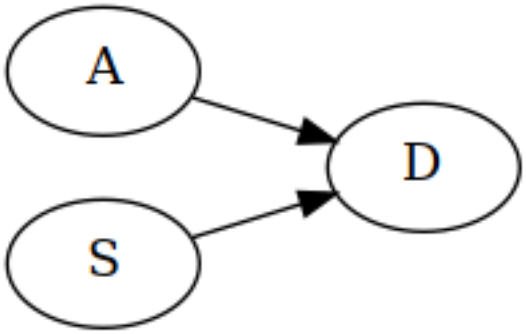
$$A = \text{academic strength},$$

$$S = \text{athletic strength},$$

and:

$$D = \text{admitted}.$$

We can represent the small causal structure as:



For simplicity, suppose academic and athletic strength are unrelated in the whole applicant population.

Now look only at admitted applicants.

Among them, learning that an applicant was not especially strong academically gives us a reason to expect some other route to admission.

In this small model, athletic strength is one such route.

Likewise, if an admitted applicant had exceptionally strong academic results, less athletic strength may have been needed for admission.

Once we restrict attention to admitted applicants, academic and athletic strength can become statistically related even though they were unrelated in the full applicant population.

Conditioning on the common effect has *created* an association.

That is the opposite of what happened in the fork.

There, conditioning on the common cause helped *remove* an association carried through that cause.

So there is no general causal rule saying that conditioning on more variables improves the analysis.

The same probabilistic operation can help remove a misleading association, remove part of the effect we are trying to measure, or create a new association.

Which of those happens depends on the causal structure. Graphs make those differences available for inspection. The *do*-notation then lets us mark the intervention whose consequences we want to follow through that structure.

What The Shapes Show

The fork, chain, and collider used the same basic graphical vocabulary.

The same probability operation, conditioning, was also available in every case.

Yet its causal significance changed with the arrangement of the arrows.

In the fork, conditioning could remove an association carried through a common cause.

In the chain, it could remove part of the effect we wanted to examine.

At a collider, it could create an association that was not present before.

The arrangement does more than display several causal claims conveniently. Once its interpretation is accepted, the structure constrains what the same probabilistic operation can tell us about an intervention.

This also makes disagreement easier to locate.

Two people may agree about the probabilities and about how conditioning works while disagreeing about whether a variable is a common cause, an intermediate variable, or a common effect.

The disagreement has moved from the calculation to the causal representation.

Making that dependency explicit does not resolve the disagreement.

It makes more of what carries the causal conclusion available for examination.

6. When Does The Answer Follow?

In the hospital example, our records came from a process in which general health helped determine treatment.

We never observed the situation represented by:

$$do(T = 1).$$

Yet, once we supplied the causal graph, the observational records were enough to determine:

$$P(R \mid do(T = 1)) = 0.65.$$

Why were they enough?

The question is whether the observations and causal assumptions already fix the intervention answer.

In this example, they did. The graph told us why general health had to be adjusted for, and the records supplied the recovery rates within each health group and the distribution of general health. Once those pieces were held in place, the result was fixed:

$$0.5 \cdot 0.9 + 0.5 \cdot 0.4 = 0.65.$$

We could not keep those inputs and freely choose another value for:

$$P(R \mid do(T = 1)).$$

When an interventional quantity can be determined from the observational information together with the causal assumptions, it is said to be *identified*.

When The Answer Is Not Fixed

Having observational information and a causal model does not guarantee that an intervention quantity will be identified.

We can see why by changing the hospital example slightly.

Our adjustment worked because general health was represented in the causal model and recorded in the data.

We knew:

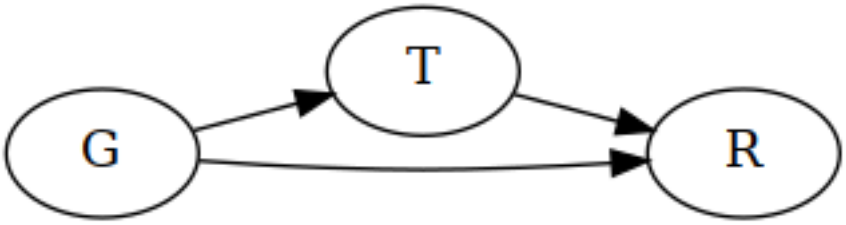
$$P(R \mid T, G),$$

$$P(R \mid T, \neg G),$$

and how common the two health conditions were in the population.

That allowed us to reconstruct the intervention result.

Now suppose the causal structure remains:



But general health was not recorded. We observe treatment and recovery, but not G.

The aggregate records can still tell us:

$$P(R \mid T)$$

and:

$$P(R \mid \neg T).$$

But our graph says that these comparisons mix the effect of treatment with the influence of general health on both treatment and recovery.

The adjustment we used before is no longer available, because the observational information needed to condition on G is missing.

And the remaining observations need not determine a unique intervention answer.

Different causal processes compatible with the causal assumptions and the same observed distribution of treatment and recovery can imply different values for:

$$P(R \mid do(T = 1)).$$

In that case, the effect is not identified from the observational information available to us under the model.

The intervention may have a determinate causal result. But the observations and causal assumptions supplied so far do not fix that fact, and more calculation using the same information cannot remove the remaining openness.

If general health was not measured, and no other observed variable or further causal assumption makes the effect recoverable in another way, we need different data or an actual intervention.

Identification marks that limit: it tells us whether the causal quantity is already fixed by the observational information under the causal assumptions we have supplied.

7. What Causality Adds

The causality essay distinguished two kinds of claims.

What happened on nights when I slept longer:

$$P(M = 1 \mid S = 1).$$

What would happen if I deliberately made longer sleep occur:

$$P(M = 1 \mid do(S = 1)).$$

These claims can look almost the same in ordinary language, but one describes an observed situation and the other an intervention.

Pearl's notation gave us a way to keep that difference explicit as the inquiry became more formal.

In simple cases, the distinction may already be easy to recognize. It becomes much harder to keep track of when observational comparisons pull in different directions. The hospital example showed why that matters.

The observed probabilities supported several different comparisons. Which one bore on the intervention question depended on how treatment, general health or blood pressure, and recovery were causally related.

The graphs made those causal claims explicit.

Where The Constraint Comes From

Once those claims were made explicit, they could begin to constrain what followed.

The arrows of the graph represented claims about what could causally affect what. Diagrams helped keep all of these of these relations visible at once.

Once those causal relations were fixed, an intervention on T had a specific effect on the representation. Setting T by intervention removed the incoming causes of T while leaving the other represented causal relations in place. The lack of incoming arrows became the definition of $do(T)$.

In the hospital example, the graph showed why the aggregate treatment groups mixed together treatment and general health, and why comparing within levels of G removed that common-cause route.

As in the investigations of logic and probability, once enough of the representation was held fixed, the result was no longer independently open.

Identification adds another kind of constraint. It asks whether the causal assumptions and observational information already supplied are enough to determine the intervention quantity.

Sometimes the answer is no. Then the failure is informative. It tells us that the intervention result is not fixed by the materials we have supplied. More calculation from those same materials cannot make one answer follow.

When the quantity is identified, the causal structure and probability information determine it.

What The Investigation Found

Applying the causal framework depended on more than the formal operations or the drawing of diagrams.

The variables had to preserve distinctions that mattered, and the arrows had to represent causal relations we found convincing enough to rely on. $Do(X)$ also had to correspond closely enough to the change we were asking about. The observational information, in turn, had to be relevant to the quantities used in the calculation.

By now, that is a familiar result.

Logic depended not just on having premises and interpretations, but on those premises, mappings, and interpretations being convincing enough for the argument at hand. Probability depended not just on assigning weights, but on the represented possibilities and their weights being appropriate to the problem. Bayesian updating added further requirements: the hypothesis space, priors, likelihoods, and representation of the evidence all had to be convincing enough for the update to bear on the inquiry.

Causal reasoning added another set of application commitments: variables, causal relations, interventions, and observational information all had to fit the problem closely enough for the result to matter.

But the investigation also returned us to something that was there before the formal machinery. We already understand, in ordinary life, that observing two things together does not tell us what will happen if we deliberately make one of them occur. The shadow of a large tree moving during daylight goes along with the Earth turning. But making that shadow move with a large lamp does not make the Earth turn. We do not confuse the two, because ordinary thinking already carries causal intuition about what produces what.

But this intuition doesn't necessarily carry over to new cases. The application of probability theory and statistical analysis can make relations visible that are difficult to interpret causally without further structure. This becomes harder when several causal relations interact, when conditioning changes what we are comparing, or when the observed numbers pull in opposite directions. In the hospital example, the aggregate and conditional comparisons did exactly that.

Formal systems can make cases available that ordinary intuition does not handle easily. With the probability formalism we could construct complicated aggregate and conditional comparisons, all of them valid as probability calculations, without probability theory itself settling which comparison answered the causal question.

Pearl's framework gave us a way to preserve that difference formally, represent causal assumptions around it, and follow consequences that were difficult to keep apart in ordinary thought. Diagrams added a formal tool the earlier investigations had not used. They kept several causal relations visible at once and made the relevant arrangements inspectable. A common cause, an intermediate variable, and a collider could all involve the same three variables and the same operation of conditioning, while changing what that operation meant for the causal question.

One formal system can make relations available that exceed what ordinary intuition can still keep apart, and another can make the distinctions needed to examine them explicit. It can hold the relevant differences steady long enough to inspect them. And it can make further questions available: which path is being opened or blocked, what changes under intervention, and whether the information already supplied is enough to determine the answer.

Constraint Without Final Authority

1. What Part II Was Looking For

An investigation does not always begin with a clear account of everything it will need.

Sometimes a problem can be stated before all of the distinctions on which its examination will depend have themselves become objects of attention. As inquiry proceeds, something that had remained in the background may become visible through the work it is already doing.

At that point we may know more clearly that something needs to be distinguished than we know exactly how the distinction should be understood.

Much of Conviction Formation Theory developed in this way.

The claim that becoming convinced is not under direct voluntary control opened the investigation, but following it repeatedly brought further distinctions and dependencies into view: orientation, the role of questions and possibilities, criteria, externalization, and participation. Recursive examination arose when the criteria and methods used in inquiry themselves became part of the problem.

Making something explicit does not settle its standing. What appeared important at first may later need qualification, division, or replacement. A representation can preserve the wrong thing, or omit something that later turns out to matter.

Still, inquiry need not wait until all of this has been settled. Sometimes the next step is to make clearer what is already carrying part of the inquiry, so that more can be asked of it.

Part II began from one such problem left by the first cycle.

We can examine convictions. We can ask what supports them, consider alternatives, compare evidence, or introduce criteria and procedures. Part I had already found that criteria can become sufficiently settled to organize inquiry, sometimes even between people whose larger orientations differ.

But those criteria remain inside conviction formation.

Logical validity has to acquire force for us, just as evidence or predictive success does. A criterion may organize an inquiry for years and later become questionable. Different criteria may also continue to matter while pulling an inquiry in different directions.

Recursive examination made this difficulty explicit.

A conviction may depend on an interpretation, an assumption, or a criterion that had remained in the background. Making such dependencies more explicit can expose further questions. And whatever we use to examine them can itself become an object of inquiry.

But a further question may be posed without becoming genuinely puzzling. We can examine a criterion while it continues to organize the inquiry. Usually only part of an orientation becomes unsettled while much of the rest continues to carry what happens next.

That gives inquiry enough local stability to continue, even while further examination remains possible.

What it does not yet give us is a criterion for assessing what happens through that examination.

Suppose an inquiry has exposed more of what a conviction depends on. Suppose a criterion has been applied carefully and an objection reconstructed. On what basis could we say that the conviction or the inquiry was now better, without simply introducing another criterion whose standing raised the same question?

The earlier notion of firmer ground gave only a partial answer. A conviction may survive examination that it had not undergone before. That can matter. It does not put different convictions on one general scale.

Part II turned to cases in which inquiry appeared to be strongly constrained while the conditions carrying that constraint could still be examined. Maybe something could be learned from them for our problem.

Logic, probability theory and Bayesian updating, and Pearl's theory of causality offered examples of such cases.

Logical conclusions can appear necessary. Probability constrains relations among possibilities even when several remain open. Causal models distinguish observed association from intervention.

These seemed like good places to look for constraint whose source could be reconstructed rather than merely asserted.

The strongest result would have been an independent criterion, something capable of governing inquiry without depending in turn on commitments that remained inside conviction formation.

Part II did not find such an independent criterion.

Logic required premises, interpretations, mappings, and a representation of the problem. Probability depended on a represented possibility space and on how weight was assigned within it. Causal reasoning depended on assumptions about causal structure and on how the situation was represented.

Any of these conditions could themselves become questionable.

But while enough of them remained convincing for the inquiry at hand, they could still constrain thinking.

2. Constraint Without Final Authority

The constraint the investigation found in logic was not quite the same as the constraint we found in probability or causality.

In logic, once enough of the representation remained fixed, a conclusion could follow or a counterexample could remain. Probability allowed several possibilities to stay open while constraining the weights they could carry. Bayesian updating constrained how those weights changed under supplied priors, likelihoods, evidence, and a model.

Causality added something different again. Under a causal model, observational information could sometimes determine an intervention quantity. In other cases it could establish that the quantity was not identified: more than one intervention answer remained compatible with what had been supplied.

The methods also had something more concrete in common. They made enough of the relevant structure available for inspection and manipulation. In the small logical cases we could enumerate the relevant possibilities and look for a counterexample. Probability let us construct and partition possibility spaces and follow how weight changed within them. Causal diagrams kept structural relations visible so that interventions and their consequences could be followed through the model.

Once enough of the arrangement remained fixed, other parts of the inquiry were no longer independently open. A conclusion followed, a counterexample remained, weights were constrained, an intervention quantity was identified, or several answers were shown still to fit.

This is the sense in which Part II found *constraint without final authority*.

The conditions carrying the constraint remained examinable. A premise could later become doubtful, or never become convincing in the first place. A probability model could omit an important possibility. Or a causal graph could represent the situation badly.

Their consequences were still constrained while they remained in place.

3. Distinctions Before Theory

Logic was built using the difference between a claim being true and a claim being false.

That difference is quite clear in many ordinary cases. The light is either on or it is not. A cup may be on the table when someone says that it is there, or the table may be empty. Someone may say that a meeting starts at ten when the schedule says eleven.

Cases like these give us enough of a distinction for inquiry to begin, even without the theory of truth being settled, or giving a criterion outside of conviction formation.

We could then understand why a counterexample mattered. An interpretation in which the premises remained true and the conclusion became false presented a problem for the proposed inference.

The formal system gave the distinction a more precise and stable role. Truth values received defined mappings, possibilities could be enumerated and inspected, and quantified logic added further structure by preserving relations among objects and properties that the smaller propositional system could not represent.

Once such structures are established and used, they can also become part of the inherited conditions of later inquiry.

Probability build on such a distinction as well.

Several possibilities can remain open while some carry more weight than others. New information can also change how strongly they bear on the inquiry. You may think one route home is more likely to be congested than another. A detective arriving at a broken shop window may consider accident, vandalism, or burglary. Seeing someone climb through the window carrying a bag of jewelry does not make every alternative impossible, but it changes how seriously they are taken.

The formal system gave this a more precise structure again, with similar consequences.

Causality builds on another distinction of this kind.

Observing what happens and deliberately changing what happens are not the same question. A doctor may notice that patients who received a treatment recovered more often than those who did not. That still leaves open what would happen if the treatment were deliberately assigned.

The formal framework gave this distinction a more precise structure. Causal graphs made assumptions about what affects what explicit, intervention was represented by changing that structure in a defined way, and identification let us ask whether the observational information and causal assumptions were enough to determine the intervention quantity.

Across the three cases, these distinctions were already doing work before any final theory of them had been settled.

They help keep apart cases that matter to the inquiry, and sometimes help give the question itself its shape.

They are also highly stable.

The distinction between a claim being true and false persists through deep disagreement about the theory of truth. Practices built on that distinction continue to develop without waiting for those disagreements to be settled.

Something similar holds for possibility and weight. Dice and card games make the distinction vivid, and it persists through different interpretations of what those weights mean. We routinely distinguish between alternatives that remain open while taking some more seriously than others. New information can change that balance without requiring us first to settle what probability ultimately means.

The causal case begins with another highly stable distinction. We routinely distinguish between cause and effect. Part I used many examples. That distinction persists even though theories of causation disagree about what causal relations are and how they should be understood.

But observing what happens and deliberately bringing something about need not be the same. In more complicated cases, the influence of different causes can become difficult to separate in the observations. Simpson's paradox is a case in point: an association can reverse when data are partitioned differently, while the causal question depends on which relations in the situation are actually doing the work.

Their stability does not place them outside examination. Nor does it magically turn them into foundations. What it does show is something narrower: theoretical disagreement need not prevent a distinction from remaining clear and usable enough to organize inquiry.

This recurring role seems worth naming.

Such a distinction will be called *operative* when it is clear and stable enough within orientation to organize an inquiry, while its theoretical standing remains unsettled and the distinction itself remains open to examination.

An inquiry need not wait for a completed theory *of* the operative distinctions it already relies on.

That helps explain how the formal systems examined here could get traction. True and false cases, differently weighted possibilities, and the difference between observation and intervention already mattered before they were given more precise formal roles.

A distinction can play different roles in an inquiry. It may be used to investigate something else while parts of its own role are made more precise, without the

inquiry thereby becoming a theory of that distinction. Classical logic can make the formal role of true and false cases precise without settling what truth is, just as probability theory can formalize weights without settling what probability ultimately means.

A theory of truth stands in a different relation to the true-false distinction. There the distinction itself, and what it amounts to, is part of what the theory is trying to explain.

Conviction Formation Theory works more like the first case. Part I relied on the operative difference between merely asserting, imagining, or acting as if something were true and actually becoming convinced. The investigation sharpened that difference and built further concepts around it without first requiring a completed theory of convincing force.

But being operative is not a warrant.

This matters for the problem Part II began with. We had asked what criteria could guide the examination and refinement of convictions when criteria themselves remained within conviction formation.

The investigations suggest that such criteria need not first become the objects of a completed theory before they can organize inquiry. Their standing can remain open while they are used, sharpened, and examined in the course of the investigation.

A distinction may be deeply embedded in orientation, widely used in a successful practice, difficult to dislodge, or clear in many ordinary cases. Those facts can help explain its persistence and force. They do not establish that the distinction is correct.

Otherwise we would simply have moved the search for final authority one step backward.

Instead of granting that authority to logic, probability, or causal theory, we would grant it to the distinctions that made those inquiries intelligible.

The investigations give us no basis for that move. What they seem to show is that finite inquiry does not always have to wait for the final theory of its starting distinctions.

Something already operative can give a question enough shape for inquiry to begin.

That does not mean it remains as it was.

4. What Finite Inquiry Can Do

Part I already gave external representation a practical role. Writing something down can preserve a possibility or comparison while attention moves elsewhere.

What had to be held in mind can remain available for another look.

Formalization can take this further.

A representation can preserve relations on which a result depends and make them available for inspection. The logic essay gave us a simple example. When the Socrates inference was represented propositionally, the repeated object and properties disappeared. The three claims became unrelated propositions. A richer representation preserved relations that the weaker one had lost.

Nothing about that made the richer representation correct merely because it was richer. It did make visible something the earlier representation could not carry.

This is one way an operative distinction can enter a more explicit arrangement. The distinction does not need to receive a final theory first. Parts of its role can become more precise while other assumptions, mappings, or relations are added around it.

More of what carries the inquiry can then become available for inspection. That gives us a limited sense of *refinement*.

A conviction need not change for the inquiry around it to become more refined. Something implicit may become visible, making it easier to locate what a result or disagreement depends on.

Refinement is not yet *improvement*.

Making more of an inquiry available for inspection does not tell us that one conviction is better than another. That still depends on the criterion by which they are compared, and Part II did not find a general criterion for settling that question.

But making more available still changes what can be examined within that inquiry.

This is where the formal investigations return us to *recursive examination*.

A distinction that helped organize a question can itself become an object of inquiry. So can a premise, a representation, or some other condition on which a result depends.

The possibility of doing this does not require everything to become unsettled.

Chapter 3 already gave us one reason. An inquiry need not possess a completed theory of every operative distinction or criterion it uses before those things can organize the inquiry.

Recursive examination gives us another. Making something available for examination does not decide what the examination will do to it. That follows from the same limitation with which Conviction Formation Theory began.

Becoming convinced is not under direct voluntary control.

We can participate in examination. We cannot choose its effect on what remains convincing.

Usually only part of what is carrying the inquiry becomes questionable. A distinction can stay operative while we examine one of its applications. A premise can continue to carry an argument while we consider what would happen if it were given up.

This is *settlement without final settlement*. What remains in place can constrain what follows. If a logical representation remains fixed, a surviving counterexample cannot simply be wished away.

The conditions can later change. That does not make their consequences arbitrary while they remain in place.

Settlement without final settlement and *constraint without final authority* describe different parts of the same local situation.

Settlement concerns what remains sufficiently in place for the inquiry to continue.

Constraint concerns what is no longer independently open while those conditions remain in place.

Something already operative can organize a question before its final theory has been settled. Representation can make more of the surrounding arrangement explicit. What becomes visible can be examined further. What becomes visible can then be examined without requiring the rest of the inquiry to be reopened.

Enough can remain settled for constraint to operate throughout.

Part I called our involvement in such processes *participation*.

We can alter some of the conditions under which conviction forms without controlling what will become convincing.

In the formal investigations, something related appeared: we could choose what to put into the representation and hold it in place, but not separately choose what followed from it, or what effect those consequences would have on conviction.

Finite inquiry does not have to wait for the final theory of everything it relies on. It can begin from distinctions clear and stable enough to organize a question, make more of what the inquiry depends on explicit, and continue from what that examination reveals. This gives us more ways to participate in inquiry without putting what will ultimately become convincing under our control.

We cannot wait until everything is finally settled. If final settlement is possible at all, it is hard to see how we could reach it except through investigation.